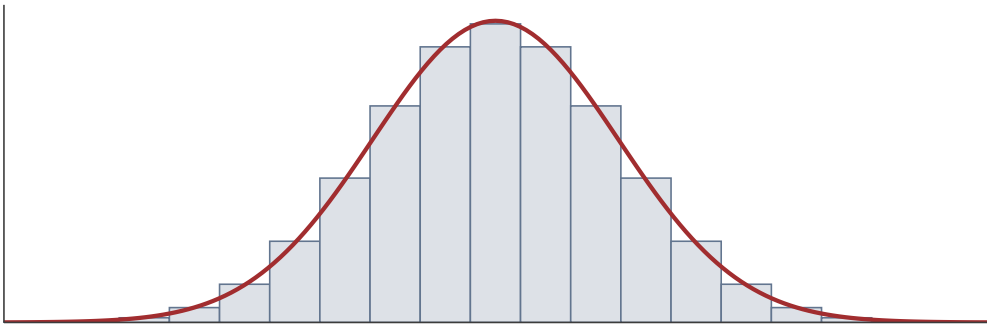
ST121 · LECTURE NOTES

Probability and Statistics I

Probability · Random variables · Joint distributions · Expected values · Limit theorems · Simulation



Binomial counts and their normal limit — Chapter 5 in one picture.

ORIGINAL NOTES

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TYPST EDITION & LECTURE NOTES

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Teaching Assistant — typeset, annotated, corrected, with practice problems

ST121: Probability and Statistics I — Lecture Notes.

Original notes, version of October 7, 2025: Copyright © 2011, 2025 Waleed A. Yousef. All Rights Reserved. Lectures follow Rice, *Mathematical Statistics and Data Analysis*, 3rd edition, Duxbury.

This Typst edition reproduces the original content — every definition, example, theorem and proof, with the original numbering — and adds, in clearly marked boxes, lecture notes, intuition, corrections, chapter summaries, practice problems by Mahmoud AboZied (Teaching Assistant). Figures taken from the original are reproduced from it; plots of densities and distributions are redrawn as vectors.

Typeset with Typst. No external packages.

About These Notes

Course objectives

- Developing rigorous treatment.
- Building intuition.
- Linking to real-life problems.

How to read this edition

The text of the original notes is set in the normal body style, in its original order. Boxes tell you what kind of material you are reading:

Definition (blue). definitions — the original numbering is kept: definitions, examples, lemmas, theorems and propositions share one running counter, as in the original.

Theorem (green). theorems, propositions, lemmas and corollaries from the original.

Example (grey). examples from the original, with their solutions.

Everything below is **added** in this edition and is not part of the original notes:

▸ **Lecture note.** explanations, connections between chapters, recipes for solving problems, and the conceptual discussions and guidance for independent study.

⋆ **Intuition.** a picture or a story that makes a formula obvious.

! Watch out. a mistake students commonly make.

Correction to the original. a typo or error in the original, and what was changed. The body text is corrected; the box records what the original said, so you can match it against the slides.

Each chapter ends with a one-page **summary** and a set of **practice problems** for independent work. Plots of densities, pmfs and cdfs are redrawn from their formulas, so they are exact; scanned figures from the original are kept where they show real data or 3D surfaces.

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CHAPTER 1

Probability

1.1 Introduction

"Probability" gives the meaning of chance or randomness; but how do we **formalize** it? "Probability" is almost everywhere:

- Genetics and bioinformatics, e.g. mutation.
- The kinetic theory of gases.
- Queuing theory — tremendous applications.
- The theory of finance.
- ⋮

▸ **Lecture note.** The whole course answers one question: **how do we reason precisely about things we cannot predict?** Chapter 1 builds the language (sample spaces, events, probability measures), Chapter 2 attaches numbers to outcomes (random variables), Chapter 3 handles several numbers at once, Chapter 4 summarises them (mean, variance, correlation) and Chapter 5 says what happens when we average many of them. Every later chapter is built on the three axioms in §1.3.

1.2 Sample Spaces

Definition 1 (Sample Space). Ω is the set of all possible outcomes. We denote each outcome by ω.

Definition 2 (Event). An event is a subset of Ω.

Example 3. Passing by 3 traffic lights, at each we either continue (c) or stop (s). Then

$$\Omega = \{ccc, ccs, csc, css, scc, scs, ssc, sss\},$$
$$A = \{sss, ssc, scc, scs\}$$

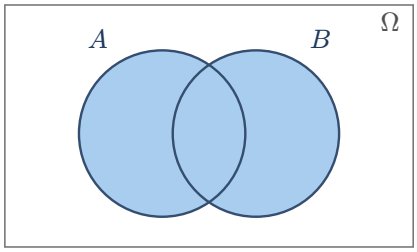
is the event of stopping at the first traffic light.

Example 4. The length of time between two successive earthquakes in a particular region:

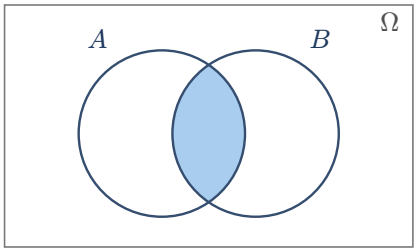
$$\Omega = \{t \mid t \geq 0\} \quad \text{(the set of nonnegative reals)},$$
$$A = \{t \mid t \geq 1 \text{ day}\}.$$

▸ **Lecture note.** Example 3 has a **finite** sample space (8 outcomes); Example 4 has an **uncountable** one (every real $t \geq 0$). The definitions are the same in both cases — that is the point of defining things with sets. Counting arguments (§1.4) only work in the finite case; the uncountable case is what forces us, in Chapter 2, to use densities and integrals.

All set-theory axioms and laws apply, because "sample space" and "events" are **sets**.



$A \cup B$



$A \cap B$

Intersection.

$$\begin{aligned} A \cap B &= \{\omega \mid \omega \in A \wedge \omega \in B\}, \\ A &= \{sss, ssc, scs, scc\} \quad \text{(stopping at first)}, \\ B &= \{sss, scs, ccs, css\} \quad \text{(stopping at third)}, \\ A \cap B &= \{sss, scs\}. \end{aligned}$$

Union.

$$\begin{aligned} A \cup B &= \{\omega \mid \omega \in A \vee \omega \in B\}, \\ A \cup B &= \{sss, ssc, scs, scc, ccs, css\}. \end{aligned}$$

Complement.

$$\begin{aligned} A^c &= \{\omega \mid \omega \notin A\}, \\ A^c &= \{ccc, ccs, csc, css\}. \end{aligned}$$

▸ **Lecture note.** Translate words into set operations before computing anything:

In words	In sets
A **and** B both happen	$A \cap B$
A **or** B (at least one) happens	$A \cup B$
A does **not** happen	A^c
A happens **but not** B	$A \cap B^c = A \setminus B$
neither A nor B	$(A \cup B)^c = A^c \cap B^c$ (De Morgan)
A and B **cannot** both happen	$A \cap B = \varnothing$ (disjoint)

1.3 Probability Measure

We define probability **rigorously**, in a way that meets what is in our minds about probability:

- probability as a **frequency** — the chance for something to happen;
- probability as a **subjective belief**.

Definition 5 (Probability Measure). A probability measure is a function P from subsets of Ω to the real numbers that satisfies

1. $P(\Omega) = 1$,
2. $\forall A \subset \Omega$, $P(A) \geq 0$,
3. if A_i and A_j are disjoint $\forall i \neq j$, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

No need for more axioms; **all the rest comes as a byproduct.**

⋆ **Intuition.** Think of P as spreading one unit of "mass" (sand) over Ω. Axiom 1: there is exactly one unit in total. Axiom 2: no region carries negative sand. Axiom 3: the sand on non-overlapping regions adds up. Every property below is a statement you could check with a bucket of sand.

1.3.1 Properties

$$\begin{aligned} P(A^c) &= 1 - P(A), \\ P(A) &\leq 1, \\ P(\varnothing) &= 0, \\ P(A) &\leq P(B), \quad \forall A \subseteq B, \\ P(A \cup B) &= P(A) + P(B) - P(A \cap B). \end{aligned}$$

Proof. **Complement.** A and A^c are disjoint and $A \cup A^c = \Omega$, so

$$P(\Omega) = P(A \cup A^c) = P(A) + P(A^c) \quad \text{(Axiom 3)} \quad \Longrightarrow \quad P(A^c) = 1 - P(A).$$

Upper bound. $P(A) = 1 - P(A^c)$ and $P(A^c) \geq 0$ (Axiom 2), so $P(A) \leq 1$.

Empty set. $P(\Omega) = P(\Omega \cup \varnothing) = P(\Omega) + P(\varnothing)$, so $P(\varnothing) = 0$.

Monotonicity. For $A \subset B$, $B = A \cup (B \setminus A)$, a disjoint union, so

$$P(B) = P(A) + P(B \setminus A) \Longrightarrow P(A) = P(B) - P(B \setminus A) \leq P(B).$$

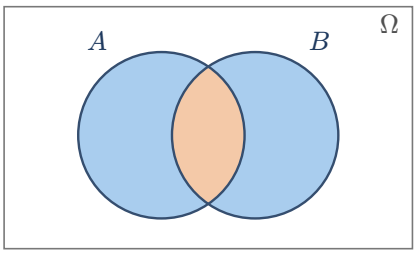
Addition rule. Write $A \cup B$ as three disjoint pieces:

$$A \cup B = \underbrace{(A \setminus (A \cap B))}_{A \cap B^c} \cup (A \cap B) \cup \underbrace{(B \setminus (A \cap B))}_{B \cap A^c}.$$

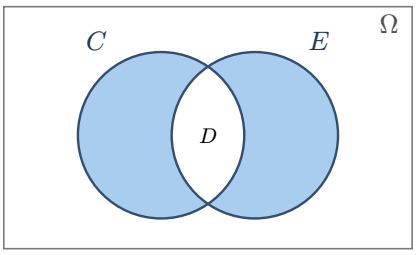
Then

$$\begin{aligned} P(A \cup B) &= P(A \setminus (A \cap B)) + P(A \cap B) + P(B \setminus (A \cap B)) \\ &= \underbrace{P(A \setminus (A \cap B)) + P(A \cap B)}_{P(A)} + \underbrace{P(A \cap B) + P(B \setminus (A \cap B))}_{P(B)} - P(A \cap B). \end{aligned}$$

□



$P(A) + P(B)$ counts the overlap twice



the parts outside the overlap D

Example 6. Tossing a coin twice: $\Omega = \{hh, ht, th, tt\}$. Assume equal probability $1/4$.

$$\begin{aligned} P(\text{head on first tossing}) &= P(\{ht, hh\}) = P(\{ht\} \cup \{hh\}) \\ &= P(\{ht\}) + P(\{hh\}) = 1/2, \\ P(\text{head on second tossing}) &= P(\{th, hh\}) = 1/2. \end{aligned}$$

Then

$$\begin{aligned} P(\text{head on first or second}) &= P\left(\underbrace{\text{head on first}}_{A} \cup \underbrace{\text{head on second}}_{B}\right) \\ &= P(\{ht, hh\} \cup \{th, hh\}) \\ &= P\left(\underbrace{\{ht, hh\}}_{A}\right) + P\left(\underbrace{\{th, hh\}}_{B}\right) - P\left(\underbrace{\{hh\}}_{A \cap B}\right) \\ &= 1/2 + 1/2 - 1/4 = 3/4, \end{aligned}$$

or directly

$$P(\{hh, ht, th\}) = 1/4 + 1/4 + 1/4 = 3/4.$$

▸ **Lecture note.** A third route, usually the fastest: the complement of "at least one head" is "no heads" $= \{tt\}$, so the answer is $1 - 1/4 = 3/4$. Whenever a question says **at least one**, try the complement first.

Example 7. An extreme case of this rule: if Bassem (B) cannot come except with Ahmed (A), then $B \subset A$ and

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A),$$

since $A \cap B = B$ and $A \cup B = A$.

1.4 Counting Methods

- Counting is beneficial for a **finite** sample space $\Omega = \{\omega_1, \omega_2, ..., \omega_N\}$.
- Denote $P(\omega_i)$ by p_i for simplicity.
- Special case: if $p_i = p$ for all i (equally likely outcomes) and $|A| = n$, then

$$P(A) = \frac{n}{N}.$$

▸ **Lecture note.** "Equally likely" is an **assumption about the model**, not a law. $P(A) = |A| \,/\, |\Omega|$ is valid only after you have chosen Ω so that its outcomes really are equally likely. The sums of two dice $\{2, ..., 12\}$ are **not** equally likely; the 36 ordered pairs are. Choose the sample space that makes counting legitimate.

1.4.1 Multiplication Principle

Proposition 8. Two experiments having m and n outcomes $\Longrightarrow N = mn$ pairs of outcomes.

Proof. The N outcomes are the entries of the $m \times n$ table $\{a_1, ..., a_m\} \times \{b_1, ..., b_n\}$. □

Proposition 9. p experiments with n_i outcomes in experiment $i \Longrightarrow N = \prod_{i=1}^{p} n_i$ total outcomes.

Proof. By induction. **Base step:** the statement is true for $p = 2$ (proven above). **Induction step:** suppose it is true for $p = q$, so the total number of outcomes is $N_q = \prod_{i=1}^{q} n_i$. For $p = q + 1$ we have two experiments, one with $\prod_{i=1}^{q} n_i$ outcomes and the second with n_{q+1}, so

$$N_{q+1} = n_{q+1} \times \prod_{i=1}^{q} n_i = \prod_{i=1}^{q+1} n_i,$$

which completes the proof. □

1.4.2 Permutations and Combinations

Let $C = \{c_1, c_2, ..., c_n\}$. How many ways are there to sample r elements?

Ordered sampling with replacement:

$$n \times n \times \cdots \times n = n^r.$$

Ordered sampling without replacement:

$$n \times (n-1) \times \cdots \times (n-r+1) = \frac{n!}{(n-r)!},$$

with the special case $r = n \Longrightarrow n!$ ways.

Example 10. If a plate has 3 letters and 3 numbers, we have $26^3 \times 10^3 = 17,576,000$ plates!

Example 11. If all letters and numbers have the same probability to occur, what is the probability that a plate has no duplicated letter or number?

$$P(A) = \frac{26 \times 25 \times 24 \times 10 \times 9 \times 8}{17,576,000} = 0.64.$$

What is the probability that a car has, at least, two digits duplicated? Very easy!!!

Solution. "At least two digits duplicated" is the complement of "all three digits different", and only the digits matter, so

$$P = 1 - \frac{10 \times 9 \times 8}{10^3} = 1 - 0.72 = 0.28.$$

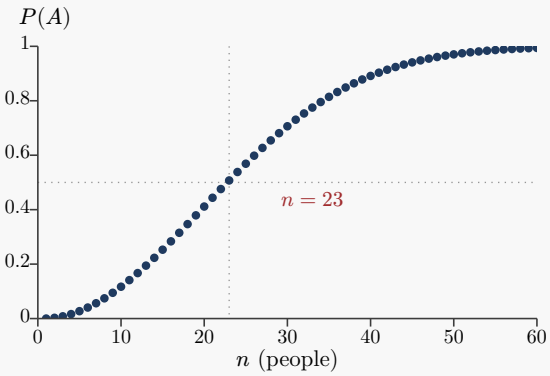
(Finding $P(A^c)$ is sometimes much easier than $P(A)$ because of the number of ways.)

Example 12 (Birthday Problem). Given n persons, what is the probability that at least two of them have the same birthday? Assume all days have the same probability and each year has only 365 days.

$$P(A^c) = \frac{365 \times 364 \times \cdots \times (365-n+1)}{365^n},$$

$$P(A) = 1 - \frac{365 \times 364 \times \cdots \times (365-n+1)}{365^n}.$$

n	$P(A)$
4	.016
16	.284
23	.507
32	.753
40	.891
56	.988



You may like to think as a "frequentist".

⋆ **Intuition.** Why so few people? With $n = 23$ there are $\binom{23}{2} = 253$ **pairs**, and any one of them can match. You are not asking "does someone share **my** birthday" (that is Example 13) — you are asking about every pair at once.

Example 13. How many persons must you ask to have a .5 chance of finding someone who shares **your** birthday?

$$P(A^c) = \frac{364^n}{365^n},$$
$$P(A) = 1 - \frac{364^n}{365^n},$$
$$n = \frac{\log(1 - P(A))}{\log(364/365)},$$

which gives $n = 253$. Why was it easier in the above example?

▸ **Lecture note.** Answer to the closing question: in Example 12 a match can happen between **any** of the $\binom{n}{2}$ pairs, so the number of chances grows like $n^2/2$; in Example 13 it must involve **you**, so it grows like n. Setting $\binom{n}{2} \approx 253$ gives $n \approx 23$ — the two answers are the same number of "chances to match".

Unordered sampling without replacement:

$$\frac{n \times (n-1) \times \cdots \times (n-r+1)}{r!} = \frac{n!}{(n-r)!\, r!} = \binom{n}{r}.$$

It is used in the binomial theorem

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k},$$

because $a^k b^{n-k}$ is obtained by summing $\binom{n}{k}$ different terms. Also,

$$2^n = \sum_{k=0}^{n} \binom{n}{k},$$

which is the number of all possible subsets.

	Order matters	Order does not matter
With replacement	n^r	$\binom{n+r-1}{r}$ (not in the notes)
Without replacement	$n!/(n-r)!$	$\binom{n}{r}$

▸ **Lecture note.** To decide which cell applies, ask two questions: **can an item be chosen twice?** (replacement) and **would swapping two chosen items give a different outcome?** (order). A committee is unordered; a president–secretary pair is ordered. The fourth cell ("stars and bars") is given for completeness — it is not used in the notes.

Example 14. A manufacturer produced n items, t out of them are defective (we do not know them). We sample m items at random; what is the probability that r out of them are defective? Why do we do this?

$$P(r \text{ defective items}) = \frac{\binom{t}{r}\binom{n-t}{m-r}}{\binom{n}{m}},$$

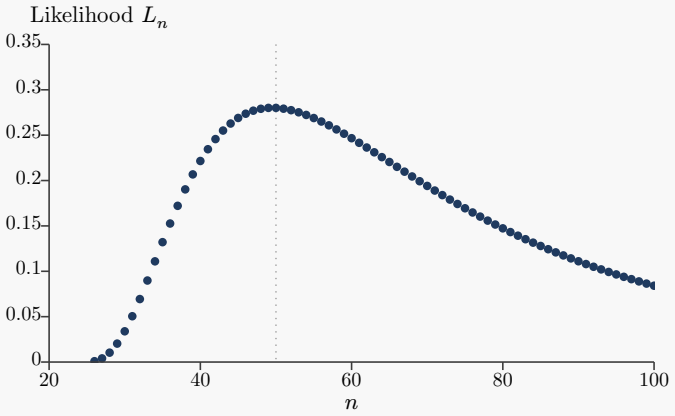
to be used in estimating t. The following example is similar, but it estimates n.

Correction to the original. The original writes $P(m$ defective items$)$ on the left. Since m is the sample size, the event is "r defective items in the sample", matching the right-hand side.

Example 15 (Capture/Recapture Method). 10 animals are captured, tagged and released. Later, 20 animals are captured; 4 of them are found to be tagged. What is the population n?

$$P(r \text{ tagged}) = \frac{\binom{t}{r}\binom{n-t}{m-r}}{\binom{n}{m}} = \frac{\binom{10}{4}\binom{n-10}{20-4}}{\binom{n}{20}}.$$

n must be the value that is most probable to happen; then maximize P.



Write L_n for this probability as a function of n, with $t = 10$, $m = 20$, $r = 4$:

$$L_n = \frac{\binom{t}{r}\binom{n-t}{m-r}}{\binom{n}{m}} = \binom{t}{r}\frac{(n-t)!\,m!\,(n-m)!}{(m-r)!\,(n-t-m+r)!\,n!}.$$

Then

$$\frac{L_n}{L_{n-1}} = \frac{(n-t)!\,(n-m)!}{(n-t-m+r)!\,n!} \times \frac{(n-1)!\,(n-t-m+r-1)!}{(n-t-1)!\,(n-m-1)!}$$
$$= \frac{(n-t)(n-m)}{n(n-t-m+r)}.$$

$L_n/L_{n-1} > 1$ if

$$(n-t)(n-m) > n(n-t-m+r)$$
$$n^2 - mn - tn + mt > n^2 - tn - mn + rn$$
$$mt > rn$$
$$n < \frac{mt}{r}.$$

So L_n increases while $n < mt/r$ and decreases afterwards. Then

$$\arg\max_n [L_n] = \frac{mt}{r} = \frac{20 \times 10}{4} = 50;$$

it makes a lot of sense since $4/20$ is related to $10/50$.

⋆ **Intuition.** The estimate just says **the sample looks like the population**: 4 of the 20 recaptured animals (20%) were tagged, so the 10 tagged animals should be about 20% of everybody, giving $n \approx 50$. This is the first **maximum likelihood estimate** you meet — the same idea runs all of Statistics II.

Proposition 16. The number of ways to partition n objects into r classes, class i having n_i objects (such that $\sum_{i=1}^{r} n_i = n$), is

$$\frac{n!}{n_1!(n-n_1)!} \cdot \frac{(n-n_1)!}{(n-n_1-n_2)!n_2!} \cdots \frac{(n-n_1-n_2-\cdots-n_{r-2})!}{(n-n_1-n_2-\cdots-n_{r-1})!n_{(r-1)!}} \times 1$$
$$= \frac{n!}{n_1!\,n_2!\cdots n_r!}.$$

This is similar to the multinomial theorem

$$(x_1 + x_2 + \cdots + x_r)^n = \sum_{n_1,\ldots,n_r} \binom{n}{n_1 \cdots n_r} x_1^{n_1} x_2^{n_2} \cdots x_r^{n_r}.$$

Correction to the original. The multinomial coefficient is printed as $\binom{n}{n_1 \cdots n_2}$ in the original; it is $\binom{n}{n_1 \cdots n_r} = n!/(n_1! \cdots n_r!)$, summed over all $n_1 + \cdots + n_r = n$.

Very important: the proposition above assumes **ordering between groups (inter)** and **non-ordering within a group (intra)**. If $\Omega = \{A, B, C, D\}$ $(n = 4)$, $n_1 = 2$, $n_2 = 2$, then the selections are $\binom{4}{2}\binom{4-2}{2} = 6$:

$$\{(\{A,B\},\{C,D\}), (\{A,C\},\{B,D\}), (\{A,D\},\{B,C\}),$$
$$(\{B,C\},\{A,D\}), (\{B,D\},\{A,C\}), (\{C,D\},\{A,B\})\}.$$

The six come in three pairs that differ only in which group is "first". If we want **non-ordering for both between-group and within-group**, the count is

$$\frac{\frac{n!}{n_1!\,n_2!\cdots n_r!}}{r!},$$

where r is the number of groups (2 above), and $r!$ is the number of ways to order r groups. (Dividing by $r!$ is valid when the groups have equal sizes, as here: $6/2! = 3$.)

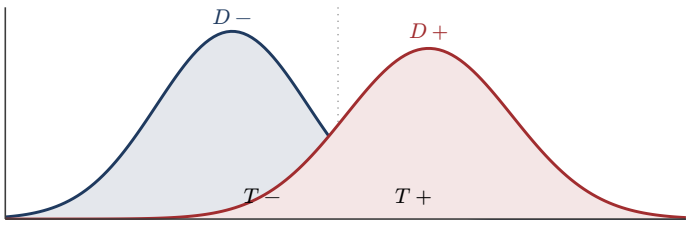
For more elaboration on counting methods, please refer to (Rosen, 2007, Ch. 5 and 7).

1.5 Conditional Probability

Consider a medical test:

- $T+$: high blood concentration (positive test),
- $T-$: low blood concentration (negative test),
- $D+$: toxicity (disease present),
- $D-$: no toxicity (disease absent).

#	$D+$	$D-$	Total
$T+$	25	14	39
$T-$	18	78	96
Total	43	92	135



The test calls a value above the threshold "$T+$". Healthy and sick people both produce a range of values, so the two populations overlap — some $D+$ fall below the threshold, some $D-$ above.

Conditioning on $T+$ changes Ω to $T+$:

$$\begin{aligned} P(D+ \mid T+) &= \frac{\#(D+ \cap T+)}{\#(T+)} = \frac{25}{39} \\ &= \frac{\#(D+ \cap T+)/\text{ Total}}{\#(T+)/\text{ Total}} = \frac{25/135}{39/135} \\ &= \frac{P(D+ \cap T+)}{P(T+)}. \end{aligned}$$

P	$D+$	$D-$	Total
$T+$	25/135	14/135	39/135
$T-$	18/135	78/135	96/135
Total	43/135	92/135	1

Definition 17. Let A and B be events, $P(B) \neq 0$. Then

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

- B acts as the new Ω.
- $P(A \mid B)$ is a normalized version of $P(A \cap B)$.
- This is a new probability measure; does it satisfy the axioms?

It does:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{+\text{ve}}{+\text{ve}} \geq 0,$$

$$P(B \mid B) = \frac{P(B \cap B)}{P(B)} = 1,$$

$$P\left(\left(\bigcup_{i=1}^{\infty} A_i\right) \mid B\right) = \frac{P\left(\left(\bigcup_{i=1}^{\infty} A_i\right) \cap B\right)}{P(B)} = \frac{P\left(\bigcup_{i=1}^{\infty} (A_i \cap B)\right)}{P(B)}$$

$$= \frac{\sum_{i=1}^{\infty} P(A_i \cap B)}{P(B)} = \sum_{i=1}^{\infty} \frac{P(A_i \cap B)}{P(B)} = \sum_{i=1}^{\infty} P(A_i \mid B),$$

where the third line uses that the $A_i \cap B$ are disjoint whenever the A_i are.

▸ **Lecture note.** Because $P(\cdot \mid B)$ is itself a probability measure, **every rule from §1.3 holds with "$\mid B$" attached**: $P(A^c \mid B) = 1 - P(A \mid B)$, the addition rule, and so on. What does **not** hold is flipping the bar: $P(A \mid B)$ and $P(B \mid A)$ are different numbers in general (25/39 vs. 25/43 in the table above). Confusing them is the single most common error in this chapter.

1.5.1 Multiplication Law

$$P(A \cap B) = P(A \mid B)P(B), \quad P(B) \neq 0,$$
$$P(A \cap B) = P(B \mid A)P(A), \quad P(A) \neq 0.$$

What is the meaning of that? To get both, get one first, then the other **given** the first.

Example 18. An urn contains 3 reds and one blue. What is the probability of selecting two red balls without replacement?

Solution 1:

$$P(2 \text{ reds}) = \frac{\#\text{ ways of selecting 2 reds}}{\#\text{ ways of selecting 2 balls}} = \frac{\binom{3}{2}}{\binom{4}{2}} = \frac{1}{2}.$$

Solution 2:

$$P(2 \text{ reds}) = P(R_1)P(R_2 \mid R_1) = \frac{3}{4} \times \frac{2}{3} = \frac{1}{2}.$$

Lemma 19 (Law of Total Probability). Let B_i, $i = 1, \ldots, n$ be such that $B_i \cap B_j = \varnothing \ \forall i \neq j$, $\bigcup_{i=1}^{n} B_i = \Omega$ (the B_i **partition** Ω), and $P(B_i) \neq 0$. Then

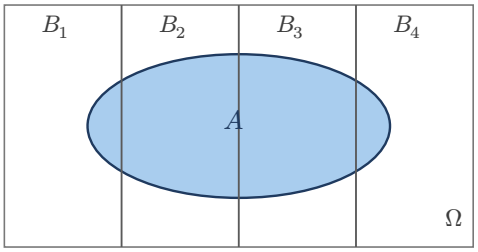
$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i).$$

Proof.

$$P(A) = P(A \cap \Omega) = P\left(A \cap \bigcup_{i=1}^{n} B_i\right) = P\left(\bigcup_{i=1}^{n} (A \cap B_i)\right)$$
$$= \sum_{i=1}^{n} P(A \cap B_i) = \sum_{i=1}^{n} P(A \mid B_i) P(B_i).$$

□

★ **Intuition.** Cut Ω into pieces $B_1, ..., B_n$. The event A is cut into the pieces $A \cap B_i$. Add the pieces back. Each piece is found with the multiplication law — the probability of **being in** B_i times the probability of A **once there**.



Example 20. In Example 18, what is the probability that a red ball is selected on the second draw?

$$P(R_2) = P(R_2 \mid R_1)P(R_1) + P(R_2 \mid B_1)P(B_1) = \frac{2}{3} \cdot \frac{3}{4} + \frac{3}{3} \cdot \frac{1}{4} = \frac{3}{4}.$$

▸ **Lecture note.** $P(R_2) = 3/4 = P(R_1)$: without peeking at the first ball, the second draw is exactly as likely to be red as the first. By symmetry, **every** draw has the same unconditional distribution — a fact worth remembering for card and urn problems.

Example 21 (Occupational Mobility). This statistic was collected by Glass and Hall (1954):

	U_2	M_2	L_2
U_1	.45	.48	.07
M_1	.05	.70	.25
L_1	.01	.50	.49

- U, M, L: upper, middle, lower levels.
- $1, 2$: father, son.
- e.g. $P(U_2 \mid U_1) = .45$ = probability a son occupies the upper level given his father did.
- Notice that $P(M_2 \mid L_1) > P(U_2 \mid M_1)$.
- What is $P(U_2)$, assuming that fathers' occupations are 10%, 40%, 50% in U, M, L respectively?

$$P(U_2) = P(U_2 \mid U_1)P(U_1) + P(U_2 \mid M_1)P(M_1) + P(U_2 \mid L_1)P(L_1)$$
$$= .45 \times .1 + .05 \times .4 + .01 \times .5 = .07.$$

- Notice that each **column** does not necessarily sum to one! Why? Where is the partition?
- Inverse problem: what is $P(U_1 \mid U_2)$? This is Bayes' rule.

Solution. Each **row** is a conditional distribution $P(\cdot \mid \text{father's level})$ and sums to 1; the partition is of the **son's** outcomes. A column collects the same son-level under three **different** conditions, so it has no reason to sum to 1. For the inverse problem,

$$P(U_1 \mid U_2) = \frac{P(U_2 \mid U_1)P(U_1)}{P(U_2)} = \frac{.45 \times .1}{.}07 \approx .64.$$

Lemma 22 (Bayes' Rule). Let A and B_i, $i = 1, ..., n$ be events, where the B_i are disjoint, **partition** Ω, and $P(B_i) \neq 0$. Then

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_{j=1}^{n} P(A \mid B_j)P(B_j)}.$$

Proof.

$$P(B_i \mid A) = \frac{P(A \cap B_i)}{P(A)} = \frac{P(A \mid B_i)P(B_i)}{\sum_{j=1}^{n} P(A \mid B_j)P(B_j)},$$

using the multiplication law in the numerator and Lemma 19 in the denominator. □

▸ **Lecture note.** A reliable three-step recipe for every Bayes question:

1. **Name the partition** (the hypotheses B_i) and write their **priors** $P(B_i)$.
2. Write the **likelihoods** $P(A \mid B_i)$ — these are what the problem usually gives.
3. Make a table of products $P(A \mid B_i)P(B_i)$; the sum is $P(A)$, and each product divided by the sum is the posterior $P(B_i \mid A)$.

Example 23 (Polygraph Test). This is a lie-detector test: $+$: polygraph positive; $-$: polygraph negative; T: person telling the truth; L: person lying. Based on Gastwirth (1987):

P	$\mid T$	$\mid L$
$+$	.14	.88
$-$	.86	.12
Sum	1	1

- Why $.14 + .88 \neq 1$ and $.86 + .12 \neq 1$, while $.14 + .86 = .88 + .12 = 1$?
- What is the meaning of $P(+ \mid T) = .14$?

Now suppose that the majority of people are telling the truth for a particular question, i.e. $P(T) = .99$. What is the probability that a person is telling the truth even if the polygraph is positive?

$$P(T \mid +) = \frac{P(+ \mid T)P(T)}{P(+ \mid T)P(T) + P(+ \mid L)P(L)} = \frac{.14 \times .99}{.14 \times .99 + .88 \times .01} = .94.$$

So most of the innocent people in a screening setup will be placed under suspicion!!

The source of the problem is:

$$P(T \mid +) = \frac{1}{1 + \frac{P(+ \mid L)P(L)}{P(+ \mid T)P(T)}} = \frac{1}{1 + \frac{.88 \times .01}{.14 \times .99}};$$

to have a small $P(T \mid +)$, either $P(+ \mid L)/P(+ \mid T)$ or $P(L)/P(T)$ should be large.

When does a very naive test ($P(+ \mid T) = P(+ \mid L)$) give a low $P(T \mid +)$?

Solution. Columns are conditional distributions (given T, given L), so **columns** sum to 1; rows mix two different conditions. $P(+ \mid T) = .14$ means 14% of truthful people still fail the test. For the naive test the ratio $P(+ \mid L)/P(+ \mid T) = 1$, so $P(T \mid +) = 1/(1 + P(L)/P(T)) = P(T)$: the test changes nothing, and $P(T \mid +)$ is low only if $P(T)$ itself was low — i.e. when most people lie anyway.

1.5.2 Bayes' Rule in Real Life: More Intuition

- $P(H)$, $P\left(\overline{H}\right) = 1 - P(H)$: the **prior** probabilities (prior knowledge).
- H: the hypothesis.
- E, $P(E)$: the evidence and its probability.
- $P(E \mid H)$, $P\left(E \mid \overline{H}\right)$: the **likelihoods**.
- $P(H \mid E)$: the **posterior** probability.

$$P(H \mid E) = \frac{P(E \mid H)P(H)}{P(E)} = \frac{P(E \mid H)P(H)}{P(E \mid H)P(H) + P\left(E \mid \overline{H}\right)P\left(\overline{H}\right)} = \frac{1}{1 + \frac{P\left(E \mid \overline{H}\right)}{P(E \mid H)} \cdot \frac{1-P(H)}{P(H)}}.$$

Example (from Pattern Recognition).

- Reading mammograms to find breast cancer (H).
- The probability of a woman having cancer is 1%: $P(H) = .01$.
- If she has breast cancer, 80% of radiologists observe it: $P(+ \mid H) = .8$.
- If she has a benign lesion, 10% of radiologists misinterpret it as a cancer: $P\left(+ \mid \overline{H}\right) = .1$.
- How much, do you think (subjectively), is $P(H \mid +)$?

When radiologists were asked about $P(H \mid +)$, the majority of them estimated it subjectively as 75%, although Bayes' rule gives

$$P(H \mid +) = \frac{P(+ \mid H)P(H)}{P(+ \mid H)P(H) + P\left(+ \mid \overline{H}\right)P\left(\overline{H}\right)} = \frac{.8 \times .01}{.8 \times .01 + .1 \times .99} = 7.5\% \quad \text{!!!}$$

⋆ **Intuition.** Think in **people**, not percentages. Take 1000 women: 10 have cancer, 8 of them test positive. Of the 990 healthy women, 99 test positive. So of $8 + 99 = 107$ positives, only 8 are sick: $8/107 \approx 7.5\%$. The radiologists anchored on $P(+ \mid H) = 80\%$ and forgot the 99 false alarms coming from the huge healthy group — this is the **base-rate fallacy**.

1.6 Independence

Motivation towards independence: if knowing B does not change the chance of A,

$$P(A) = P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad \text{then} \quad P(A \cap B) = P(A)P(B).$$

Definition 24. Events A and B are said to be **independent** if $P(A \cap B) = P(A)P(B)$.

Are disjoint events independent?

▸ **Lecture note.** **No** — almost the opposite. If $A \cap B = \varnothing$ with $P(A), P(B) > 0$, then $P(A \cap B) = 0 \neq P(A)P(B)$. Disjoint events are strongly **dependent**: learning that A happened tells you for certain that B did not. "Disjoint" is about the sets; "independent" is about the probabilities. You cannot see independence on a Venn diagram.

Definition 25 (Independence Generalized). Events A_i, $i = 1, ..., n$ are said to be **independent** (or **mutually independent**) if for any subcollection $A_{i_1}, ..., A_{i_m}$,

$$P\left(\bigcap_{j=1}^{m} A_{i_j}\right) = \prod_{j=1}^{m} P\left(A_{i_j}\right),$$

for which the definition above is a special case.

Example 26. A backup system:

- has 3 mirrored hard drives;
- the probability that one fails (F) is p;
- what is the probability that the system fails?

$$P(\text{system fails}) = P(F_1 \cap F_2 \cap F_3) = P(F_1)P(F_2)P(F_3) = p^3.$$

If $p = .001$, then $p^3 = 10^{-9}$.

! Watch out. p^3 is correct only if the drives fail **independently**. Drives in the same box share a power supply, a fire, a flood and a software bug — failures are positively dependent, and the true risk can be orders of magnitude larger than 10^{-9}. Independence is an assumption you must be able to defend.

Definition 27 (Weaker Independence). Events A_i, $i = 1, ..., n$ are said to be **pair-wise independent** if

$$P\left(A_i \cap A_j\right) = P(A_i)P\left(A_j\right) \quad \forall i \neq j.$$

Example 28 (Counter Example). Tossing a coin twice: A_1 = head on first, A_2 = head on second, A_3 = exactly one head in both. Clearly

$$P(A_i) = \frac{1}{2},$$

$$P(A_1 \cap A_2) = P(\{hh\}) = \frac{1}{4} = P(A_1)P(A_2),$$

$$P(A_1 \cap A_3) = P(\{ht\}) = \frac{1}{4} = P(A_1)P(A_3),$$

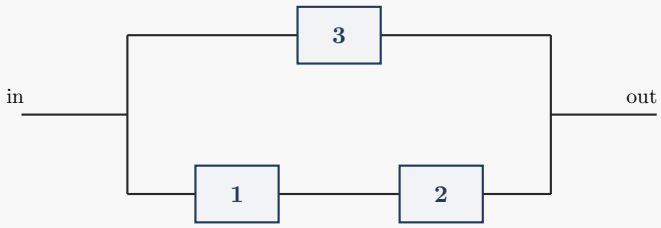
$$P(A_2 \cap A_3) = P(\{th\}) = \frac{1}{4} = P(A_2)P(A_3);$$

hence they are pairwise independent. But

$$P\left(\bigcap_{i=1}^{3} A_i\right) = P(\varnothing) = 0 \neq \prod_{i=1}^{3} P(A_i) = \frac{1}{8}.$$

Example 29. Consider the circuit in the figure. Event A_i is "the ith relay works". The A_i are independent with probability p each. Then the event C = "passing current" has

$$\begin{aligned} P(C) &= P(A_3 \cup (A_1 \cap A_2)) \\ &= P(A_3) + P(A_1 \cap A_2) - P(A_1 \cap A_2 \cap A_3) \\ &= p + p^2 - p^3. \end{aligned}$$



▸ **Lecture note.** Two building blocks solve every circuit of this type. **Series** (all must work): $P = \prod p_i$. **Parallel** (at least one must work): $P = 1 - \prod(1 - p_i)$ — the complement trick again. Here relays 1–2 are in series (p^2), in parallel with relay 3: $1 - (1-p)(1-p^2) = p + p^2 - p^3$, the same answer.

Chapter summary

- **Axioms:** $P(\Omega) = 1$, $P(A) \geq 0$, countable additivity for disjoint events. Everything else — complement rule, $P(\varnothing) = 0$, monotonicity, the addition rule — is derived.
- **Equally likely outcomes:** $P(A) = |A| \,/\, |\Omega|$. Count with n^r, $n!/(n-r)!$, $\binom{n}{r}$, and $n!/(n_1!\cdots n_r!)$.
- **Conditional probability:** $P(A \mid B) = P(A \cap B)/P(B)$ is a probability measure on the new sample space B.
- **Multiplication law** $P(A \cap B) = P(A \mid B)P(B)$; **total probability** $P(A) = \sum_i P(A \mid B_i)P(B_i)$; **Bayes** $P(B_i \mid A) = P(A \mid B_i)P(B_i)/P(A)$.
- **Independence:** $P(A \cap B) = P(A)P(B)$. Mutual independence needs **every** subcollection; pairwise is weaker (Example 28). Disjoint $\neq$ independent.
- **Habits:** try the complement for "at least one"; draw a tree or table for Bayes; think in 1000 people when a posterior looks surprising.

Practice problems — Chapter 1

1.1 Show that $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$ using only the two-set addition rule.

1.2 Five cards are dealt from a standard 52-card deck. Find the probability of (a) a flush (all five of the same suit), (b) exactly one pair.

1.3 A fair die is rolled 4 times. Find the probability of at least one six.

1.4 In a class, 60% study for the exam. Of those who study, 90% pass; of those who don't, 40% pass. A student passed. What is the probability they studied?

1.5 Show that if A and B are independent, so are A and B^c.

1.6 A coin with $P(\text{head}) = p$ is tossed until the first head. What is the probability that more than k tosses are needed?

1.7 In Example 29, compute $P(A_3 \mid C)$: given that current passes, how likely is it that relay 3 works?

CHAPTER 2

Random Variables (r.v.)

A random variable is essentially a **number** attached to each outcome:

Definition 30. A random variable X is a mapping from Ω to $\mathbb{R} \cup \{\infty, -\infty\}$:

$$X : \Omega \to [-\infty, \infty].$$

Example 31. Consider tossing a coin three times:

$$\Omega = \{hhh, hht, hth, htt, thh, tht, tth, ttt\}.$$

- What is the number of heads?
- What is the number of tails?

Each question defines a random variable: for instance $X(hth) = 2$ heads, and $Y(hth) = 1$ tail. Note $X + Y = 3$ for every ω.

▸ **Lecture note.** A random variable is neither random nor a variable — it is a **function**. The randomness is in which ω occurs; once ω is fixed, $X(\omega)$ is a fixed number. Writing $\{X = 2\}$ is shorthand for the **event** $\{\omega \in \Omega : X(\omega) = 2\}$, so $P(X = 2)$ is the probability measure of Chapter 1 applied to that event. Nothing new is needed.

2.1 Discrete r.v.

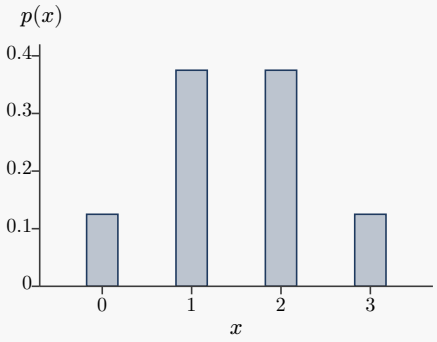
Definition 32. Consider the (possibly infinite) partition A_i, $i = 1, ..., n$ of Ω. We call X a **discrete** r.v. if it takes a value $x_k \in [-\infty, \infty]$, $k = 1, ..., n$, whenever A_k happens. More compactly,

$$X = \sum_{i=1}^{n} x_i I_{A_i}, \qquad I_{A_i} = \begin{cases} 1, \text{if } \omega \in A_i \\ 0, \text{if } \omega \notin A_i. \end{cases}$$

The probability of the discrete r.v. is called the **probability mass function** (pmf): $p(x_k) = P(X = x_k) = P(A_k)$.

Example 33. In the previous example, assume the coin is fair, and X is the number of heads. What is the pmf of X? Here $0 \leq X \leq 3$ and

$$P(X=0) = P(\{ttt\}) = \frac{1}{8},$$
$$P(X=1) = P(\{htt, tht, tth\}) = \frac{3}{8},$$
$$P(X=2) = P(\{thh, hth, hht\}) = \frac{3}{8},$$
$$P(X=3) = P(\{hhh\}) = \frac{1}{8}.$$



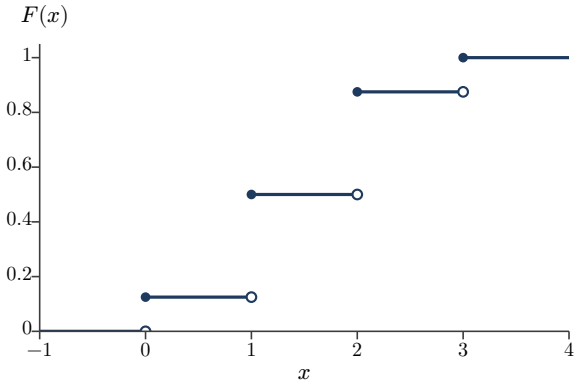
What is $P(X=4)$? Prove!

Solution. $\{X=4\} = \{\omega : X(\omega) = 4\} = \varnothing$, since no outcome has four heads. By the property $P(\varnothing) = 0$ of §1.3, $P(X=4) = 0$.

Definition 34. The **cumulative distribution function** (cdf) — sometimes called the **distribution function** (df) — is defined as

$$F(x) = P(X \le x).$$

We may write F_X so as not to confuse it with, e.g., F_Y.



Lemma 35. Any cdf F has:
- $F(x_1) \le F(x_2)\ \forall x_1 \le x_2$ (monotonically non-decreasing),
- $F(-\infty)\ (= \lim_{x\to-\infty} F(x)) = 0$,
- $F(\infty)\ (= \lim_{x\to\infty} F(x)) = 1$,
- $F(x) = F(x^+)$ (continuous from the right).

Proof. Omitted in the original. The first property is monotonicity of P: if $x_1 \le x_2$ then $\{X \le x_1\} \subseteq \{X \le x_2\}$. The other three follow from continuity of P along monotone sequences of events, which is a consequence of Axiom 3. □

▸ **Lecture note.** For a discrete r.v. the cdf is a **staircase**: flat between the possible values, jumping by $p(x_k)$ at x_k. The filled dot sits on the **upper** step — that is right-continuity: $F(1) = P(X \le 1) = 1/2$ **includes** the jump at 1. Reading probabilities off a cdf: $P(a < X \le b) = F(b) - F(a)$, and $P(X = a) = F(a) - F(a^-)$ is the jump size.

2.1.1 Bernoulli(p)

The r.v. X is Bernoulli if

$$P(X=x)=\begin{cases}p, & x=1\\ 1-p, & x=0.\end{cases}$$

Note that $P(1)+P(0)=1$. If we toss a fair coin and $A=\{\text{head}\}$, then $I_A\sim\text{Bernoulli}(0.5)$.

- Plot the pmf of Bernoulli(p).
- Repeating an experiment many times under this pmf, what does the data look like?

▸ **Lecture note.** The pmf is two bars, of heights $1-p$ at 0 and p at 1. Data from it is a long string of 0s and 1s, e.g. for $p=0.3$: `0 1 0 0 0 1 0 1 0 0 …`, with the **fraction** of 1s settling near 0.3 as the string grows. That last sentence is the Law of Large Numbers (Chapter 5) — the bridge between a pmf on paper and data in the world.

2.1.2 Binomial(n,p)

If X is the number of successes in n independent trials, each with success probability p, then

$$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k},\quad 0\le k\le n.$$

Note that

$$\sum_{k=0}^{n}\binom{n}{k}p^k(1-p)^{n-k}=(p+(1-p))^n=1,$$

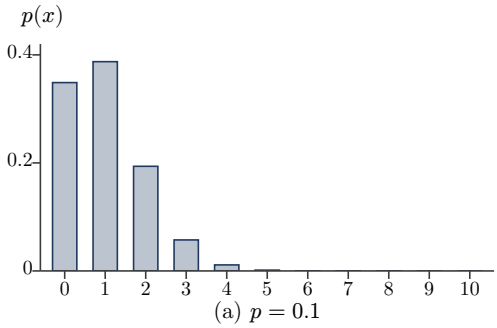
by the binomial theorem. Also, observe that

$$X=\sum_{i=1}^{n}I_i,$$

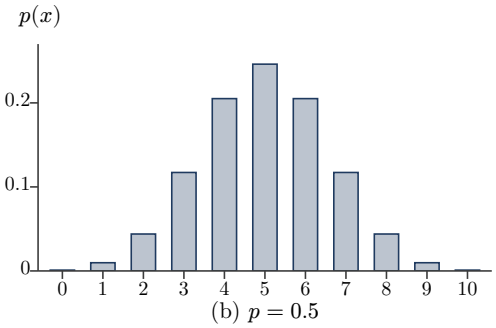
where $I_i\sim\text{Ber}(p)$, and the I_i are independent (definition coming soon).

▸ **Lecture note.** **Why** $\binom{n}{k}$? Any particular sequence with k successes (say $SSFSF\ldots$) has probability $p^k(1-p)^{n-k}$ by independence; there are $\binom{n}{k}$ ways to choose **which** k of the n trials are the successes; the sequences are disjoint, so add. **When to use it:** a fixed number n of trials, two outcomes each, the same p each time, independence. If any of the four fails (e.g. sampling without replacement), it is not binomial.

A plot of the pmf of $\text{Bin}(10,p)$, $p=.1,.5$ respectively. Repeating an experiment many times under one of those pmfs, what does the data look like?



(a) $p=0.1$



(b) $p=0.5$

Example 36 (Tay-Sachs Disease).

- A disease common among people of Jewish or eastern European extraction.
- If a couple are both carriers, one in every four of their children is affected.
- If they have four children, then

$$P(k) = \binom{4}{k}(.25)^k(1-.25)^{4-k}, \quad 0 \le k \le 4.$$

k	0	1	2	3	4
$P(k)$	.316	.422	.211	.047	.004

Correction to the original. The original says "one of every four children of theirs is a **carrier**". For two carrier parents, 1/4 is the probability a child is **affected** (inherits two copies); the chance of being a carrier is 1/2. The binomial computation is unchanged.

Example 37 (Error Correction).

- The error in sending a single bit is .1.
- Remedy: the system sends it 5 times.
- The receiver takes a majority vote.
- The probability of receiving one bit correctly is

$$P(\#\text{ errors} \le 2) = \sum_{k=0}^{2}\binom{5}{k}(.1)^k(.9)^{5-k} = .9914.$$

▸ **Lecture note.** Repetition buys reliability: one transmission is right 90% of the time, five-with-majority 99.14%. The price is five times the bandwidth. Try $n = 3$ yourself (.972) — the gain from 3 to 5 is much smaller than from 1 to 3.

2.1.3 Geometric(p)

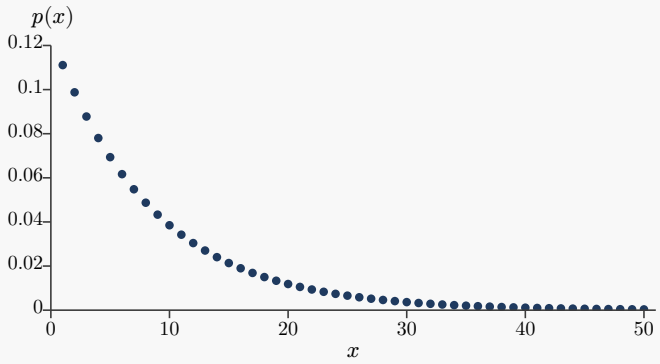
Constructed from infinitely many independent Bernoulli trials. X = number of trials to get the first success; hence

$$P(X = k) = (1-p)^{k-1}p, \quad k \ge 1.$$

Note that

$$\sum_{k=1}^{\infty} p(1-p)^{k-1} = p\sum_{k=1}^{\infty}(1-p)^{k-1} = p \cdot \frac{1}{p} = 1.$$

Example 38. If the probability of winning a draw is $p = 1/9$, then the number of draws necessary for winning is Geometric(1/9).



▸ **Lecture note.** The geometric pmf is always largest at $k = 1$ — the single most likely moment for the first success is **right now**, even when p is small. Its tail is $P(X > k) = (1-p)^k$ (the first k trials all fail), which gives the memoryless property $P(X > k+m \mid X > k) = P(X > m)$ — the discrete twin of the exponential in §2.2.1.

2.1.4 NBinomial(r, p)

A generalization of the geometric distribution. X = number of trials necessary to get the first r successes. Then the last trial must be a success, and

$$P(X = k) = \underbrace{p}_{\text{last trial}} \times \underbrace{\binom{k-1}{r-1} p^{r-1}(1-p)^{(k-1)-(r-1)}}_{(r-1) \text{ successes in } (k-1) \text{ trials}} = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad r \le k.$$

Prove that

$$\sum_{k=r}^{\infty} \binom{k-1}{r-1} p^r (1-p)^{k-r} = 1.$$

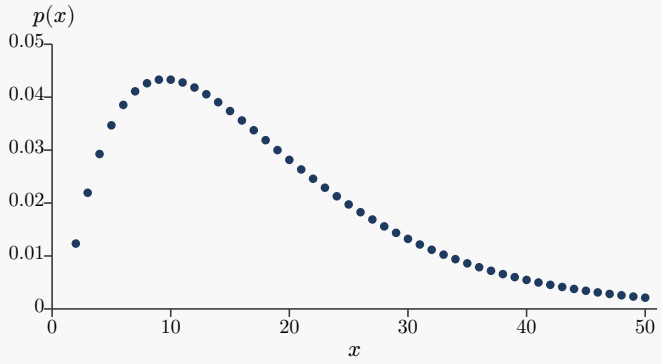
Proof. Put $q = 1-p$ and $j = k-r$. The negative binomial series $\sum_{j=0}^{\infty} \binom{j+r-1}{r-1} q^j = (1-q)^{-r}$ (differentiate the geometric series $r-1$ times) gives

$$\sum_{k=r}^{\infty} \binom{k-1}{r-1} p^r q^{k-r} = p^r \sum_{j=0}^{\infty} \binom{j+r-1}{r-1} q^j = p^r (1-q)^{-r} = 1.$$

□

Example 39. Continuing the previous example, the number of draws necessary to get the second winning draw is NBin$(2, 1/9)$. Then

$$P(k) = (k-1)(1/9)^2 (8/9)^{k-2}.$$



2.1.5 Hypergeometric(n, r, m)

Recall the capture/recapture method. Draw m objects from a total of n with r labeled; then X, the number of selected labeled objects, has

$$P(X = k) = \frac{\binom{r}{k}\binom{n-r}{m-k}}{\binom{n}{m}}, \quad 0 \le k \le m.$$

Prove that (this was a HW problem)

$$\sum_{k=0}^{m} \frac{\binom{r}{k}\binom{n-r}{m-k}}{\binom{n}{m}} = 1.$$

▸ **Lecture note.** Hypergeometric = binomial **without replacement**. When n is large compared with m, removing a few objects barely changes the proportion r/n, and Hypergeometric$(n, r, m) \approx$ Bin$(m, r/n)$.

2.1.6 Poisson(λ)

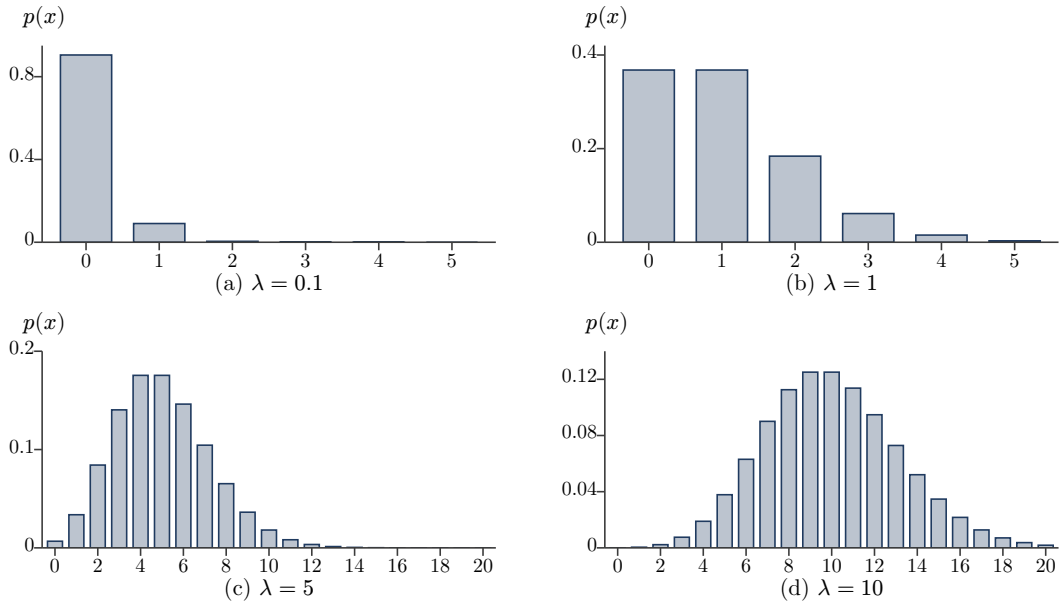
A Poisson r.v. X is defined by

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, ...$$

Note that

$$\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} e^{-\lambda} = e^{\lambda} e^{-\lambda} = 1.$$

The following is a plot of the Poisson pmf at $\lambda = .1, 1, 5, 10$ respectively.



(a) $\lambda = 0.1$ (b) $\lambda = 1$ (c) $\lambda = 5$ (d) $\lambda = 10$

The Poisson r.v. can be seen as a limiting process for the binomial, with $n \to \infty$, $p \to 0$, $np = \lambda$:

$$P(k) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} = \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

$$= \left(\frac{\lambda^k}{k!}\right) \underbrace{\left(\frac{n!}{(n-k)! n^k}\right)}_{\to 1} \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{\to e^{-\lambda}} \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-k}}_{\to 1},$$

$$\lim_{n \to \infty} p(k) = \frac{\lambda^k}{k!} e^{-\lambda}.$$

Example 40. In dice rolling, with $n = 100$ rolls, $P((6, 6)) = 1/36$. If we consider the Poisson approximation with $np = 100 \times \frac{1}{36} = 2.78$:

k	Binomial Pr	Poisson Pr
0	0.0598	0.0622
1	0.1708	0.1727
2	0.2416	0.2399
3	0.2255	0.2221

4	0.1562	0.1542
5	0.0857	0.0857
6	0.0388	0.0397
7	0.0149	0.0157
8	0.0049	0.0055
9	0.0014	0.0017
10	0.0004	0.0005
11	0.0001	0.0001

▸ **Lecture note.** Read the Poisson as the **law of rare events**: many independent opportunities (n large), each unlikely (p small), a moderate expected count $\lambda = np$. You no longer need n and p separately — only their product. That is why it fits typos per page, calls per minute, decays per second: nobody knows "n", but everybody can measure the average rate λ.

Poisson r.v. and continuous time:

- Consider independent events in time T.
- Subdivide T into many small intervals Δt.
- An event happens in Δt with small probability p.
- Two events cannot happen in the same Δt.

Applications:

- Modeling traffic in general:
  - incoming calls in telephone systems;
  - light traffic (but cars are not independent).
- Modeling alpha-particle emission.

Example 41 (Telephone Calls).

- Model the calls per minute as Poisson(λ).
- $\lambda = .5$/ min.

$$P(k) = \frac{\lambda^k}{k!} e^{-\lambda},$$

$$P(\text{no calls / min}) = P(0) = e^{-.5} = 0.607,$$

$$P(\text{just one call / min}) = P(1) = .5e^{-.5} = 0.303.$$

Notice that if the period is, e.g., 2 min, then $X = X_{1\text{ min}} + X_{1\text{ min}}$; we will prove later that the sum of independent Poisson(λ_1) and Poisson(λ_2) is Poisson($\lambda_1 + \lambda_2$).

2.2 Continuous r.v.

Instead of the pmf, we define the **probability density function** (pdf) f so that

$$P(a < X < b) = \int_a^b f(x)\,\mathrm{d}x.$$

Notice that:

$$\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1,$$

$$P(X = c) = \int_{c}^{c} f(x)\,\mathrm{d}x = 0,$$

$$P(a < X < b) = P(a \le X < b) = P(a < X \le b).$$

More elaboration:

$$P\left(x - \frac{\delta}{2} \le X \le x + \frac{\delta}{2}\right) = \int_{x-\delta/2}^{x+\delta/2} f(u)\,\mathrm{d}u \approx \delta f(x),$$

equivalently,

$$P(x \le X \le x + \mathrm{d}x) = f(x)\,\mathrm{d}x.$$

> **! Watch out.** $f(x)$ is **not** a probability — it is a probability **per unit length**, and it can exceed 1 (the exponential with $\lambda = 2$ has $f(0) = 2$). Only areas under f are probabilities. And every single point has probability zero, which is why $<$ and $\le$ can be swapped freely for continuous r.v.s — but **never** for discrete ones.

CDF:

$$F(x) = \int_{-\infty}^{x} f(u)\,\mathrm{d}u.$$

In very theoretical probability, what is defined first is F as before,

$$F(x) = P(X \le x),$$

then, if it is differentiable, the density f is defined as

$$f(x) = F'(x).$$

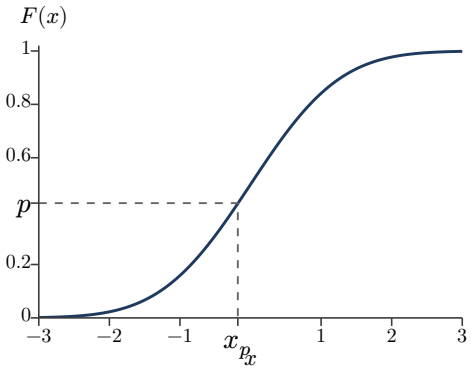
It is clear that

$$P(a \le X \le b) = F(b) - F(a) = \int_{a}^{b} f(x)\,\mathrm{d}x.$$

Inverse of the CDF (F^{-1}):

> **Definition 42.** The pth **quantile** is defined as the value x_p of the r.v. that satisfies $F(x_p) = p$.

- If F is monotonically (strictly) increasing, the pth quantile is unique (see figure).
- $F^{-1}(.5)$ is the **median**.
- $F^{-1}(.25)$ and $F^{-1}(.75)$ are the lower and upper **quartiles**.



> **Example 43.** Suppose $F(x) = x^2$, $0 \le x \le 1$. Then

$$x_p^2 = p, \quad x_p = \sqrt{p},$$

$$x_{.5} = \sqrt{.5} = .707, \quad x_{.25} = \sqrt{.25} = .5, \quad x_{.75} = \sqrt{.75} = .866.$$

2.2.1 Uniform(a, b)

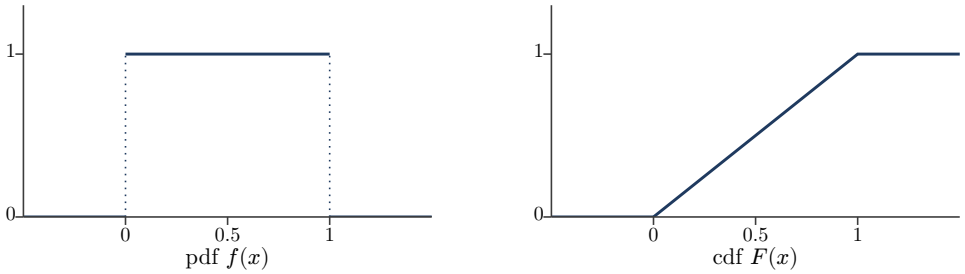
$$f(x) = \begin{cases} \frac{1}{b-a}, a \le x \le b \\ 0, \quad \text{otherwise.} \end{cases}$$

- Finite support ($\int_{-\infty}^{\infty} f(t)\,\mathrm{d}t \stackrel{?}{=} 1$).
- A non-informative distribution.
- CDF (for Uniform$(0, 1)$):

$$F(x) = \begin{cases} 0, x < 0 \\ x, 0 \le x \le 1 \\ 1, x > 1. \end{cases}$$

- Plot the pdf of Uniform$(0, 1)$.
- Repeating an experiment many times under this pdf, what does the data look like?
- HW: write and draw a discrete version Uniform(n), $X = 1, ..., n$. Write down 30 numbers drawn from Uniform(10) using your mind first, then using Matlab. Bring them next time! :)

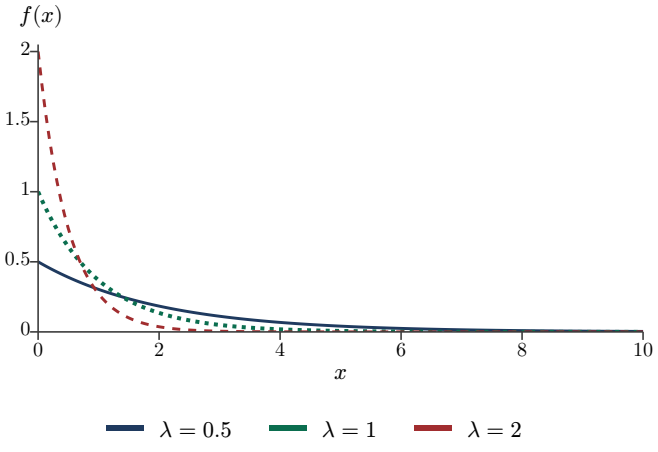
▸ **Lecture note.** For general (a, b): $F(x) = (x - a)/(b - a)$ on $[a, b]$, and $P(c < X < d) = (d - c)/(b - a)$ for $a \le c < d \le b$ — probability is just **relative length**.



2.2.2 Exponential(λ)

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, x \ge 0 \\ 0, \quad\quad x < 0. \end{cases}$$

Note the effect of λ (the rate) on the pdf.



$$\int_{-\infty}^{\infty} f(t)\,\mathrm{d}t \stackrel{?}{=} 1.$$

CDF:

$$F(x) = \int_{-\infty}^{x} f(u)\,\mathrm{d}u = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0. \end{cases}$$

Median:

$$1 - e^{-\lambda x_{.5}} = \frac{1}{2} \quad \Longrightarrow \quad x_{.5} = \frac{\log 2}{\lambda}.$$

Application: modeling life time T.

$$P(T > t + s \mid T > s) = \frac{P(T > t + s \cap T > s)}{P(T > s)} = \frac{P(T > t + s)}{P(T > s)}$$
$$= \frac{1 - \left(1 - e^{-\lambda(t+s)}\right)}{1 - (1 - e^{-\lambda s})} = e^{-\lambda t}.$$

Notice that:

- $e^{-\lambda t} = P(T > t)$ (unconditional);
- $e^{-\lambda t}$ is not a function of s!!! This is called the **memoryless property**;
- this is not good modeling for human life time. Why?
- Probably good for electronic components.

▸ **Lecture note.** **Why not for humans:** memorylessness says a 90-year-old has the same chance of living ten more years as a newborn. Humans **age** — their failure rate increases with time. Electronic components (to a first approximation) do not wear out; they fail from random shocks, which arrive at a constant rate regardless of age. When the hazard is constant, the life is exponential — and the next box proves that the converse also holds.

Connection with Poisson(λ):

- t_0: the time of an event;
- T: the time between two successive events;
- λ: events per unit time (the Poisson parameter).

$$P(T > t) = P(\text{no events in } (t_0, t_0 + t)) = (\lambda t)^k \frac{e^{-\lambda t}}{k!}\bigg|_{k=0} = e^{-\lambda t}.$$

⋆ **Intuition.** One process, two descriptions. Count the events in a window of length t: that count is Poisson(λt). Measure the gap between consecutive events: that gap is Exponential(λ). "No event for t time units" is the same statement as "zero events in the window", which is why the two formulas match.

Example 44 (Marshall et al. (1990)).

- Blocking a frog's nerve channel (passing current) using the M-Sux drug.
- The drug closes the channel: T is the duration of opening.
- Effect of the drug on the opening time.
- Set $\lambda = 1/\tau$:

$$f(t) = \frac{1}{\tau} e^{-t/\tau}.$$

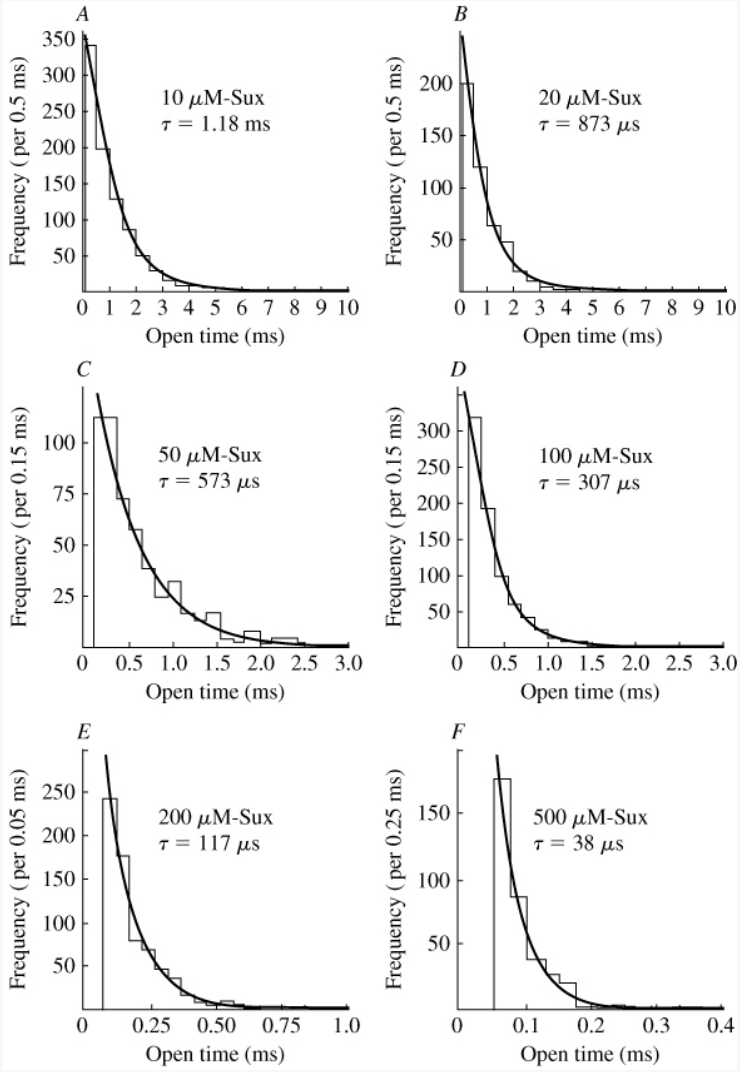


Figure 1: Histograms of open times at six M-Sux concentrations with fitted exponential densities (Rice, 3rd ed.).

Observations:

- $T \sim \text{Exponential}(\lambda)$;
- $\tau(= 1/\lambda)$ **decreases** as the amount of M-Sux increases.

Correction to the original. The original says τ **increases** with the amount of M-Sux, but the panels go $10\,\mu M \to \tau = 1.18$ ms, $20 \to 873\,\mu s$, ..., $500 \to 38\,\mu s$: the mean open time falls as the dose rises — more drug, faster blocking.

Exponential $\iff$ Memorylessness. We showed exponential $\implies$ memoryless. Conversely, set

$$P(T > t + s \mid T > s) = P(T > t) \quad \forall s \geq 0.$$

Then

$$\frac{P(T > t+s \cap T > s)}{P(T > s)} = P(T > t)$$
$$\frac{P(T > t+s)}{P(T > s)} = P(T > t)$$
$$1 - F(t+s) = (1 - F(s))(1 - F(t))$$
$$F(s)(1 - F(t)) = F(t+s) - F(t)$$
$$\frac{F(s)}{s}(1 - F(t)) = \frac{F(t+s) - F(t)}{s}$$
$$(1 - F(t)) \lim_{s\to 0} \frac{F(s)}{s} = \lim_{s\to 0} \frac{F(t+s) - F(t)}{s},$$

where (using $F(0) = 0$ for a lifetime)

$$\lim_{s\to 0} \frac{F(s)}{s} = F' \frac{s}{1}\Big|_{s=0} = F'(0) = f(0) \quad \text{(call it } \lambda\text{)}.$$

Hence

$$(1 - F(t))\lambda = F'(t) = -(1 - F(t))'$$
$$-\lambda t + c = \log(1 - F(t))$$
$$F(t) = 1 - e^{-\lambda t + c}.$$

To find the constant c: $f(t) = \lambda e^{-\lambda t + c}$, so $f(0) = \lambda = \lambda e^c$ and $c = 0$:

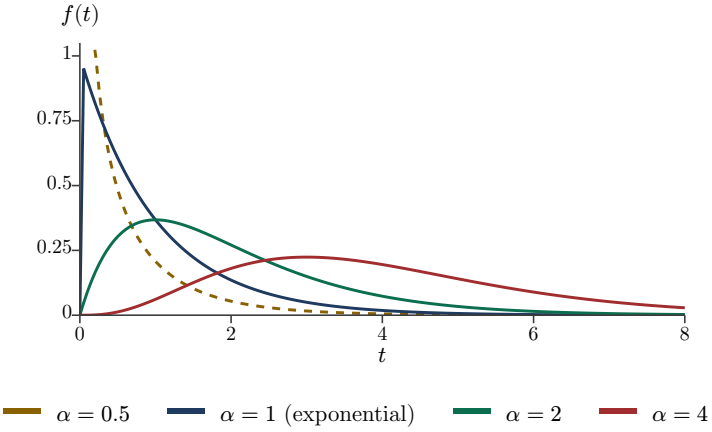
$$F(t) = 1 - e^{-\lambda t}, \quad f(t) = \lambda e^{-\lambda t}.$$

2.2.3 Gamma(α, λ)

$$f(t) = \frac{\lambda^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\lambda t}, \quad 0 \le t; \quad \alpha, \lambda > 0,$$
$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t}\, \mathrm{d}t,$$
$$\int_{-\infty}^{\infty} f(t)\, \mathrm{d}t \stackrel{?}{=} 1.$$

- α and λ are the **shape** and **rate** parameters ($1/\lambda$ is the scale).
- See the Mathematica notebook.
- Prove that $\alpha = 1$ gives Exponential(λ).

Solution. $\Gamma(1) = \int_0^\infty e^{-t}\, \mathrm{d}t = 1$, so with $\alpha = 1$: $f(t) = \lambda^1 t^0 e^{-\lambda t}/1 = \lambda e^{-\lambda t}$. The normalisation follows from the substitution $u = \lambda t$: $\int_0^\infty \lambda^\alpha t^{\alpha-1} e^{-\lambda t}\, \mathrm{d}t = \int_0^\infty u^{\alpha-1} e^{-u}\, \mathrm{d}u = \Gamma(\alpha)$.



Gamma densities with $\lambda = 1$ — added figure.

▸ **Lecture note.** Useful facts: $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ (integrate by parts), so $\Gamma(n) = (n-1)!$ for integers, and $\Gamma(1/2) = \sqrt{\pi}$. For integer $\alpha = n$, Gamma(n, λ) is the waiting time until the nth Poisson event — the continuous analogue of the negative binomial. Small $\alpha < 1$ piles mass near 0; large α looks more and more like a normal curve.

Example 45 (Earthquake Pattern).

- Very erratic, and difficult to model.
- Find the pdf of T, the time separating a sequence of small earthquakes.
- Exponential is not good because of the memorylessness property.
- Gamma (▲) fits the data better than exponential (•): $\alpha = .509$, $\lambda = .0015$.
- With these values, one can show that there is a large probability that the next earthquake immediately follows any given one; and this probability decreases with time (of course; why?).

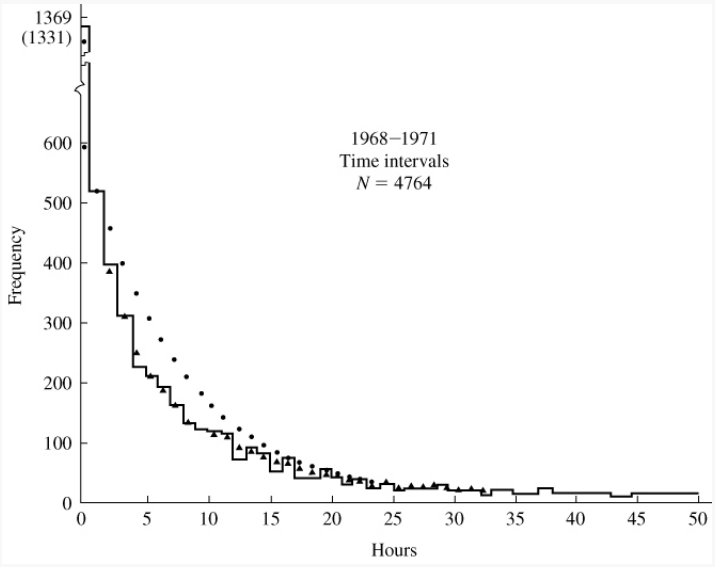
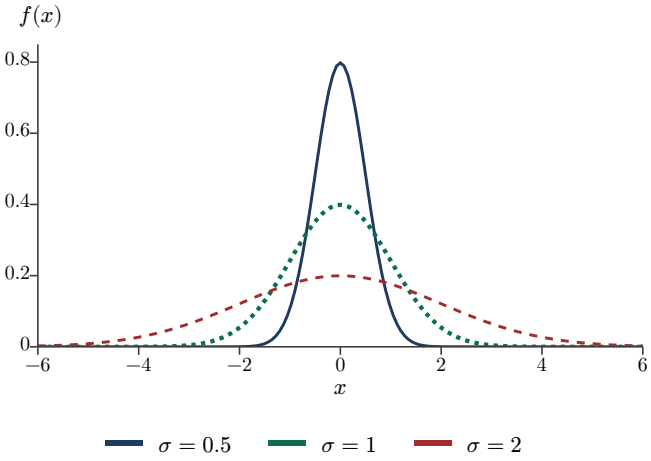


Figure 2: Times between successive earthquakes, 1968–1971, $N = 4764$, with fitted gamma (▲) and exponential (•) frequencies (Rice, 3rd ed.).

▸ **Lecture note.** **Why?** With $\alpha < 1$, the gamma density is infinite at 0 and decreasing: short gaps are very likely. Physically, earthquakes come in **clusters** (aftershocks) — one quake raises the chance of another soon after. The exponential cannot express that, because its hazard rate is constant; the gamma with $\alpha < 1$ has a **decreasing** hazard, which is exactly clustering.

2.2.4 Normal(μ, σ^2) (our ever friend)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(x-\mu)^2/\sigma^2}, \quad \sigma > 0, \qquad \int_{-\infty}^{\infty} f(t)\,\mathrm{d}t \stackrel{?}{=} 1.$$



- **Normal**, because it is normal (statisticians).
- **Gaussian**, after Carl Friedrich Gauss, in measuring errors (applied scientists).
- **Bell**, because it has a bell shape (some other parties).
- Symmetric around μ.
- No closed-form CDF; it is called Φ (for $N(0,1)$).
- Again: repeating an experiment many times under this pdf, what does the data look like? How does this apply to the next example?

Proof. (Normalisation, $\mu = 0$, $\sigma = 1$.) Let $I = \int_{-\infty}^{\infty} e^{-x^2/2}\,\mathrm{d}x$. Then, in polar coordinates,

$$I^2 = \iint e^{-(x^2+y^2)/2}\,\mathrm{d}x\,\mathrm{d}y = \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2} r\,\mathrm{d}r\,\mathrm{d}\theta = 2\pi,$$

so $I = \sqrt{2\pi}$. The general case follows by the substitution $z = (x-\mu)/\sigma$. □

Example 46 (Veitch and Wilks (1985)). Fitting the amplitude of ice-cracking noise in the Arctic to a normal distribution.

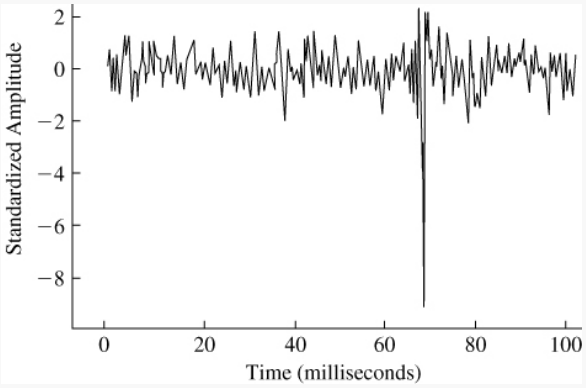


Figure 3: Standardised amplitude vs. time.

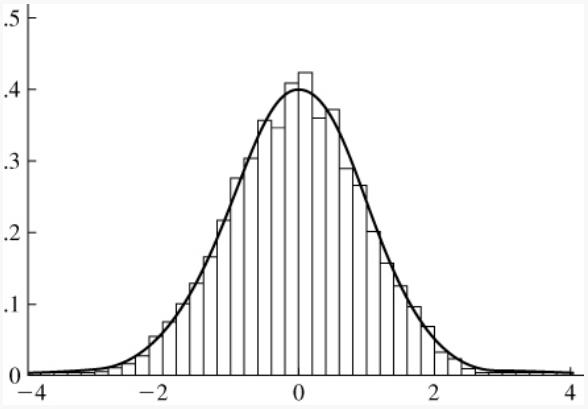


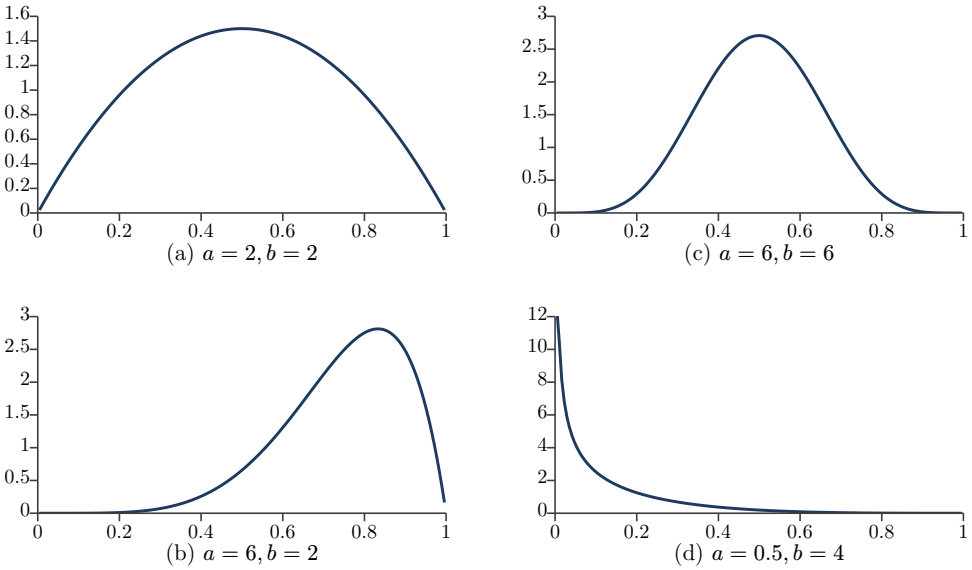
Figure 4: Histogram of amplitudes with the fitted normal density.

2.2.5 Beta(a, b)

$$f(x) = \frac{1}{B(a,b)} x^{a-1}(1-x)^{b-1}, \quad 0 \le x \le 1, \qquad B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}, \qquad \int_{-\infty}^{\infty} f(t)\,\mathrm{d}t \overset{?}{=} 1.$$

- Prove that $a = 1$, $b = 1$ gives Uniform$(0,1)$.
- Important in the Bayesian approach (later).
- See the Mathematica notebook for the shape parameters.

Correction to the original. The original writes $f(x) = \text{Beta}(a,b)x^{a-1}(1-x)^{b-1}$ with $\text{Beta}(a,b) = \Gamma(a+b)/(\Gamma(a)\Gamma(b))$ — i.e. it names the **normalising constant** "Beta". The standard notation is the Beta **function** $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, which appears in the **denominator**. The density is the same.



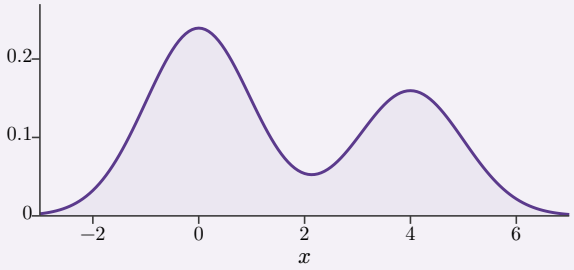
Solution. $B(1,1) = \Gamma(1)\Gamma(1)/\Gamma(2) = 1$, so $f(x) = x^0(1-x)^0 = 1$ on $[0,1]$: exactly Uniform$(0,1)$.

Example 47 (Mixtures). We toss a coin, where …

▸ **Lecture note.** The original stops mid-sentence here. The standard completion: toss a coin with $P(\text{head}) = \pi$. On a head, draw X from density f_1; on a tail, from f_2. By the law of total probability (Lemma 19) applied to $\{X \le x\}$,

$$F(x) = \pi F_1(x) + (1-\pi)F_2(x), \qquad f(x) = \pi f_1(x) + (1-\pi)f_2(x).$$

For example, $f_1 = N(0,1)$ and $f_2 = N(4,1)$ with $\pi = 0.6$ give a **bimodal** density — something no single normal can produce. Mixtures model populations made of sub-groups (e.g. heights of men and women together).



2.3 Functions of r.v.

Theorem 48. If $Y = g(X)$, and g is monotonically increasing (or decreasing), then g^{-1} exists and

$$f_Y(y) = \left|\frac{1}{\mathrm{d}y/\,\mathrm{d}x}\right| f_X(g^{-1}(y)).$$

Meaning first:

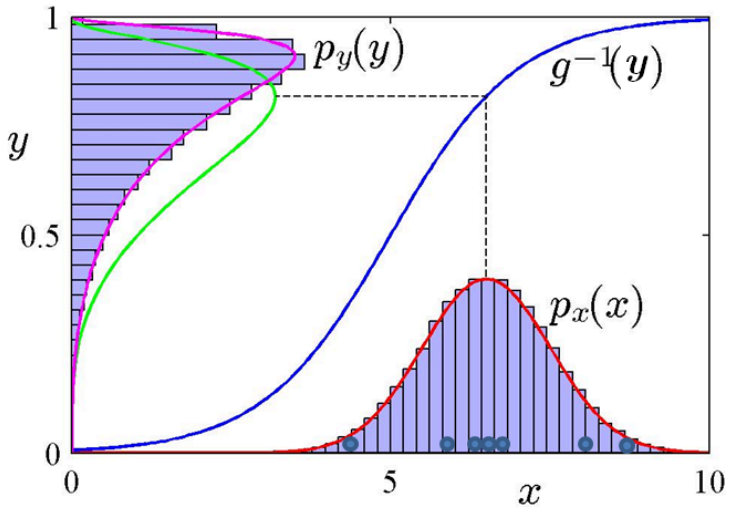


Figure 5: A density $p_{x(x)}$ pushed through an increasing map g. Each bar of the x-histogram lands in a y-bar whose **width** is stretched or squeezed by the slope of g; the height changes to keep the area. The magenta curve is $p_{y(y)}$; the green curve is the naive (wrong) answer $p_{x(g^{-1}(y))}$ without the slope factor.

"Not a proof." At (x, y), where $y = g(x)$ (or $x = g^{-1}(y)$):

$$P(y < Y \le y + \mathrm{d}y) = P(x < X \le x + \mathrm{d}x),$$
$$f_Y(y)\,|\,\mathrm{d}y| = f_X(g^{-1}(y))\,|\,\mathrm{d}x|.$$

The "not a proof" is complete.

Proof. For increasing g:

$$P(Y \le y) = P(X \le x), \quad y = g(x) \implies F_Y(y) = F_X(x).$$

Differentiate with respect to y:

$$f_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y}F_X(x) = \frac{\mathrm{d}x}{\mathrm{d}y}\frac{\mathrm{d}}{\mathrm{d}x}F_X(x) = \frac{1}{\mathrm{d}y/\,\mathrm{d}x}f_X(x) = \frac{1}{\mathrm{d}y/\,\mathrm{d}x}f_X(g^{-1}(y)).$$

Similarly, if g is monotonically decreasing, the only difference is $P(Y > y) = P(X \le x)$, and we reach

$$f_Y(y) = \frac{-1}{\mathrm{d}y/\,\mathrm{d}x}f_X(g^{-1}(y)).$$

Since $\mathrm{d}y/\,\mathrm{d}x < 0$ in that case, both formulas are, in general,

$$f_Y(y) = \left|\frac{1}{\mathrm{d}y/\,\mathrm{d}x}\right| f_X(g^{-1}(y)).$$

□

▸ **Lecture note.** The four-step recipe for any transformation problem:

1. Write the pdf of X **and its support**.
2. Find the support of $Y = g(X)$ and check that g is monotone on it.
3. Solve $y = g(x)$ for $x = g^{-1}(y)$ and compute $|\mathrm{d}x/\,\mathrm{d}y|$.
4. $f_Y(y) = f_X(g^{-1}(y))|\mathrm{d}x/\,\mathrm{d}y|$ on the support of Y, zero elsewhere.

Forgetting step 2 (the new support) is the most common way to lose marks.

Take care if g^{-1} has two values, e.g. $X = Z^2$. You can also do it from scratch:

$$P(X \le x) = P\big(-\sqrt{x} \le Z \le \sqrt{x}\big) = F_Z\big(\sqrt{x}\big) - F_Z\big(-\sqrt{x}\big),$$

$$f_X(x) = \left|\frac{\mathrm{d}}{\mathrm{d}x}\sqrt{x}\right| f_Z\big(\sqrt{x}\big) + \left|\frac{\mathrm{d}}{\mathrm{d}x}\sqrt{x}\right| f_Z\big(-\sqrt{x}\big) = \frac{1}{2}x^{-1/2} f_Z\big(\sqrt{x}\big) + \frac{1}{2}x^{-1/2} f_Z\big(-\sqrt{x}\big).$$

Now, if $Z \sim N(0,1)$, then it is symmetric and

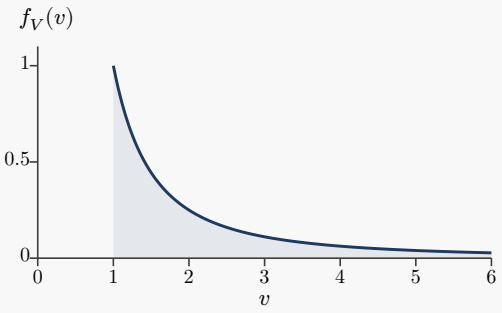
$$f_X(x) = x^{-1/2} f_Z\big(\sqrt{x}\big) = x^{-1/2}\frac{1}{\sqrt{2\pi}}\exp\left[-\frac{1}{2}z^2\right]\Big|_{z=\sqrt{x}} = \frac{1}{\sqrt{2\pi}}x^{-1/2}e^{-x/2}, \quad x > 0,$$

which is the pdf of $\mathrm{Gamma}\big(\frac{1}{2}, \frac{1}{2}\big)$. It is also called the **chi-square** (χ^2) distribution with 1 degree of freedom.

Example 49 (Solving by Theorem).

- $U \sim \mathrm{Uniform}(0,1)$, $V = 1/U$.
- First, the pdf of U, its domain, V, its domain, and the function g.
- Apply the theorem to get f_V, then draw it:

$$f_V(v) = f_U\left(\frac{1}{v}\right)\left|\frac{\mathrm{d}}{\mathrm{d}v}\left(\frac{1}{v}\right)\right| = 1 \times \frac{1}{v^2}, \quad v > 1.$$



Special case: $Y = aX + b$.

$$F_Y(y) = P(Y < y) = P(aX + b < y) = P\left(X < \frac{y-b}{a}\right) = F_X\left(\frac{y-b}{a}\right), \quad a > 0,$$

$$f_Y(y) = \frac{1}{a} f_X\left(\frac{y-b}{a}\right).$$

If $a < 0$, similarly:

$$F_Y(y) = P\left(X > \frac{y-b}{a}\right) = 1 - F_X\left(\frac{y-b}{a}\right),$$

$$f_Y(y) = -\frac{1}{a} f_X\left(\frac{y-b}{a}\right).$$

In general $\forall a \neq 0$ (why?):

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right).$$

We will see later (a HW problem in the Extra Materials) the condition for the maxima of f_Y to coincide with the maxima of f_X (the case of the green pdf above).

For the case $X \sim N(\mu, \sigma^2)$:

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right) = \frac{1}{|a|}\frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}(x-\mu)^2/\sigma^2\right]\Big|_{x=\frac{y-b}{a}}$$

$$= \frac{1}{|a|\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{y-(b+a\mu)}{a\sigma}\right)^2\right],$$

which is $N(b+a\mu, (a\sigma)^2)$.

Interestingly: if

$$Z = \frac{X-\mu}{\sigma} = \frac{1}{\sigma}X - \frac{\mu}{\sigma},$$

then Z has the standard normal density: $Z \sim N(0,1)$.

Application: finding $P(x_0 < X < x_1)$ for any normal r.v. requires knowing only the CDF of Z:

$$P(x_0 < X < x_1) = P\left(\frac{x_0-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{x_1-\mu}{\sigma}\right) = P\left(\frac{x_0-\mu}{\sigma} < Z < \frac{x_1-\mu}{\sigma}\right)$$

$$= \Phi\left(\frac{x_1-\mu}{\sigma}\right) - \Phi\left(\frac{x_0-\mu}{\sigma}\right).$$

Take care; e.g.

$$x_0 < X \iff \frac{x_0-\mu}{\sigma} < \frac{X-\mu}{\sigma}, \quad \text{therefore} \quad P(x_0 < X) = P\left(\frac{x_0-\mu}{\sigma} < \frac{X-\mu}{\sigma}\right).$$

But (show why by events and axioms):

$$x_0 < X \Longrightarrow x_0^2 < X^2 \quad (x_0 \geq 0), \quad \text{therefore} \quad P(x_0 < X) \leq P(x_0^2 < X^2).$$

Solution. For $x_0 \geq 0$, the event $\{x_0 < X\}$ is a **subset** of $\{x_0^2 < X^2\}$ (every ω with $X(\omega) > x_0 \geq 0$ also has $X(\omega)^2 > x_0^2$), but not equal: $X < -x_0$ is in the second event only. Monotonicity of P (§1.3) gives the inequality. An **increasing** map preserves events exactly ($\iff$); a non-monotone map only gives $\Longrightarrow$.

Example 50 (IQ Test Scores).

- It is found that $X \sim N(100, 15^2)$.
- What is the probability that $X \in [120, 130]$?

$$\Pr = P(120 < X < 130) = P\left(\frac{120-100}{15} < \frac{X-100}{15} < \frac{130-100}{15}\right)$$

$$= P(1.33 < Z < 2) = \Phi(2) - \Phi(1.33) = .9772 - .9082 = .069.$$

So only 7% of students score in that range.

Example 51 (σ and μ).

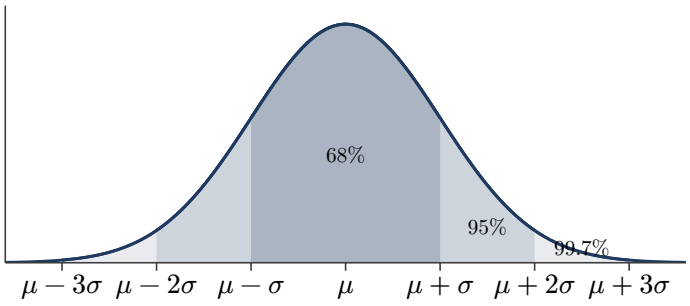
$$P(|X-\mu| < \sigma) = P(-\sigma < X-\mu < \sigma) = P\left(-1 < \frac{X-\mu}{\sigma} < 1\right)$$

$$= P(-1 < Z < 1) = \Phi(1) - \Phi(-1) = .68.$$

Similarly,

$$P(|X-\mu|<2\sigma)=\Phi(2)-\Phi(-2)=.9545,\quad P(|X-\mu|<3\sigma)=\Phi(3)-\Phi(-3)=.9973.$$

So almost all the probability is contained in $[-3\sigma,3\sigma]$ around μ. Also, $[-2\sigma,2\sigma]$ is a good approximation.



The 68–95–99.7 rule — added figure.

▸ **Lecture note.** Use symmetry to read negative values: $\Phi(-z)=1-\Phi(z)$, so $P(|Z|<z)=2\Phi(z)-1$. Standardise **first**, look up **second**: every normal question becomes a Z question.

2.3.1 Uniform Distribution & r.v. Generators

Proposition 52. Let $Z=F_X(X)$; then Z is distributed as $\text{Uniform}(0,1)$.

Proof. First, draw the problem, then notice that $0\le Z\le 1$. Then

$$F_Z(z)=P(Z\le z)=P(F_X(X)\le z)=P(X\le F_X^{-1}(z))=F_X(F_X^{-1}(z))=z,$$

which is the cdf of $\text{Uniform}(0,1)$. If we want to prove it by using the theorem:

$$f_Z(z)=f_X(F_X^{-1}(z))\left|\frac{\mathrm{d}}{\mathrm{d}z}F_X^{-1}(z)\right|=f_X(F_X^{-1}(z))\times\frac{1}{f_X(F_X^{-1}(z))}=1.$$

□

Proposition 53. Let $U\sim\text{Uniform}(0,1)$, and let $X=F^{-1}(U)$ (for some cdf F). Then $F_X=F$.

Proof. First, draw the problem.

$$F_X(x)=P(X\le x)=P(F^{-1}(U)\le x)=P(U\le F(x))=F(x),$$

which is used to generate a r.v. if we know its distribution F_X and have access only to a uniform r.v. generator. □

Example (inverse-cdf sampling — added). To simulate Exponential(λ): solve $u=F(x)=1-e^{-\lambda x}$ to get $x=-\log(1-u)/\lambda$. So `X = -log(1 - rand()) / lambda` produces exponential data from a uniform generator. (Since $1-U$ is also uniform, `-log(rand()) / lambda` works too.)

Chapter summary

- A **random variable** is a function $X : \Omega \to \mathbb{R}$; probabilities about X are probabilities of events $\{\omega : X(\omega) \in ...\}$.
- **cdf** $F(x) = P(X \le x)$: non-decreasing, 0 at $-\infty$, 1 at ∞, right-continuous. Discrete: a staircase with jumps $p(x_k)$. Continuous: $f = F'$, and $P(a < X < b) = \int_a^b f$.

Distribution	pmf / pdf	Story
Ber(p)	$p^x(1-p)^{1-x}$, $x \in \{0,1\}$	one trial
Bin(n,p)	$\binom{n}{k}p^k(1-p)^{n-k}$	successes in n trials
Geom(p)	$(1-p)^{k-1}p$, $k \ge 1$	trials to 1st success
NBin(r,p)	$\binom{k-1}{r-1}p^r(1-p)^{k-r}$	trials to rth success
HGeom(n,r,m)	$\binom{r}{k}\binom{n-r}{m-k}/\binom{n}{m}$	labeled in a sample, no replacement
Poisson(λ)	$\lambda^k e^{-\lambda}/k!$	rare events at rate λ
U(a,b)	$1/(b-a)$ on $[a,b]$	no information
Exp(λ)	$\lambda e^{-\lambda x}$, $x \ge 0$	memoryless waiting time
Gamma(α,λ)	$\lambda^\alpha x^{\alpha-1}e^{-\lambda x}/\Gamma(\alpha)$	waiting for α events
$N(\mu,\sigma^2)$	$e^{-(x-\mu)^2/2\sigma^2}/\left(\sigma\sqrt{2\pi}\right)$	sums of many small effects
Beta(a,b)	$x^{a-1}(1-x)^{b-1}/B(a,b)$	a random proportion

- **Transformations:** $f_Y(y) = f_X(g^{-1}(y))|\mathrm{d}x/\,\mathrm{d}y|$ for monotone g; for non-monotone g, work through the cdf. $Y = aX + b$: $f_Y(y) = f_X((y-b)/a)/|a|$.
- **Standardising:** $Z = (X-\mu)/\sigma \sim N(0,1)$; $P(|X-\mu| < k\sigma) = .68, .95, .997$.
- $F_X(X) \sim \mathrm{U}(0,1)$ and $F^{-1}(U) \sim F$ — the basis of simulation.

Practice problems — Chapter 2

2.1 A die is rolled twice; let X be the larger of the two numbers. Find the pmf and sketch the cdf of X.

2.2 A multiple-choice exam has 10 questions with 4 options each. A student guesses all of them. Find the probability of at least 5 correct answers.

2.3 The number of typos per page of a book is Poisson(0.5). Find the probability that a given page has (a) no typos, (b) at least two typos.

2.4 Let $f(x) = cx^2$ on $[0,2]$. Find c, the cdf, the median, and $P(X > 1)$.

2.5 A light bulb lasts Exponential with mean 1000 hours. Given that it has lasted 500 hours, what is the probability it lasts at least another 1000?

2.6 $X \sim N(70, 10^2)$. Find $P(X > 85)$ and the value x such that $P(X < x) = 0.95$.

2.7 Let $X \sim \mathrm{U}(0,1)$ and $Y = -2\log X$. Find f_Y and name the distribution.

2.8 Let $Z \sim N(0,1)$ and $Y = e^Z$ (the **lognormal**). Find f_Y.

CHAPTER 3

Joint Distributions

3.1 Introduction: Examples for Joint Distributions

- The number of predators and the number of prey for a particular species in ecology.
- Height and weight of a particular category of people.
- A model for the joint distribution of age and length in a population of fish.

Motivation by a very simple example: the two-point distributions on $\{(0,0),(1,1)\}$ and on $\{(1,0),(0,1)\}$ (each point with probability $1/2$) are different joint distributions. However, in both, X and Y each have the same marginal distribution — Bernoulli$(1/2)$.

▸ **Lecture note.** This is the whole reason the chapter exists. In the first case $X = Y$ always; in the second $X = 1 - Y$ always — opposite relationships, identical marginals. **The marginals do not determine the joint.** To talk about how two quantities move together you need their joint distribution.

Another example: the height and weight of some species of fish are reported for 100 fish. How are they related to each other?

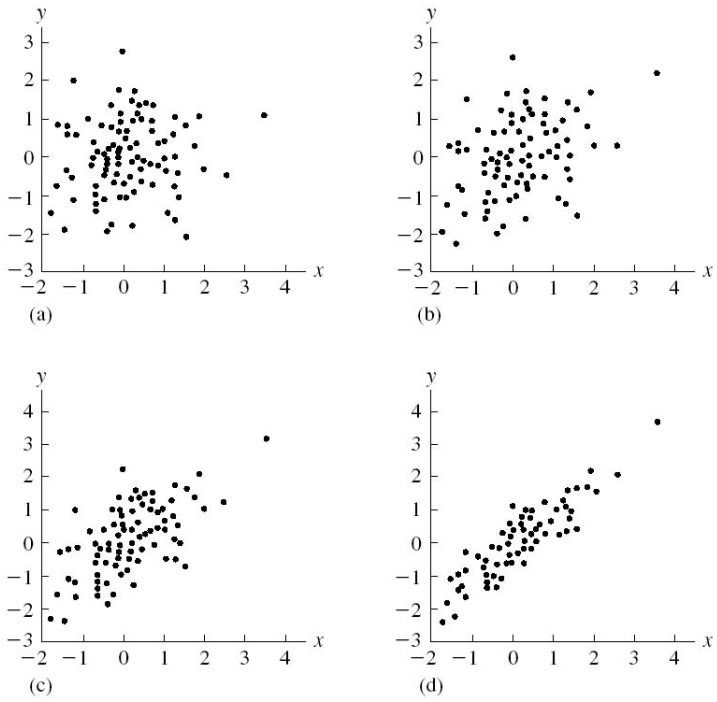


Figure 6: Four scatterplots of 100 pairs (x, y) with identical marginals but increasing dependence, from (a) none to (d) strong (Rice, 3rd ed.).

Revision: the relation between F, f and p. For both discrete and continuous r.v.s:

$$P(X \le x) = F_X(x),$$
$$P(x_1 < X \le x_2) = P(X \in (x_1, x_2]) = F_X(x_2) - F_X(x_1), \quad x_1 < x_2.$$

Discrete only:

$$F_X(x_k) = \sum_{i=-\infty}^{k} p(x_i), \qquad p_X(x_k) = F_X(x_k) - F_X(x_{k-1}).$$

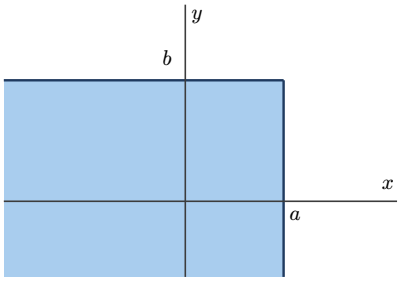
Continuous only:

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\,\mathrm{d}u, \qquad f_X(x) = F_X'(x).$$

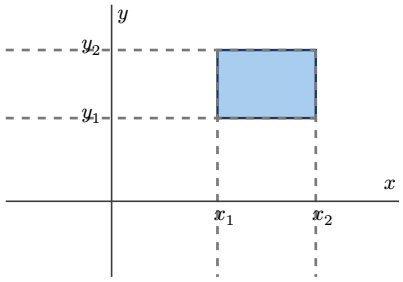
Definition 54. The **joint distribution function** of two r.v.s is defined as

$$F_{XY}(x, y) = P(X \le x, Y \le y).$$

When there is no confusion we can write just F instead of F_{XY}.



$F(a, b) = P(X \le a, Y \le b)$: the quadrant below-left of (a, b)



a rectangle by inclusion–exclusion of four quadrants

Therefore, for any rectangle in the space,

$$P(x_1 < X \le x_2, y_1 < Y \le y_2) = F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1).$$

⋆ **Intuition.** Start with the big quadrant at (x_2, y_2). Cut away the quadrant below y_1 and the one left of x_1. The corner quadrant at (x_1, y_1) was cut away **twice**, so add it back once — the addition rule of Chapter 1, in two dimensions.

It can be proven that (proof omitted) any more complex area than rectangles, e.g. a circle, can be determined by limits of rectangles. Hence, F can determine the probability of any region in the space.

Definition 55 (Generalization). The joint distribution function of several r.v.s, in p-dimensional space, is defined as

$$F_{X_1 \ldots X_p} = P(X_1 \le x_1, \ldots, X_p \le x_p).$$

3.2 Discrete r.v.

Definition 56. The joint pmf of two r.v.s:

$$p(x_i, y_j) = P(X = x_i, Y = y_j).$$

Example 57. Coin tossing 3 times: $\Omega = \{hhh, hht, hth, htt, thh, tht, tth, ttt\}$. X and Y are the number of heads in the 1st toss and in all 3 tosses, respectively.

$X \setminus Y$	0	1	2	3
0	$\frac{1}{8}$ $\{ttt\}$	$\frac{2}{8}$ $\{tht, tth\}$	$\frac{1}{8}$ $\{thh\}$	0 $\{\}$
1	0 $\{\}$	$\frac{1}{8}$ $\{htt\}$	$\frac{2}{8}$ $\{hht, hth\}$	$\frac{1}{8}$ $\{hhh\}$

$$p_Y(0) = P(Y=0) = P(\{(Y=0, X=0)\} \cup \{(Y=0, X=1)\})$$
$$= P(Y=0, X=0) + P(Y=0, X=1) = \frac{1}{8},$$
$$p_Y(1) = P(Y=1) = P(Y=1, X=0) + P(Y=1, X=1) = \frac{3}{8}.$$

Correction to the original. The original says "X, Y: # heads in 1st and 3 tosses respectively" — Y counts heads in **all three** tosses (it ranges over $0, ..., 3$), not in the third toss.

Definition 58. The **marginal pmf** is defined as

$$p_{X_1}(x_1) = \sum_{x_2} p(x_1, x_2);$$

more generally,

$$p_{X_1}(x_1) = \sum_{x_2, \dots, x_p} p(x_1, \dots, x_p);$$

and more generally still,

$$p_{X_1, \dots, X_r}(x_1, \dots, x_r) = \sum_{x_{r+1}, \dots, x_p} p(x_1, \dots, x_p).$$

▸ **Lecture note.** "Marginal" is literal: in a joint table, sum each row (or column) and write the total **in the margin**. In Example 57, the column sums are $p_Y = (1/8, 3/8, 3/8, 1/8)$ — the Bin$(3, 1/2)$ of Chapter 2 — and the row sums are $p_X = (1/2, 1/2)$.

Notice that:

$$F_{X_1, \dots, X_p}(x_1, \dots, x_p) = P(X_1 \le x_1, \dots, X_p \le x_p) = \sum_{a_1=-\infty}^{x_1} \cdots \sum_{a_p=-\infty}^{x_p} p(a_1, \dots, a_p).$$

3.2.1 Multinomial Distribution

- A generalization of Binomial(n, p).
- n independent trials.
- Each can result in one of r types of outcomes (c.f. 2 types in Binomial(n, p)).
- Each type has a probability p_i, $\sum_i p_i = 1$ (c.f. $p, 1-p$).
- The outcome is $N_1 = n_1, \dots, N_r = n_r$, $\sum_i N_i = n$ (c.f. $N = k$),
- each such sequence with probability $p_1^{n_1} \cdots p_r^{n_r}$.
- For example: n students, each can get A, B, C, D, or F, with probabilities $p_1, \dots, p_5$. What is the probability that n_1 get A, n_2 get B, ..., n_5 fail?
- The number of these choices (from §1.4.2) is $n!/(n_1! \cdots n_r!)$. Therefore

$$p(n_1,...,n_r) = \frac{n!}{n_1!\cdots n_r!} p_1^{n_1}\cdots p_r^{n_r} = \binom{n}{n_1\cdots n_r} \prod_{i=1}^{r} p_i^{n_i}.$$

Special case $r = 2$:

$$p(n_1,n_2) = \frac{n!}{n_1!n_2!} p_1^{n_1} p_2^{n_2} = \frac{n!}{n_1!(n-n_1)!} p_1^{n_1}(1-p_1)^{n-n_1} = \binom{n}{n_1} p_1^{n_1}(1-p_1)^{n-n_1} \equiv \text{Binomial}(n,p_1).$$

Notice that the marginal, e.g. $p_{N_1}(n_1)$, is very difficult to obtain by

$$p_{N_1}(n_1) = \sum_{n_2,\ldots,n_r} p(n_1,...,n_r), \quad \text{or in general} \quad p_{N_i}(n_i) = \sum_{n_1,\ldots,n_{i-1},n_{i+1},\ldots,n_r} p(n_1,...,n_r).$$

However, it can be obtained at once by noticing that

$$N_i \sim \text{Binomial}(n,p_i), \qquad p_{N_i}(n_i) = \binom{n}{n_i} p_i^{n_i}(1-p_i)^{n-n_i}.$$

⋆ **Intuition.** To find the marginal of N_1 ("how many A's?"), **merge** the other categories into one: "A" versus "not A". Each trial is now a success with probability p_1 — a binomial, no summation needed. Merging categories is a powerful trick: e.g. $N_1 + N_2 \sim \text{Bin}(n, p_1 + p_2)$.

Example 59 (Histogram).

- n independent observations, e.g. $n = 100$, from Uniform$(0,1)$.
- Partition the interval $[0,1]$ into 10 equal bins.
- n_i is the number of observations falling in the ith bin, i.e.

$$n_i = \sum_{j=1}^{n} I_{X_j \in i^{\text{th}} \text{ bin}}.$$

- Therefore, $N_1,...,N_{10}$ are Multinomial$(100, p_1,...,p_{10})$, $p_i = 0.1$ (one tenth of the interval $[0,1]$).
- Notice the fluctuations in the following histograms; each corresponds to a new set of 100 observations.

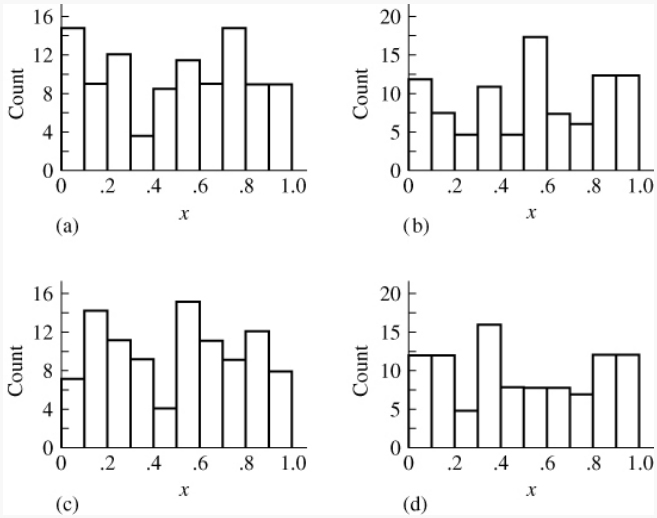


Figure 7: Four histograms, each of 100 new Uniform$(0,1)$ observations (Rice, 3rd ed.).

Later we will see, very interestingly, that with increasing n:

- we can increase the number of bins, i.e. decrease the width of each bin;
- the histogram stabilizes ("converges");
- the histogram "converges" to the original pdf, regardless of whether it is uniform or not.

▸ **Lecture note.** How big are the fluctuations? Each bar is $\mathrm{Bin}(100, 0.1)$, with mean 10 and standard deviation $\sqrt{100 \cdot 0.1 \cdot 0.9} = 3$ (Chapter 4). So bars between 4 and 16 are entirely normal — exactly what the four histograms show. A histogram of 100 points is a rough sketch of the pdf, not a photograph.

3.3 Continuous r.v.

Definition 60. If the joint cdf of two r.v.s is differentiable, then their joint pdf is defined as

$$f_{XY}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x,y);$$

and therefore

$$F_{XY}(x,y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{XY}(u,v)\,\mathrm{d}u\,\mathrm{d}v,$$

and it can also be shown that (see extra material)

$$P((X,Y) \in A) = \iint_A f_{XY}(u,v)\,\mathrm{d}u\,\mathrm{d}v.$$

Again: for very small δ_x and δ_y,

$$P\big(x \le X \le x+\delta_x, y \le Y \le y+\delta_y\big) = \int_x^{x+\delta_x} \int_y^{y+\delta_y} f_{XY}(u,v)\,\mathrm{d}u\,\mathrm{d}v \approx f_{XY}(x,y)\delta_x\delta_y.$$

Marginal:

$$F_X(x) = F_{XY}(x,\infty) = \int_{-\infty}^{x} \int_{-\infty}^{\infty} f_{XY}(u,v)\,\mathrm{d}v\,\mathrm{d}u, \qquad f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,\mathrm{d}y.$$

Generalization:

$$F\big(x_1,...,x_p\big) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_p} f\big(u_1,...,u_p\big)\,\mathrm{d}u_1 \cdots \mathrm{d}u_p,$$

$$P\big((X_1,...,X_p) \in A\big) = \int \cdots \int_A f\big(u_1,...,u_p\big)\,\mathrm{d}u_1 \cdots \mathrm{d}u_p,$$

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f\big(x_1,x_2,...,x_p\big)\,\mathrm{d}x_2 \cdots \mathrm{d}x_p.$$

Example 61.

$$f(x,y) = \frac{12}{7}(x^2+xy), \quad 0 \le x \le 1, \quad 0 \le y \le 1.$$

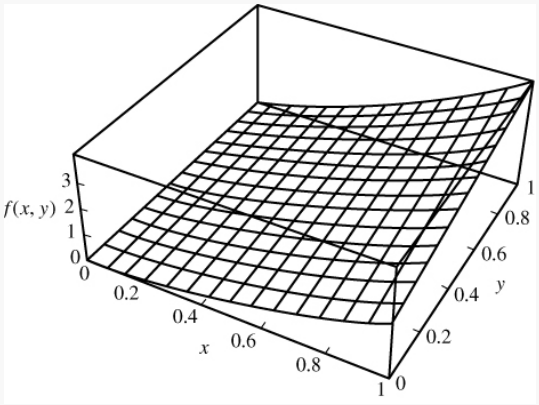


Figure 8: The density surface $f(x, y)$.

First make sure that it integrates to one:

$$\int_{x=0}^{x=1}\int_{y=0}^{y=1} f(x,y)\,\mathrm{d}y\,\mathrm{d}x = \frac{12}{7}\int_0^1 \left(x^2+\frac{x}{2}\right)\mathrm{d}x = \frac{12}{7}\left(\frac{1}{3}+\frac{1}{4}\right) = 1.$$

Next, find $P(X > Y)$. What does it mean when I pick a pair (x, y) from the population?

$$P(X>Y) = \frac{12}{7}\int_0^1\int_0^x (x^2+xy)\,\mathrm{d}y\,\mathrm{d}x = \frac{12}{7}\int_0^1 \frac{3x^3}{2}\,\mathrm{d}x = \frac{9}{14}.$$

Next, find f_X:

$$f_X(x) = \frac{12}{7}\int_0^1 (x^2+xy)\,\mathrm{d}y = \frac{12}{7}\left(x^2+\frac{x}{2}\right), \quad 0\le x\le 1.$$

▸ **Lecture note. The meaning of $P(X > Y)$:** if you sample many pairs from this population, about $9/14 \approx 64\%$ of them have the first coordinate larger. **The method** for any $P((X, Y) \in A)$: (1) draw the support of f, (2) shade A, (3) read the integration limits off the picture — the inner integral runs across a strip, the outer one moves the strip. Step 3 without step 1 is where most errors come from.

Example 62. Consider the density

$$f(x,y) = \begin{cases} \lambda^2 e^{-\lambda y}, & 0\le x\le y,\ \lambda>0 \\ 0, & \text{otherwise.}\end{cases}$$

Take care: it can be rewritten as

$$f(x,y) = \begin{cases} \lambda^2 e^{-\lambda y} I_{0\le x} I_{x\le y}, & \lambda>0 \\ 0, & \text{otherwise.}\end{cases}$$

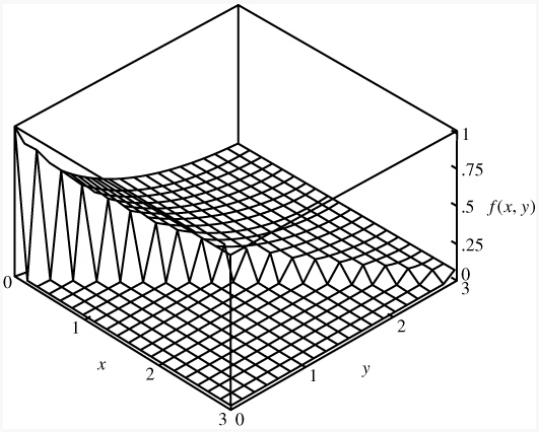


Figure 9: The density surface.

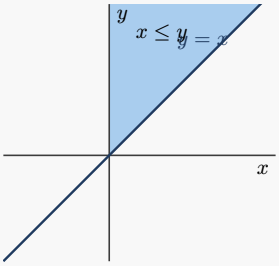


Figure 10: Its support: the wedge $0 \le x \le y$.

$$f_X(x) = \int_x^\infty \lambda^2 e^{-\lambda y}\,\mathrm{d}y = \lambda e^{-\lambda x}, \quad x \ge 0,$$

which is Exponential(λ);

$$f_Y(y) = \int_0^y \lambda^2 e^{-\lambda y}\,\mathrm{d}x = \lambda^2 y e^{-\lambda y}, \quad y \ge 0,$$

which is Gamma$(2, \lambda)$.

Example 63. A point is chosen randomly in a disk of radius 1. Then

$$f_{XY}(x,y) = \begin{cases} \frac{1}{\pi}, x^2 + y^2 \le 1 \\ 0, \text{ otherwise.} \end{cases}$$

What is $P(R \le r)$? In general, if $X = (X_1, ..., X_p)$ is uniformly distributed over a region A, then

$$f_X(x_1, ..., x_p) = \frac{1}{|A|}, \qquad P(X \in B) = \int_B f_X\,\mathrm{d}x = \frac{|B|}{|A|}.$$

So

$$P(R \le r) = P(X^2 + Y^2 \le r^2) = \frac{\pi r^2}{\pi} = r^2.$$

Later, we can do it differently: $R = \sqrt{X^2 + Y^2}$, which is a function of 2 r.v.s (transformation). What is f_X?

$$f_X(x) = \frac{1}{\pi} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \mathrm{d}y = \frac{2}{\pi}\sqrt{1-x^2}, \quad -1 \le x \le 1.$$

Curse of dimensionality: for p-dimensional spheres, $P(R \le r) = r^p$, which means that the data will be on the surface!!

Correction to the original. The original writes $P(R \le r) = P(x^2 + y^2 \le r)$; the event is $X^2 + Y^2 \le r^2$, and "$r = x^2 + y^2$" should be $r = \sqrt{x^2 + y^2}$.

▸ **Lecture note. Curse of dimensionality, in numbers:** the probability that a uniform point in the unit ball lies in the outer 5% shell is $1 - 0.95^p$: 10% for $p = 2$, 40% for $p = 10$, 99.4% for

$p = 100$. In high dimensions **almost all the volume is near the surface** — "the centre is empty". This is why nearest-neighbour methods struggle on high-dimensional data.

3.4 Independent r.v.

Definition 64. The r.v.s $X_1, ..., X_n$ are said to be **independent** if

$$F_{X_1 ... X_n}(x_1, ..., x_n) = F_{X_1}(x_1) \cdots F_{X_n}(x_n), \quad \forall x_1, ..., x_n.$$

For two r.v.s (for intuition),

$$F_{X_1X_2}(x_1, x_2) = F_{X_1}(x_1)F_{X_2}(x_2),$$
$$P(X_1 \le x_1, X_2 \le x_2) = P(X_1 \le x_1)P(X_2 \le x_2),$$

which is the independence of two events (from Ch. 1).

In particular, for continuous r.v.s, if $F_{X_1X_2}(x_1, x_2) = F_{X_1}(x_1)F_{X_2}(x_2)$, then

$$f_{X_1X_2}(x_1, x_2) = \frac{\partial^2 \left[F_{X_1}(x_1)F_{X_2}(x_2)\right]}{\partial x_1 \partial x_2} = f_{X_1}(x_1)f_{X_2}(x_2).$$

Also, if $f_{X_1X_2}(x_1, x_2) = f_{X_1}(x_1)f_{X_2}(x_2)$, then

$$\begin{aligned} F_{X_1X_2}(x_1, x_2) &= \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} f_{X_1}(u_1)f_{X_2}(u_2)\, \mathrm{d}u_2\, \mathrm{d}u_1 \\ &= \int_{-\infty}^{x_1} f_{X_1}(u_1)\, \mathrm{d}u_1 \int_{-\infty}^{x_2} f_{X_2}(u_2)\, \mathrm{d}u_2 = F_{X_1}(x_1)F_{X_2}(x_2). \end{aligned}$$

Also, it is easy to show that if X_1 and X_2 are independent, then

$$P(X_1 \in A, X_2 \in B) = P(X_1 \in A)P(X_2 \in B).$$

The proof can start easily from the pdf, or more generally from the cdf (see extra materials).

Lemma 65. If X and Y are independent r.v.s, then the r.v.s $G = g(X)$ and $H = h(Y)$ are independent as well.

Proof. Omitted in the original. Sketch: $\{G \le a, H \le b\} = \{X \in g^{-1}((-\infty, a]), Y \in h^{-1}((-\infty, b])\}$, and the product rule for $P(X \in A, Y \in B)$ above splits it into $P(G \le a)P(H \le b)$. □

Example 66 (cont. Ex. 62).

$$f_{XY}(x, y) = \begin{cases} \lambda^2 e^{-\lambda y}, & 0 \le x \le y,\ \lambda > 0 \\ 0, & \text{otherwise.} \end{cases}$$

Take care: it **looks** like it factors; however, it does not, since

$$f(x, y) = \lambda^2 e^{-\lambda y} I_{0 \le x} I_{x \le y},$$

and the indicator $I_{x \le y}$ ties x and y together. The marginals were

$$f_X(x) = \lambda e^{-\lambda x}, \quad 0 \le x, \qquad f_Y(y) = \lambda^2 y e^{-\lambda y}, \quad 0 \le y.$$

They are not independent, since $f_X f_Y \ne f_{XY}$.

! **Watch out.** **Factorisation test:** X and Y are independent iff $f(x,y) = g(x)h(y)$ **for all** (x,y) — including the support. If the support is not a rectangle (a triangle, a disk, a diamond), the variables are dependent no matter how the formula looks: knowing x changes the **range** of y.

Example 67. Suppose that a point is selected uniformly from a unit square centred at 0. Then

$$f_{XY}(x,y) = \begin{cases} 1, -\frac{1}{2} \le x \le \frac{1}{2}, -\frac{1}{2} \le y \le \frac{1}{2} \\ 0, \text{otherwise} \end{cases} = I_{-\frac{1}{2}\le x\le\frac{1}{2}} I_{-\frac{1}{2}\le y\le\frac{1}{2}};$$

then it factors and X and Y are independent (check by finding f_X and f_Y, proving that each is uniform and that $f_{XY} = f_X f_Y$).

However, if X and Y are uniformly distributed over the diamond (the square rotated by 45°), then let's draw it and formalize it:

$$f_{XY}(x,y) = \begin{cases} 1, |y \pm x| \le 1/\sqrt{2} \\ 0, \text{otherwise} \end{cases} = I_{|y\pm x|\,\le 1/\sqrt{2}},$$

which does not factor; hence they are not independent. Check by finding f_X and f_Y:

$$f_X(x) = \begin{cases} \int_{-x-1/\sqrt{2}}^{x+1/\sqrt{2}} \mathrm{d}y, -1/\sqrt{2} \le x \le 0 \\ \int_{x-1/\sqrt{2}}^{-x+1/\sqrt{2}} \mathrm{d}y, 0 \le x \le 1/\sqrt{2} \end{cases} = \begin{cases} 2x + \sqrt{2}, & -1/\sqrt{2} \le x \le 0 \\ -2x + \sqrt{2}, 0 \le x \le 1/\sqrt{2} \end{cases}$$

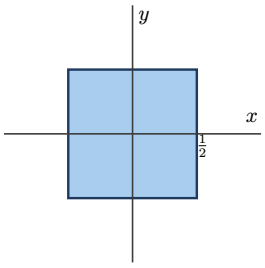
$$= -2\,|x| + \sqrt{2}, \quad |x| \le 1/\sqrt{2};$$

similarly

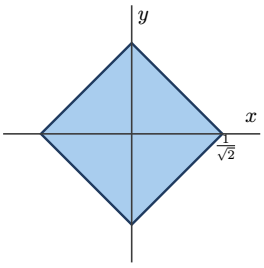
$$f_Y(y) = -2\,|y| + \sqrt{2}, \quad |y| \le 1/\sqrt{2}.$$

We see that $f_{XY} \neq f_X f_Y$.

Correction to the original. The original says the diamond comes from "rotating a square 90°"; a 90° rotation maps a square to itself. The diamond is the square rotated by 45° (its area stays 1, so the density stays 1).



square — independent



diamond — dependent

3.5 Conditional Distributions

3.5.1 The Discrete Case

Definition 68. The discrete conditional pmf is defined as

$$P_{Y \mid X}(Y = y_j \mid X = x_i) = \frac{P_{XY}(X = x_i, Y = y_j)}{P_X(X = x_i)},$$

where $P(X = x_i) \neq 0$. And hence,

$$P_{XY}(x, y) = P_{Y \mid X}(y \mid x) P_X(x), \qquad P_Y(y) = \sum_x P_{XY}(x, y) = \sum_x P_{Y \mid X}(y \mid x) P_X(x),$$

which is nothing but the law of total probability.

Notice that conditional probability was proven to be a true probability measure; hence the conditional pmf is a genuine pmf.

Example 69 (cont. Ex. 57).

$X \setminus Y$	0	1	2	3
0	$\frac{1}{8}$ $\{ttt\}$	$\frac{2}{8}$ $\{tht, tth\}$	$\frac{1}{8}$ $\{thh\}$	0 $\{\}$
1	0 $\{\}$	$\frac{1}{8}$ $\{htt\}$	$\frac{2}{8}$ $\{hht, hth\}$	$\frac{1}{8}$ $\{hhh\}$

To find $P_{X \mid Y=1}$:

$$P_{X \mid Y=1}(0 \mid 1) = \frac{P_{XY}(x = 0, y = 1)}{P_Y(y = 1)} = \frac{P_{XY}(0, 1)}{P_{XY}(0, 1) + P_{XY}(1, 1)} = \frac{2/8}{2/8 + 1/8} = \frac{2}{3},$$

$$P_{X \mid Y=1}(1 \mid 1) = \frac{1/8}{2/8 + 1/8} = \frac{1}{3}.$$

▸ **Lecture note.** Conditioning on $Y = 1$ means **keep only the $Y = 1$ column and rescale it to sum to 1**. Here the column is $(2/8, 1/8)$, rescaled to $(2/3, 1/3)$. Makes sense: given exactly one head in three tosses, it is in the first toss with probability $1/3$.

Example 70 (Imperfect Counter).

- A particle counter detects each particle with probability p.
- The number of incoming particles per unit time is Poisson(λ).
- N is the true number of particles, X is the number counted; therefore

$$X \mid N = n \sim \text{Binomial}(n, p), \qquad P(X = k \mid N = n) = \binom{n}{k} p^k (1 - p)^{n-k};$$

$$N \sim \text{Poisson}(\lambda), \qquad P(N = n) = \frac{\lambda^n e^{-\lambda}}{n!}.$$

$$P(X=k) = \sum_{n=k}^{\infty} P(X=k \mid N=n)P(N=n) = \sum_{n=k}^{\infty} \frac{\lambda^n e^{-\lambda}}{n!}\binom{n}{k}p^k(1-p)^{n-k}$$
$$= \frac{(\lambda p)^k}{k!}e^{-\lambda}\sum_{n=k}^{\infty}\lambda^{n-k}\frac{(1-p)^{n-k}}{(n-k)!} = \frac{(\lambda p)^k}{k!}e^{-\lambda}\sum_{j=0}^{\infty}\frac{\lambda^j(1-p)^j}{j!}$$
$$= \frac{(\lambda p)^k}{k!}e^{-\lambda}e^{\lambda(1-p)} = \frac{(\lambda p)^k}{k!}e^{-\lambda p},$$

so $X \sim \text{Poisson}(\lambda p)$.

⋆ **Intuition. Thinning a Poisson process gives a Poisson process.** Particles arrive at rate λ; each survives detection independently with probability p; the survivors arrive at rate λp. The same holds for customers who buy, emails that are spam, etc.

3.5.2 The Continuous Case

Definition 71. The conditional density $f_{Y \mid X}(y \mid x)$ is defined as

$$f_{Y \mid X}(y \mid x) = \frac{f_{XY}(x,y)}{f_X(x)}.$$

For intuition:

$$f_{Y \mid X}(y \mid x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{P(x \le X \le x+\mathrm{d}x, y \le Y \le y+\mathrm{d}y)/(\mathrm{d}x\,\mathrm{d}y)}{P(x \le X \le x+\mathrm{d}x)/\,\mathrm{d}x},$$

$$f_{Y \mid X}(y \mid x)\,\mathrm{d}y = \frac{P(x \le X \le x+\mathrm{d}x, y \le Y \le y+\mathrm{d}y)}{P(x \le X \le x+\mathrm{d}x)} = P(y \le Y \le y+\mathrm{d}y \mid x \le X \le x+\mathrm{d}x).$$

Alternatively:

$$f_{Y \mid X}(y \mid x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{f_{XY}(x,y)}{\int_{-\infty}^{\infty} f_{XY}(x,y)\,\mathrm{d}y}.$$

So the denominator is just a normalizing factor, and indeed $f_{Y \mid X}(y \mid x)$ is a pdf and integrates to one. **Let's try playing with the Mathematica notebook.**

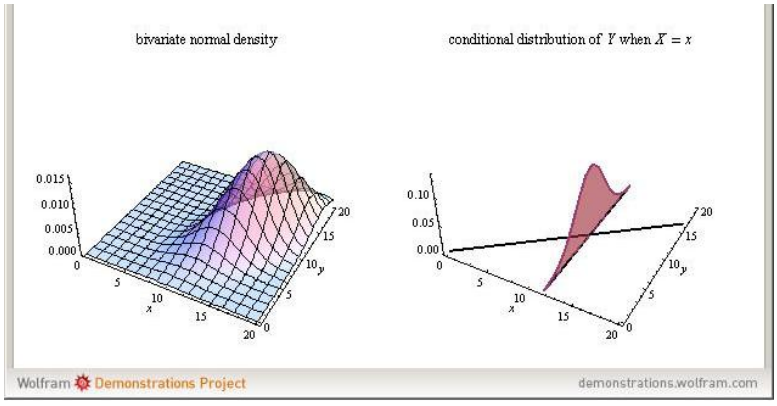


Figure 11: A bivariate normal density (left) and the conditional density of Y when $X = x$ (right) — a slice of the surface, rescaled to area 1 (Wolfram Demonstrations Project).

Analogously to the discrete case:

$$f_{XY}(x,y) = f_{Y\mid X}(y\mid x)f_X(x), \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,\mathrm{d}x = \int_{-\infty}^{\infty} f_{Y\mid X}(y\mid x)f_X(x)\,\mathrm{d}x,$$

which can be interpreted as a law of total probability for the continuous case.

Example 72 (cont. Ex. 62).

$$f(x,y) = \begin{cases}\lambda^2 e^{-\lambda y}, 0\le x\le y, \lambda>0\\ 0, \quad \text{otherwise}\end{cases}, \quad f_X(x) = \lambda e^{-\lambda x},\ 0\le x, \quad f_Y(y) = \lambda^2 y e^{-\lambda y},\ 0\le y.$$

$$f_{Y\mid X}(y\mid x) = \frac{\lambda^2 e^{-\lambda y}}{\lambda e^{-\lambda x}} = \lambda e^{-\lambda(y-x)}, \quad 0\le x\le y, \quad \lambda>0,$$

so $Y\mid X \sim x + \text{Exponential}(\lambda)$ (an exponential shifted to start at x);

$$f_{X\mid Y}(x\mid y) = \frac{\lambda^2 e^{-\lambda y}}{\lambda^2 y e^{-\lambda y}} = \frac{1}{y}, \quad 0\le x\le y,$$

so $X\mid Y \sim \text{Uniform}(0,y)$.

Notice that we can generate (X,Y) by generating x followed by $y\mid x$, or by generating y followed by $x\mid y$.

Correction to the original. The original writes $Y\mid X\sim\text{Exponential}(\lambda)$ and $X\mid Y\sim\text{Uniform}(0,1/y)$. The conditional density $\lambda e^{-\lambda(y-x)}$ lives on $y\ge x$: it is an exponential **shifted by x** ($Y-X\mid X\sim\text{Exp}(\lambda)$). And the density $1/y$ on $[0,y]$ is $\text{Uniform}(0,y)$; $1/y$ is its **height**, not its endpoint.

⋆ **Intuition.** Example 62 is the Poisson process again: X = time of the first event, Y = time of the second. Given the first at x, the wait for the second is a fresh exponential (memorylessness). Given the second at y, the first is equally likely to have been anywhere in $[0,y]$.

3.5.3 Bayesian Inference

- Coin tossing $\sim\text{Bernoulli}(\theta)$, where θ is unknown.
- Number of heads $x\sim\text{Binomial}(n,\theta)$.
- Given x and n, what is θ?
- The conditional probability of x is $f_{X\mid\Theta}(x\mid\theta)$.
- The prior probability of Θ is $f_\Theta(\theta)$.
- The posterior probability of Θ is $f_{\Theta\mid X}(\theta\mid x)$.

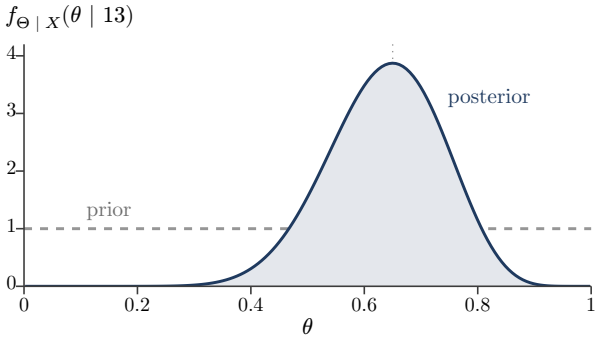
Therefore, we proceed very analogously to Bayes' rule in Ch. 1:

$$f_{\Theta\mid X}(\theta\mid x) = \frac{f_{X,\Theta}}{f_X} = \frac{f_{X\mid\Theta}f_\Theta}{\int f_{X,\Theta}\,\mathrm{d}\theta} = \frac{f_{X\mid\Theta}f_\Theta}{\int f_{X\mid\Theta}f_\Theta\,\mathrm{d}\theta} = \underbrace{\frac{\text{Const}(x)}{1/\int f_{X\mid\Theta}f_\Theta\,\mathrm{d}\theta}}\, f_{X\mid\Theta}f_\Theta$$

$$= \text{Const}(x)\cdot\binom{n}{x}\theta^x(1-\theta)^{n-x}\cdot 1 = \text{Const}'(x)\,\theta^x(1-\theta)^{n-x},$$

which is the density of $\text{Beta}(x+1,n-x+1)$, and $\text{Const}'(x)$ has to be equal to $1/B(x+1,n-x+1)$. Anyway, this is a normalizing factor and has no significance on the shape of $f_{\Theta\mid X}(\theta\mid x)$.

Given $n=20$, $x=13$, $f_{\Theta\mid X}(\theta\mid x)$ is drawn as follows, and it is unlikely that $\theta<0.4$.



Interpretation:

- Prior to observation: $\Theta \sim \text{Uniform}(0, 1)$ (the no-information case).
- Posterior to observations: $\Theta \sim \text{Beta}(x + 1, n - x + 1)$.
- The posterior represents my **belief** after observations, based on my **subjective belief** prior to observations.
- We can estimate θ by the value that maximizes the posterior, as was done in Ch. 1.

▸ **Lecture note.** The posterior peaks at $\hat{\theta} = x/n = 13/20 = 0.65$ — the observed proportion, exactly the "most probable value" logic of the capture–recapture example. As n grows the posterior becomes narrower: with $n = 200$, $x = 130$, almost all its mass lies in $[0.58, 0.72]$. Data overwhelm the prior.

3.6 Functions of Jointly Distributed r.v.s

Revision: a single function of a single r.v.

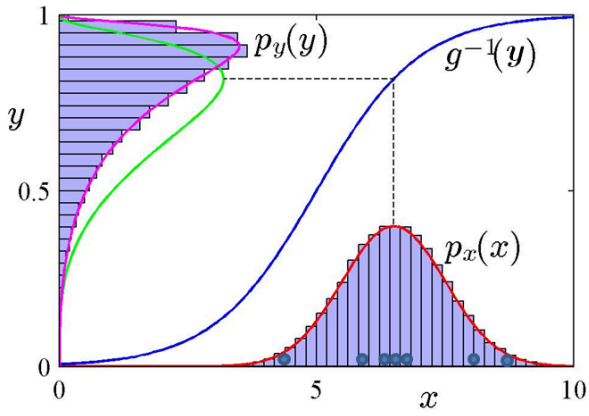
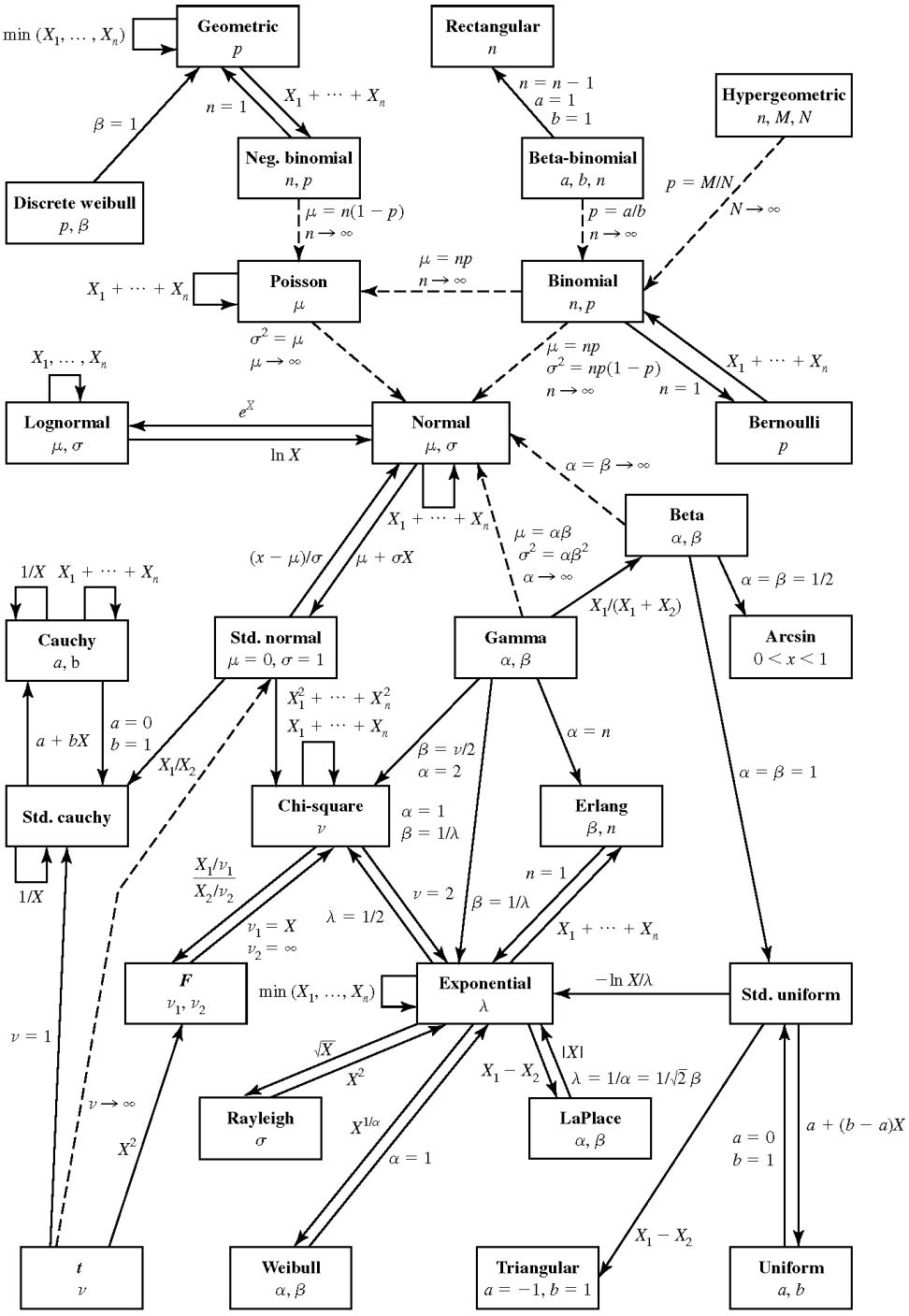


Figure 12: Recall Theorem 48.

- X is discrete: $p_Y(y) = p_X(g^{-1}(y))$.
- X is continuous: $f_Y(y)\,|\mathrm{d}y| = f_X(g^{-1}(y))\,|\mathrm{d}x|$.



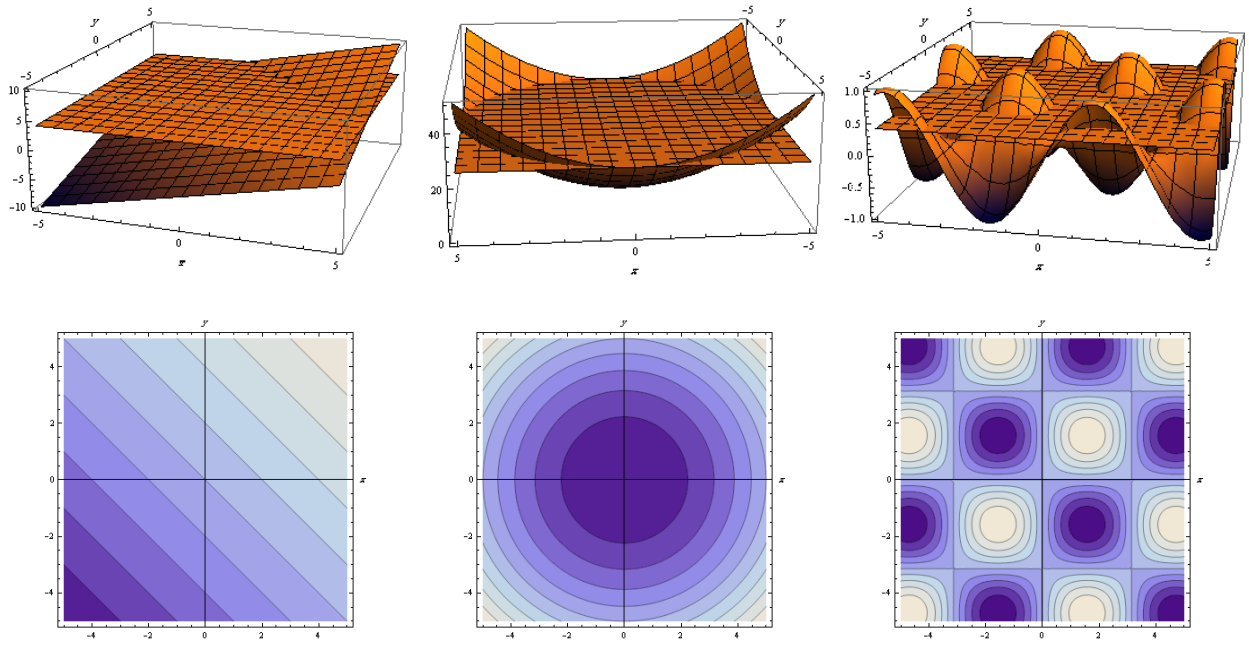
Adapted from Leemis, L. M. (1986), "Relationships among common univariate distributions," *The American Statistician,* Vol. 40, No. 2, pp. 143–146. With permission.

Figure 13: Relationships among common univariate distributions. Adapted from Leemis, L. M. (1986), "Relationships among common univariate distributions", *The American Statistician* 40(2), 143–146.

3.6.1 Single Function of Jointly Distributed r.v.s

- Let's see the examples below.
- In each example we draw the 3D function $Z = f(X, Y)$.
- We cut it at some level, $Z = \text{const}$,
- we get the 3D line $f(X, Y) = \text{const}$, $Z = \text{const}$,
- and project it down to the X–Y plane to get the contour plot $f(X, Y) = \text{const}$, $Z = 0$.

$Z = X + Y$, $Z = 3$ $Z = X^2 + Y^2$, $Z = 25$ $Z = \sin(X)\sin(Y)$, $Z = 0.4$



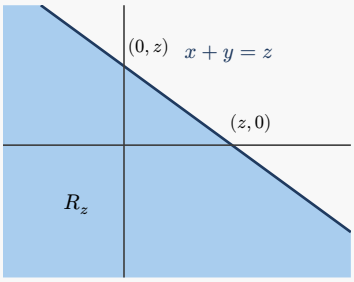
▸ **Lecture note.** Why bother with the surfaces? Because $\{Z \le z\}$ is the region of the plane **under the contour at height z**, and $F_Z(z) = P(Z \le z)$ is the integral of f_{XY} over that region. Every example below is: draw the region $\{g(x,y) \le z\}$, integrate, differentiate in z.

Example 73 (Sums). $Z = X + Y$.

For X, Y discrete:

$$P(Z = z) = P(X + Y = z) = P(\text{points on the line } X + Y = z),$$

$$p_Z(z) = \sum_{x=-\infty}^{\infty} p_{XY}(x, z - x).$$



E.g. integers $1 \le X \le 3$, $1 \le Y \le 3$, $p_{XY}(x,y) = 1/9$. Let's draw it:

$x \setminus y$	1	2	3
1	2	3	4
2	3	4	5
3	4	5	6

The cells show $z = x + y$; each diagonal (one colour) is one line $x + y = z$.

$$p_Z(2) = \sum_{x=1}^{1} p(x, 2-x) = \frac{1}{9}, \quad p_Z(3) = \sum_{x=1}^{2} p(x, 3-x) = \frac{2}{9}, \quad p_Z(4) = \sum_{x=1}^{3} p(x, 4-x) = \frac{3}{9},$$

$$p_Z(5) = \sum_{x=2}^{3} p(x, 5-x) = \frac{2}{9}, \quad p_Z(6) = \sum_{x=3}^{3} p(x, 6-x) = \frac{1}{9}.$$

What about $Z = 2X + Y$ on the same region (HW)?

The continuous case has to be done through the CDF:

$$F_Z(z) = P(Z \le z) = P(R_z) = \iint_{R_z} f_{XY}(x,y)\,\mathrm{d}x\,\mathrm{d}y = \int_{-\infty}^{\infty}\int_{-\infty}^{z-x} f_{XY}(x,y)\,\mathrm{d}y\,\mathrm{d}x.$$

Substitution: $y = v - x$, $\mathrm{d}y = \mathrm{d}v$, $v = y + x$; the limit $y = z - x$ becomes $v = z$:

$$f_Z(z) = \frac{\mathrm{d}}{\mathrm{d}z}\int_{-\infty}^{\infty}\int_{-\infty}^{z} f_{XY}(x, v-x)\,\mathrm{d}v\,\mathrm{d}x = \frac{\mathrm{d}}{\mathrm{d}z}\int_{-\infty}^{z}\int_{-\infty}^{\infty} f_{XY}(x, v-x)\,\mathrm{d}x\,\mathrm{d}v$$
$$= \int_{-\infty}^{\infty} f_{XY}(x, z-x)\,\mathrm{d}x.$$

If the region of f_{XY} is more complicated, the integration limits will change and we have to do everything from scratch (see the next application).

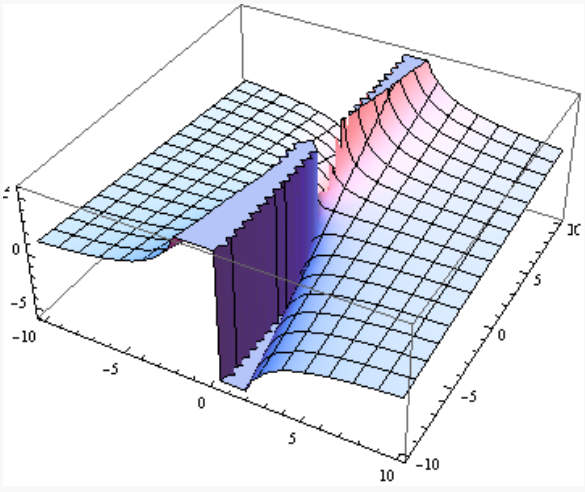
Application:

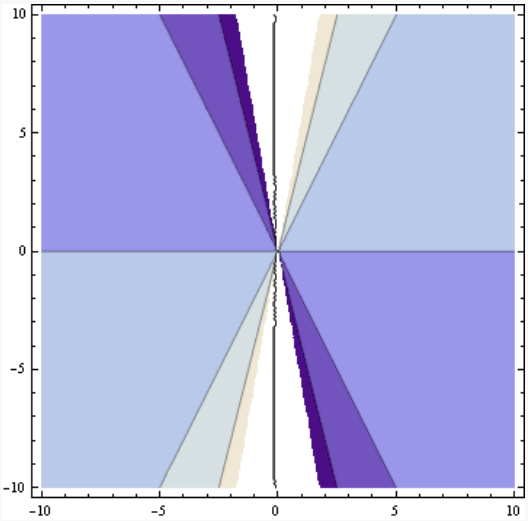
- The lifetime of a system with two independent components (one a spare for the other) is $S = T_1 + T_2$.
- Both T_1 and T_2 are Exponential(λ).
- Draw to find that $f_{T_1T_2}$ is non-zero over the region $0 \le T_1 \le S$ and $0 \le T_2 = S - T_1$.

$$f_S(s) = \int_{-\infty}^{\infty} f_{T_1T_2}(t, s-t)\,\mathrm{d}t = \int_{-\infty}^{\infty} f_{T_1}(t) f_{T_2}(s-t)\,\mathrm{d}t$$
$$= \int_{-\infty}^{0} + \int_{0}^{s} \left(\lambda e^{-\lambda t}\right)\left(\lambda e^{-\lambda(s-t)}\right)\mathrm{d}t + \int_{s}^{\infty} = \int_{0}^{s} \lambda^2 e^{-\lambda s}\,\mathrm{d}t = \lambda^2 s e^{-\lambda s}.$$

▸ **Lecture note.** $\lambda^2 s e^{-\lambda s}$ is Gamma$(2,\lambda)$ — the same as f_Y in Example 62 (where Y was the time of the second event). In general, a sum of n independent Exponential(λ) is Gamma(n,λ). The integral $\int f_X(t) f_Y(s-t)\,\mathrm{d}t$ is called a **convolution**; the outer pieces vanish because $f_{T_1}(t) = 0$ for $t < 0$ and $f_{T_2}(s-t) = 0$ for $t > s$.

Example 74 (Quotients). $Z = Y/X$.





There is no need to draw the 3D function every time; it suffices to draw the X–Y regions. Then

$$Z = \frac{Y}{X}, \qquad Z \le z \equiv \frac{Y}{X} \le z \equiv \begin{cases} Y \ge zX, X < 0 \\ Y \le zX, X > 0 \end{cases}$$

$$F_Z(z) = \int_{-\infty}^{0}\int_{xz}^{\infty} f_{XY}(x,y)\,\mathrm{d}y\,\mathrm{d}x + \int_{0}^{\infty}\int_{-\infty}^{xz} f_{XY}(x,y)\,\mathrm{d}y\,\mathrm{d}x.$$

Substitution: $y = xv$, $v = y/x$, $\mathrm{d}y = x\,\mathrm{d}v$; $y = xz \Rightarrow v = z$; $y = \infty \Rightarrow v = -\infty$ for $x < 0$; $y = -\infty \Rightarrow v = -\infty$ for $x > 0$:

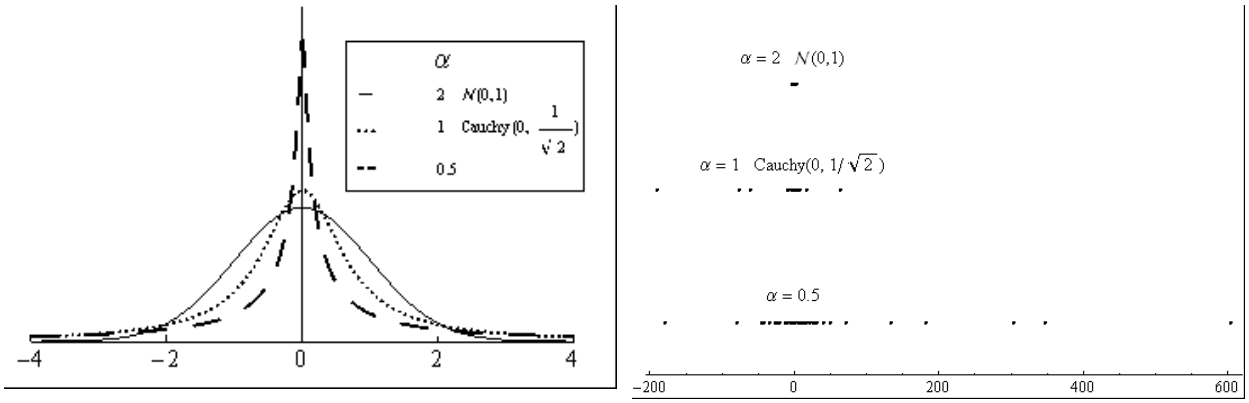
$$\begin{aligned} F_Z(z) &= \int_{-\infty}^{0}\int_{z}^{-\infty} x f_{XY}(x,xv)\,\mathrm{d}v\,\mathrm{d}x + \int_{0}^{\infty}\int_{-\infty}^{z} x f_{XY}(x,xv)\,\mathrm{d}v\,\mathrm{d}x \\ &= \int_{-\infty}^{0}\int_{-\infty}^{z} (-x) f_{XY}(x,xv)\,\mathrm{d}v\,\mathrm{d}x + \int_{0}^{\infty}\int_{-\infty}^{z} x f_{XY}(x,xv)\,\mathrm{d}v\,\mathrm{d}x \\ &= \int_{-\infty}^{z} \mathrm{d}v \left[\int_{-\infty}^{0} (-x) f_{XY}(x,xv)\,\mathrm{d}x + \int_{0}^{\infty} x f_{XY}(x,xv)\,\mathrm{d}x\right], \end{aligned}$$

$$f_Z(z) = \int_{-\infty}^{0} (-x) f_{XY}(x,xz)\,\mathrm{d}x + \int_{0}^{\infty} x f_{XY}(x,xz)\,\mathrm{d}x = \int_{-\infty}^{\infty} |x|\; f_{XY}(x,xz)\,\mathrm{d}x.$$

Application: $Z = Y/X$, and $X, Y \sim N(0,1)$ independent.

$$\begin{aligned} f_Z(z) &= 2\int_{0}^{\infty} x f_{XY}(x,xz)\,\mathrm{d}x = 2\int_{0}^{\infty} x \left(\frac{1}{\sqrt{2\pi}} e^{-x^2/2}\right)\left(\frac{1}{\sqrt{2\pi}} e^{-(xz)^2/2}\right)\mathrm{d}x \\ &= \frac{1}{\pi}\int_{0}^{\infty} x e^{-x^2(z^2+1)/2}\,\mathrm{d}x = \frac{1}{\pi(z^2+1)}\left(-e^{-x^2(z^2+1)/2}\right)\Big|_{0}^{\infty} = \frac{1}{\pi(z^2+1)}. \end{aligned}$$

This is the density of the **Cauchy** r.v. It can also be obtained by a univariate transformation!



Left: $N(0,1)$ ($\alpha = 2$), Cauchy ($\alpha = 1$) and a heavier-tailed stable law ($\alpha = 0.5$). Right: samples from each — note the horizontal scale.

▸ **Lecture note.** The Cauchy looks like a normal with fatter tails, but it is a different animal: its tails are so heavy that it has **no mean**. Averaging Cauchy samples does not settle down — the average of n Cauchy variables is again Cauchy, as spread out as one. The right panel shows why: the occasional enormous value (hundreds of units out) dominates any average. Keep it in mind when you meet the Law of Large Numbers in Chapter 5 — that law needs a finite mean.

3.6.2 p Functions of p r.v.s (Space Transformation)

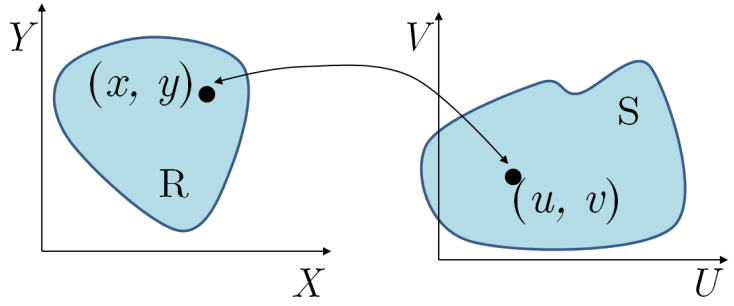


Figure 14: A region R in the (x, y) plane is mapped to a region S in the (u, v) plane.

- Transforming the whole space, not just a single variable $U = f(X, Y)$:

$$(X, Y) \to (U, V), \quad U = g_1(X, Y), \quad V = g_2(X, Y), \quad X = h_1(U, V), \quad Y = h_2(U, V).$$

- So the joint density f_{UV} is of interest.

Notice: if we want only f_U, $U = f(X, Y)$, then:

1. either do as before;
2. or make up a variable $V = g_2(X, Y)$, find f_{UV}, then $f_U = \int f_{UV}\, \mathrm{d}v$.

Theorem 75. Suppose that X and Y are two jointly distributed r.v.s with pdf f_{XY}, mapped **onto** U and V by

$$u = g_1(x, y), \quad v = g_2(x, y), \quad x = h_1(u, v), \quad y = h_2(u, v),$$

where h_1, h_2, g_1, g_2 are continuous and have continuous first derivatives. Then

$$f_{UV}(u, v) = f_{XY}(x, y)\ |J|^{-1}\ (x, y), \quad J(x, y) = \begin{vmatrix} \frac{\partial g_1}{\partial x} & \frac{\partial g_1}{\partial y} \\ \frac{\partial g_2}{\partial x} & \frac{\partial g_2}{\partial y} \end{vmatrix}, \quad (x, y) = (h_1(u, v), h_2(u, v)).$$

Correction to the original. The original prints $f_{UV} = f_{XY} J^{-1}$; the density uses the **absolute value** $|J|^{-1}$ (an area is never negative — compare $|1/(\mathrm{d}y/\mathrm{d}x)|$ in Theorem 48). The theorem statement also says "X and Y … with pdf f_{UV}"; it is f_{XY}.

"Not a proof":

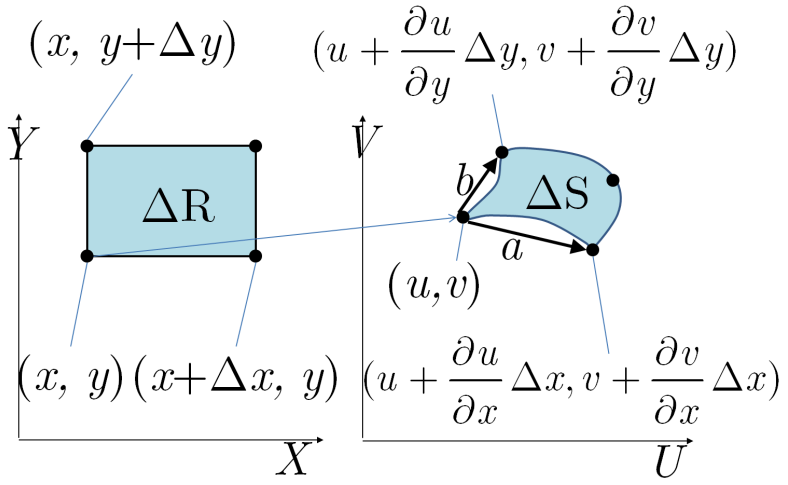


Figure 15: A small rectangle ΔR maps to an (approximate) parallelogram ΔS spanned by a and b.

$$a = \left(u + \frac{\partial u}{\partial x}\Delta x, v + \frac{\partial v}{\partial x}\Delta x\right) - (u, v) = \left(\frac{\partial u}{\partial x}, \frac{\partial v}{\partial x}\right)\Delta x,$$
$$b = \left(u + \frac{\partial u}{\partial y}\Delta y, v + \frac{\partial v}{\partial y}\Delta y\right) - (u, v) = \left(\frac{\partial u}{\partial y}, \frac{\partial v}{\partial y}\right)\Delta y.$$

From elementary vector calculus, it is known that

$$\sin\theta = \frac{|b \times a|}{\|a\|\|b\|}, \qquad \cos\theta = \frac{b \cdot a}{\|a\|\|b\|}.$$

Therefore, the area of the parallelogram is

$$A = \|a\|\|b\| \sin\theta = |a \times b| = \left|\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}\right| \Delta x \Delta y = |J(x, y)|\ \Delta x \Delta y.$$

Hence,

$$P(S) = P(R),$$
$$f_{UV}(u, v)\ |J(x, y)|\ \Delta x \Delta y = f_{XY}(x, y)\Delta x \Delta y,$$
$$f_{UV}(u, v) = f_{XY}(x, y)\ |J|^{-1}\ (x, y),$$

where $(x, y) = (h_1(u, v), h_2(u, v))$. The (rigorous) proof is out of scope and omitted.

Example 76 (Polar System). If X and Y have density f_{XY}, what is the pdf of a point selected at radius r and angle θ?

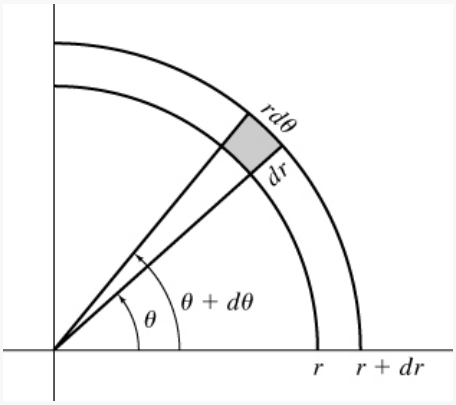


Figure 16: The polar area element has sides $\mathrm{d}r$ and $r\,\mathrm{d}\theta$.

Let's draw the regions, the transformed regions, and the transformation functions h, g:

$$(X, Y) \to (R, \Theta), \quad 0 \le R, \quad 0 \le \Theta \le 2\pi, \quad R = \sqrt{X^2 + Y^2},$$

$$\Theta = \begin{cases} \tan^{-1}(Y/X), & 0 < X \\ \tan^{-1}(Y/X) + \pi, & X < 0 \\ \frac{\pi}{2}\ \mathrm{sign}(Y), & X = 0, Y \neq 0 \\ 0, & X = 0, Y = 0 \end{cases} \qquad X = R\cos\Theta, \quad Y = R\sin\Theta,$$

$$J = \left|\begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix}\right| = \left|\begin{vmatrix} \frac{x}{\sqrt{x^2+y^2}} & \frac{y}{\sqrt{x^2+y^2}} \\ \frac{-y}{x^2+y^2} & \frac{x}{x^2+y^2} \end{vmatrix}\right| = \frac{1}{\sqrt{x^2 + y^2}} = \frac{1}{r},$$

$$f_{R\Theta}(r, \theta) = r f_{XY}(r\cos\theta, r\sin\theta).$$

Example 77 (Rayleigh Density). Suppose that X and Y are independent standard normal; then

$$f_{XY}(x,y) = f_X(x)f_Y(y) = \frac{1}{2\pi}\exp\left[-\frac{1}{2}(x^2+y^2)\right],$$

$$f_{R\Theta}(r,\theta) = r f_{XY}(r\cos\theta, r\sin\theta) = \frac{1}{2\pi} r \exp\left[-\frac{1}{2}r^2\right],$$

$$f_R(r) = \int_0^{2\pi} \frac{1}{2\pi} r \exp\left[-\frac{1}{2}r^2\right] \mathrm{d}\theta = r\exp\left[-\frac{1}{2}r^2\right], \quad 0 \le r,$$

$$f_\Theta(\theta) = \int_0^{\infty} \frac{1}{2\pi} r \exp\left[-\frac{1}{2}r^2\right] \mathrm{d}r = \frac{1}{2\pi}, \quad 0 \le \theta \le 2\pi.$$

Notice that:

- The independence of R and Θ is obvious from the separability of $f_{R\Theta}(r,\theta)$, or from $f_{R\Theta}(r,\theta) = f_R(r)f_\Theta(\theta)$.
- It makes a lot of sense (from the symmetry of the problem).
- Symmetry is consistent with $\Theta \sim \text{Uniform}(0, 2\pi)$.
- R is called a **Rayleigh** r.v.

▸ **Lecture note.** Run Example 77 backwards and you get the **Box–Muller** method for simulating normals: take $\Theta = 2\pi U_1$ and $R = \sqrt{-2\log U_2}$ (inverse cdf of the Rayleigh, $F_R(r) = 1 - e^{-r^2/2}$, by Proposition 53); then $X = R\cos\Theta$ and $Y = R\sin\Theta$ are two independent $N(0,1)$. It also gives the normalisation $\int e^{-x^2/2}\,\mathrm{d}x = \sqrt{2\pi}$ proved in Chapter 2.

3.7 Extrema and Order Statistics

Suppose that $X_1, ..., X_n$ are **continuous** i.i.d. from cdf F. What is the pdf of

$$U = \max(X_i), \quad V = \min(X_i), \quad i = 1, ..., n?$$

What is the meaning of that for observations?

$$X \overset{\text{Sample}_1}{\to} x_1, x_2, ..., x_n, \qquad X \overset{\text{Sample}_2}{\to} x_1, x_2, ..., x_n, \qquad \vdots$$

Then the r.v.s $X_1, ..., X_n$ as well as, of course, X, are i.i.d.

▸ **Lecture note.** "i.i.d." = **independent and identically distributed**: each X_i is a fresh, independent copy of the same X. Picture a table whose rows are repeated samples and whose columns are $X_1, ..., X_n$. The maximum U is a new column computed row by row — and it has its own distribution, different from F.

PDF derivation:

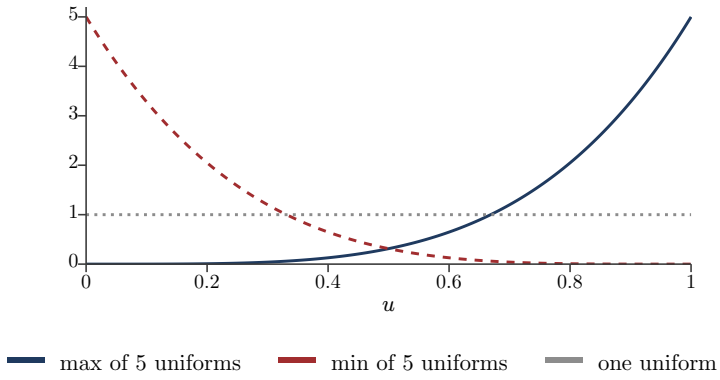
$$F_U(u) = P(U \le u) = P(X_1 \le u, ..., X_n \le u) = P(X_1 \le u)P(X_2 \le u)\cdots P(X_n \le u) = [F(u)]^n,$$
$$f_U(u) = nf(u)[F(u)]^{n-1};$$

$$1 - F_V(v) = P(V > v) = P(X_1 > v, ..., X_n > v) = [P(X > v)]^n,$$
$$F_V(v) = 1 - [1 - F(v)]^n, \qquad f_V(v) = nf(v)[1 - F(v)]^{n-1}.$$

More intuition by studying $F_U(u)$ for Uniform$(0,1)$:

$$f_U(u) = nu^{n-1}, \qquad f_V(v) = n(1-v)^{n-1}, \quad 0 \le u, v \le 1.$$

Correction to the original. The original writes $f_U(u) = n(1-u)^{n-1}$ for the uniform case; that is the density of the **minimum** V. The maximum has $f_U(u) = nu^{n-1}$ — it piles up near 1, as a maximum should.



Example 78.

- n independent components in series.
- Each has lifetime $T \sim \text{Exponential}(\lambda)$.
- The system fails when any component fails.
- V is the system lifetime; what is f_V?

$$V = \min(T_i), \quad f_V(v) = nf_T(v)[1-F_T(v)]^{n-1} = n\big(\lambda e^{-\lambda v}\big)\big(1-\big(1-e^{-\lambda v}\big)\big)^{n-1} = n\lambda e^{-n\lambda v}.$$

Therefore $V \sim \text{Exponential}(n\lambda)$: it decays faster, concentrated at low values (makes a lot of sense). If they are connected in parallel, then

$$U = \max(T_i), \quad f_U(u) = nf_T(u)[F_T(u)]^{n-1} = n\lambda e^{-\lambda u}\big(1-e^{-\lambda u}\big)^{n-1}.$$

Theorem 79 (Order Statistics). The density of the kth-order statistic $X_{(k)}$ is

$$f_{(k)}(x) = \frac{n!}{(k-1)!(n-k)!} f_X(x) F^{k-1}(x)[1-F(x)]^{n-k}.$$

Proof. $P(X \le x) = F(x)$, so the number of observations $\le x$ is $\text{Binomial}(n, F(x))$:

$$P(j \text{ observations} \le x) = \binom{n}{j} F^j(x)(1-F(x))^{n-j}.$$

$$F_{(k)}(x) = P(\text{at least } k \text{ observations} \le x) = \sum_{j=k}^{n} \binom{n}{j} F^j(x)[1-F(x)]^{n-j},$$

$$f_{(k)}(x) = \sum_{j=k}^{n} \binom{n}{j} jf(x)F^{j-1}(x)[1-F(x)]^{n-j} - \sum_{j=k}^{n} \binom{n}{j} F^j(x)(n-j)f(x)[1-F(x)]^{n-j-1}$$

$$= \binom{n}{k} kf(x)F^{k-1}(x)[1-F(x)]^{n-k} + \sum_{j=k+1}^{n} \binom{n}{j} jf(x)F^{j-1}(x)[1-F(x)]^{n-j}$$

$$- \sum_{j=k}^{n-1} \binom{n}{j} F^j(x)(n-j)f(x)[1-F(x)]^{n-j-1}.$$

Re-index the middle sum with $j \to j+1$:

$$f_{(k)}(x) = \binom{n}{k} k f(x) F^{k-1}(x)[1-F(x)]^{n-k} + \sum_{j=k}^{n-1} \Bigg[\underbrace{\binom{n}{j+1}(j+1)}_{=n\binom{n-1}{j}} - \underbrace{\binom{n}{j}(n-j)}_{=n\binom{n-1}{j}} \Bigg] f(x) F^{j}(x)[1-F(x)]^{n-j-1}.$$

The bracket vanishes term by term (the sum telescopes), leaving

$$f_{(k)}(x) = \frac{n!}{(k-1)!(n-k)!} f_X(x) F^{k-1}(x)[1-F(x)]^{n-k},$$

which completes the proof. □

⋆ **Intuition.** A one-line reading of the formula: for $X_{(k)}$ to sit at x, one observation must land at x (density $f(x)$), $k-1$ below it (F^{k-1}) and $n-k$ above it ($[1-F]^{n-k}$); the multinomial coefficient $n!/((k-1)!\,1!\,(n-k)!)$ counts the ways to choose which observations play which role. $k=n$ and $k=1$ recover f_U and f_V.

Example 80. Find $f_{(k)}$ when $X \sim \text{Uniform}(0,1)$. Here $f(x)=1$, $F(x)=x$, so

$$f_{(k)}(x) = \frac{n!}{(k-1)!(n-k)!} f(x) F^{k-1}(x)[1-F(x)]^{n-k} = \frac{n!}{(k-1)!(n-k)!} x^{k-1}(1-x)^{n-k}, \quad 0 \le x \le 1.$$

Notice: $X_{(k)} \sim \text{Beta}(k, n-k+1)$.

Chapter summary

- **Joint cdf** $F(x,y) = P(X \le x, Y \le y)$; rectangles by inclusion–exclusion. **Joint pmf/pdf** $p(x,y)$, $f(x,y) = \partial^2 F/\partial x \partial y$; $P((X,Y) \in A) = \iint_A f$.
- **Marginals:** sum or integrate out the other variable. Marginals do **not** determine the joint.
- **Multinomial:** $n!/(n_1!\cdots n_r!)\prod p_i^{n_i}$; each $N_i \sim \mathrm{Bin}(n, p_i)$.
- **Independence:** $F = F_X F_Y \iff f = f_X f_Y$ — the factorisation must include the support. Functions of independent r.v.s are independent.
- **Conditional:** $f_{Y \mid X} = f_{XY}/f_X$ — a slice, renormalised. Total probability: $f_Y = \int f_{Y \mid X} f_X \,\mathrm{d}x$. Bayes: posterior $\propto$ likelihood $\times$ prior.
- **Sums:** $f_{X+Y}(z) = \int f(x, z-x)\,\mathrm{d}x$ (convolution if independent). **Quotients:** $f_{Y/X}(z) = \int |x|\ f(x, xz)\,\mathrm{d}x$.
- **Change of variables:** $f_{UV} = f_{XY}/\ |J|$; polar coordinates give $|J| = 1/r$.
- **Extremes:** $F_{\max} = F^n$, $F_{\min} = 1-(1-F)^n$; kth order statistic $n!/((k-1)!(n-k)!) f F^{k-1}(1-F)^{n-k}$.

Practice problems — Chapter 3

3.1 (X,Y) has joint pmf $p(x,y) = c(x+y)$ for $x, y \in \{0,1,2\}$. Find c, the marginals, and $P(X = 1 \mid Y = 2)$. Are X and Y independent?

3.2 $f(x,y) = 2$ on the triangle $0 \le y \le x \le 1$. Find f_X, f_Y, $f_{Y \mid X}$ and $P(X + Y \le 1)$.

3.3 X and Y are independent Exponential(1). Find the density of $Z = X + Y$ and of $W = X/(X+Y)$.

3.4 A fair die is rolled 12 times. Find the probability that each face appears exactly twice.

3.5 $X_1, \ldots, X_5$ are i.i.d. $\mathrm{U}(0,1)$. Find $P(\max X_i \le 0.8)$ and the density of the median $X_{(3)}$.

3.6 X, Y i.i.d. $N(0,1)$. Use polar coordinates to find $P(X^2 + Y^2 \le 4)$.

CHAPTER 4

Expected Values

▸ **Lecture note.** A distribution is a whole function; often we want a few numbers that summarise it. This chapter builds them: the **mean** (where is it?), the **variance** (how spread out?), the **covariance and correlation** (how do two r.v.s move together?), **conditional expectation** (the best prediction of one from another), and the **moment-generating function**, which packs all the moments into one function and turns sums of independent r.v.s into products.

4.1 The Expected Value of a r.v.

Definition 81. If X is a discrete r.v. with pmf $p(x)$, the **expected value** (mean) is defined as

$$E(X) = \sum_i x_i p(x_i),$$

provided that $\sum_i |x_i|\ p(x_i) < \infty$; otherwise, it is undefined.

- $E(X)$ or μ_X.
- It is a weighted sum.
- It is also the centre of mass.
- It is also the point of balance.
- All the above in the mathematical sense: $\sum_i x_i p(x_i)$.

⋆ **Intuition.** Put a weight $p(x_i)$ at each position x_i on a weightless ruler. $E(X)$ is where the ruler balances on your finger. And, looking ahead to Chapter 5: if you repeat the experiment many times and average the results, the average settles at $E(X)$. "Expected" does not mean "most likely" — a die's expected value is 3.5, which never occurs.

Example 82 (Geometric(p)).

$$p_X(k) = p(1-p)^{k-1}, \quad 1 \le k, \quad q = 1-p,$$

$$E(X) = \sum_{k=1}^{\infty} kpq^{k-1} = p\sum_{k=1}^{\infty} kq^{k-1} = p\sum_{k=1}^{\infty} \frac{\mathrm{d}}{\mathrm{d}q} q^k = p\frac{\mathrm{d}}{\mathrm{d}q}\sum_{k=1}^{\infty} q^k$$

$$= p\frac{\mathrm{d}}{\mathrm{d}q}\frac{q}{1-q} = \frac{p}{(1-q)^2} = \frac{1}{p}.$$

▸ **Lecture note.** $E(X) = 1/p$: if a win has probability $1/9$ (Example 38), you wait 9 draws on average. Interchanging $\mathrm{d}/\mathrm{d}q$ and $\sum$ is allowed because a power series can be differentiated term by term inside its radius of convergence ($|q| < 1$).

Example 83 (Poisson(λ)).

$$E(X) = \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} e^{-\lambda} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} e^{\lambda} = \lambda.$$

Compare this to the peak, which occurs at $k = \lfloor \lambda \rfloor$ (coincidence). (See the Poisson pmf plots in §2.1.5.)

Example 84 (Decisions in Life).

- Deciding A results in a loss (−ve) or a gain (+ve),
- with probabilities p and $(1-p)$:

$$L_A = \begin{cases} L_1, p \\ L_2, (1-p) \end{cases} \qquad E(L_A) = L_1 p + L_2 (1-p).$$

- Most times $L_2 = 0$.
- Travel (A) or not (A')?
- Calculate $E(L_A)$ and $E(L_{A'})$.
- The loss and gain are subjective.
- The self-consciousness of p is subjective too.
- The decision ultimately differs across people.
- "Acquire the most beneficial and prevent the most harmful."

Definition 85. If X is a continuous r.v., then

$$E(X) = \int_{-\infty}^{\infty} x f_X(x)\, \mathrm{d}x,$$

if $\int_{-\infty}^{\infty} |x|\ f_X(x)\, \mathrm{d}x < \infty$; otherwise, it is undefined.

Example 86 (Gamma(α, λ)).

$$E(X) = \int_0^{\infty} x \frac{\lambda^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}\, \mathrm{d}x = \frac{\lambda^{\alpha}}{\Gamma(\alpha)} \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} \int_0^{\infty} \underbrace{\frac{\lambda^{\alpha+1}}{\Gamma(\alpha+1)} x^{\alpha} e^{-\lambda x}}_{\text{pdf of Gamma}(\alpha+1,\lambda)}\, \mathrm{d}x$$

$$= \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)} \frac{1}{\lambda} = \frac{\alpha}{\lambda}.$$

Notice that Exponential(λ) is Gamma($1, \lambda$); hence its mean is $1/\lambda$.

▸ **Lecture note.** The trick in Example 86 is worth naming: **multiply and divide to manufacture a pdf, then use that a pdf integrates to 1.** It replaces a hard integral by a ratio of normalising constants, and it works for almost every named distribution.

Example 87 (Normal(μ, σ^2)).

$$E(X) = \int_{-\infty}^{\infty} x \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2} \right] \mathrm{d}x.$$

Substitute $z = x - \mu$:

$$E(X) = \int_{-\infty}^{\infty} \underbrace{z \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{z^2}{2\sigma^2}\right]}_{\text{odd (antisymmetric)}} \mathrm{d}z + \mu \int_{-\infty}^{\infty} \underbrace{\frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{z^2}{2\sigma^2}\right]}_{\text{pdf of Normal}} \mathrm{d}z = 0 + \mu = \mu.$$

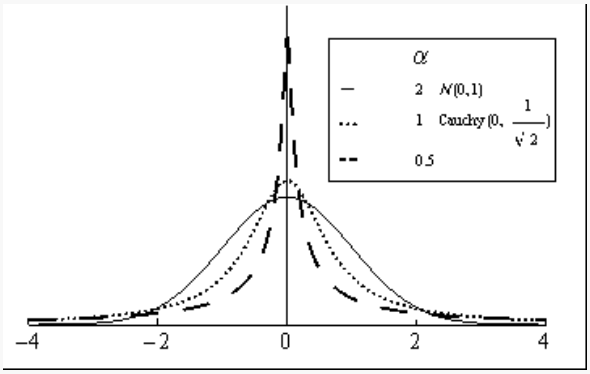
Example 88 (Cauchy).

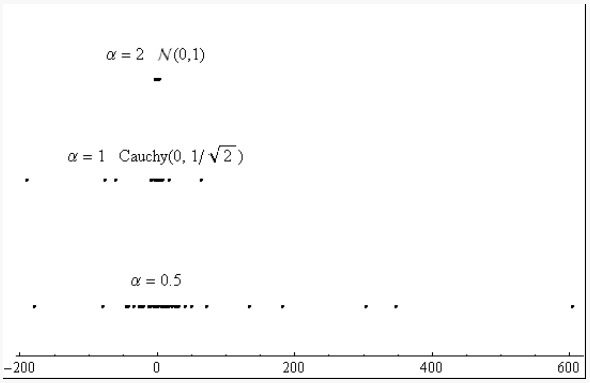
$$E(X) = \int_{-\infty}^{\infty} x \frac{1}{\pi} \frac{1}{1+x^2} \mathrm{d}x = \infty - \infty \quad \text{undefined!!}$$

Why $E(X) \neq 0$? Because

$$\int_{-\infty}^{\infty} |x| \frac{1}{\pi} \left(\frac{1}{1+x^2}\right) \mathrm{d}x = \infty.$$

We will see later that this has a serious impact on the sample mean $\frac{1}{n}\sum_{i=1}^{n} x_i$. Please recall the following figures:





! Watch out. "The Cauchy is symmetric about 0, so its mean is 0" is **wrong**. Symmetry only gives $\int_{-a}^{a} = 0$ for each finite a; but $\int_0^{\infty} x/(\pi(1+x^2))\,\mathrm{d}x$ diverges like log, so the two halves are $+\infty$ and $-\infty$, and their "difference" depends on how you take the limits. A mean exists only when $E|X| < \infty$.

Theorem 89 (Markov's Inequality). If X is a r.v. with $P(X \geq 0) = 1$, then

$$P(X \geq t) \leq \frac{E(X)}{t} \quad \forall t > 0.$$

Proof.

$$E(X) = \int x f(x)\,\mathrm{d}x = \underbrace{\int_{x<t} x f(x)\,\mathrm{d}x}_{\geq 0} + \int_{x \geq t} x f(x)\,\mathrm{d}x$$

$$\geq \int_{x \geq t} x f(x)\,\mathrm{d}x \geq t \int_{x \geq t} f(x)\,\mathrm{d}x = tP(X \geq t).$$

Intuition: $P(X \geq kE(X)) \leq 1/k$. □

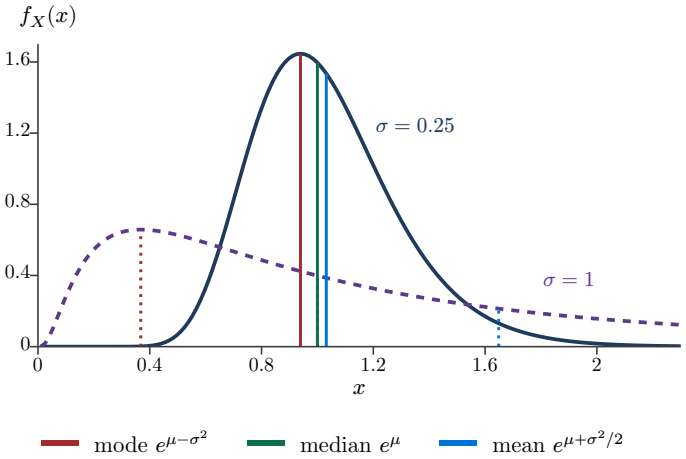
▸ **Lecture note.** Markov in words: **a non-negative quantity cannot often be much bigger than its mean.** If the average salary is 5000, at most 1/4 of people can earn 20 000 or more

— otherwise those people alone would push the average above 5000. Crude, but it needs nothing except $X \geq 0$ and the mean.

4.1.1 Mean, Median, and Mode

- They do not have to coincide.
- Any of them can be to the left/right of another.
- More on that when discussing skewness.
- Example: the lognormal density

$$f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left[-\frac{(\log x - \mu)^2}{2\sigma^2}\right], \quad 0 < x.$$



Correction to the original. The original lognormal density is missing the factor $1/x$ and writes $(-\mu + \log x)^2$; the density of $X = e^Y$, $Y \sim N(\mu, \sigma^2)$, is $f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp[-(\log x - \mu)^2/2\sigma^2]$ (Theorem 48 with $|\mathrm{d}y/\,\mathrm{d}x| = 1/x$). The plot above uses $\mu = 0$.

4.1.2 Expectations of Functions of r.v.

Theorem 90. Suppose that $Y = g(X)$; then for discrete X,

$$E(Y) = \sum_x g(x)p(x), \quad \text{if} \sum_x |g(x)|\ p(x) < \infty;$$

and for continuous X,

$$E(Y) = \int_{-\infty}^{\infty} g(x)f(x)\,\mathrm{d}x, \quad \text{if} \int |g(x)|\ f(x)\,\mathrm{d}x < \infty.$$

This is much easier than finding $f_Y(y) = f_X(g^{-1}(y))/\ |\mathrm{d}y/\,\mathrm{d}x|$ and then $E(Y) = \int y f_Y(y)\,\mathrm{d}y$.

Proof. For the discrete case, let $A_i = \{x : g(x) = y_i\}$, so $p(Y = y_i) = p(A_i)$:

$$E(Y) = \sum_i y_i p(y_i) = \sum_i y_i p(A_i) = \sum_i y_i \sum_{x\in A_i} p(x) = \sum_i \sum_{x\in A_i} y_i p(x) = \sum_x g(x)p(x).$$

Not a proof (continuous case): $E(Y) = \int y\underline{f_Y(y)\,\mathrm{d}y} = \int g(x)\underline{f_X(x)\,\mathrm{d}x}$, since the underlined probabilities are equal by Theorem 48. □

Example 91 (Kinetic Energy).

- The velocity of a gas molecule is a r.v. with

$$f_X(x) = \frac{\sqrt{2/\pi}}{\sigma^3} x^2 \exp\left[-\frac{1}{2}\frac{x^2}{\sigma^2}\right], \quad x \geq 0.$$

- The kinetic energy is $Y = \frac{1}{2}mX^2$.
- What is the mean kinetic energy?

$$E(Y) = \int_0^\infty \frac{1}{2} m x^2 f_X(x)\,\mathrm{d}x = \frac{m}{\sqrt{2\pi}\sigma^3} \int_0^\infty x^4 \exp\left[-\frac{1}{2}\frac{x^2}{\sigma^2}\right] \mathrm{d}x.$$

Change variables: $u = x^2/2\sigma^2$, $x = \sqrt{2}\sigma u^{1/2}$, $\mathrm{d}x = \frac{\sigma}{\sqrt{2}} u^{-1/2}\,\mathrm{d}u$, $\int_0^\infty \Rightarrow \int_0^\infty$:

$$E(Y) = \frac{2m\sigma^2}{\sqrt{\pi}} \int_0^\infty u^{3/2} \exp[-u]\,\mathrm{d}u = \frac{2m\sigma^2}{\sqrt{\pi}} \Gamma(3/2+1) = \frac{2m\sigma^2}{\sqrt{\pi}} 1.5 \times .5 \times \Gamma(.5) = \frac{3}{2} m\sigma^2.$$

Notice we have used $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ and $\Gamma(.5) = \sqrt{\pi}$.

▸ **Lecture note.** This is the **Maxwell–Boltzmann** speed distribution. In physics $\sigma^2 = kT/m$, so the result reads $E(Y) = \frac{3}{2}kT$: the average kinetic energy of a gas molecule depends only on temperature — a famous result of statistical mechanics, derived here with one change of variables.

Theorem 92 (Generalization). Suppose $X_1, ..., X_n$ are jointly distributed r.v.s, and $Y = g(X_1, ..., X_n)$. Then

1. for the discrete case,

$$E(Y) = \sum_{x_1,...,x_n} g(x_1, ..., x_n) p(x_1, ..., x_n),$$

   provided that $\sum |g(x_1, ..., x_n)|\ p(x_1, ..., x_n) < \infty$;
2. for the continuous case,

$$E(Y) = \int \cdots \int g(x_1, ..., x_n) f(x_1, ..., x_n)\,\mathrm{d}x_1 \cdots \mathrm{d}x_n,$$

   provided that $\int \cdots \int |g(x_1, ..., x_n)|\ f(x_1, ..., x_n)\,\mathrm{d}x_1 \cdots \mathrm{d}x_n < \infty$.

Corollary 93. If X and Y are independent r.v.s, then

$$E(g(X)h(Y)) = E(g(X))E(h(Y)),$$

and in particular $E(XY) = E(X)E(Y)$.

4.1.3 Expectations of Linear Combinations of r.v.

Theorem 94. Suppose $X_1, ..., X_n$ are jointly distributed — **NOT NECESSARILY INDEPENDENT** — and $E(X_i)$ exists, and $Y = a + \sum_i b_i X_i$. Then

$$E(Y) = a + \sum_i b_i E(X_i).$$

Proof. For the continuous case:

$$\begin{aligned} E(Y) &= \int \cdots \int \left(a + \sum_i b_i x_i \right) f(x_1, ..., x_n)\, \mathrm{d}x_1 \cdots \mathrm{d}x_n \\ &= \int \cdots \int a f(x_1, ..., x_n)\, \mathrm{d}x_1 \cdots \mathrm{d}x_n + \sum_i b_i \int \cdots \int x_i f(x_1, ..., x_n)\, \mathrm{d}x_1 \cdots \mathrm{d}x_n \\ &= a + \sum_i b_i \int x_i f(x_i)\, \mathrm{d}x_i = a + \sum_i b_i E(X_i). \end{aligned}$$

□

Example 95 (Mean of Binomial). $Y \sim \text{Binomial}(n, p)$, $p(k) = \binom{n}{k} p^k (1-p)^{n-k}$, so

$$E(Y) = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k},$$

which is not straightforward. However, thanks to indicator variables, $\text{Binomial}(n, p) \sim \sum_{i=1}^{n} \text{Bernoulli}(p)$:

$$Y = \sum_i I_i, \quad E(Y) = E\left(\sum_i I_i \right) = \sum_{i=1}^{n} E(I_i) = n(p \times 1 + (1-p) \times 0) = np.$$

Example 96 (Coupon Collection).

- Collect n equally likely different coupons.
- The number of trials required is a r.v.
- Getting the pmf may be hard; what about the mean?

$$X = \sum_{r=1}^{n} X_r, \quad X_r = \#\text{ trials to get the } r^{\text{th}} \text{ coupon after getting } r-1,$$

$$X_r \sim \text{Geometric}\left(\frac{n-(r-1)}{n} \right), \quad E(X_r) = \frac{1}{p} = \frac{n}{n-r+1},$$

$$E(X) = \sum_{r=1}^{n} \frac{n}{n-r+1} = \frac{n}{n} + \frac{n}{n-1} + \frac{n}{n-2} + \cdots + \frac{n}{1} = n \sum_{r=1}^{n} \frac{1}{r}.$$

E.g., for $n = 10$, $E(X) = 29.3$.

⋆ **Intuition.** The first few coupons come fast; the last one is the killer — with one missing, each purchase succeeds with probability only $1/n$, so it alone takes n purchases on average. Since $\sum_{r=1}^{n} 1/r \approx \log n + 0.577$, $E(X) \approx n \log n$: for the 32 football stickers in a pack series, about $32 \times 4.06 \approx 130$ purchases.

4.2 Variance and Standard Deviation

Definition 97. If X is a r.v. with $E(X)$, then

$$\sigma^2 \equiv \text{Var}(X) = E[(X - E(X))^2], \qquad \sigma \equiv \text{sd}(X) = \sqrt{\text{Var}(X)},$$

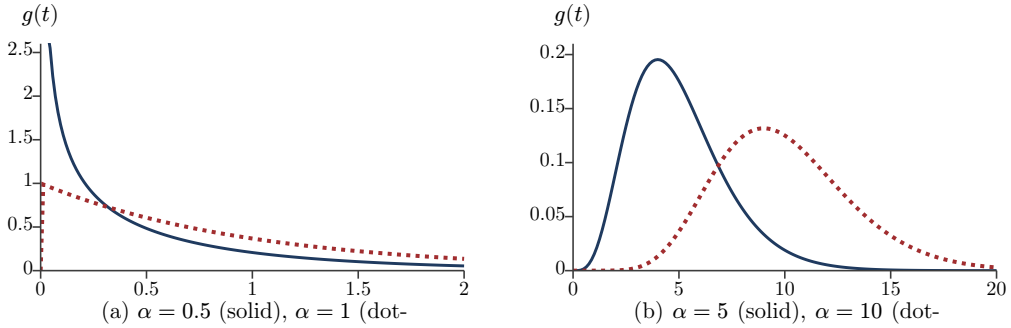
provided that $E[(X - E(X))^2] < \infty$.

Intuition:

- We need some measure of "dispersion".

- The SD has the same units as X, so it is more meaningful.
- What is the pdf of $Y = X - E(X)$?
- We could have defined it as $\mathrm{Var}_2(X) = E(|X - E(X)|)$, which is called the **mean absolute deviation**.

▸ **Lecture note.** Why squares, not absolute values? Squares are smooth (differentiable), they add nicely for independent sums (Corollary 113), and they lead to the normal distribution through the CLT. The price: variance weighs large deviations heavily, so a few outliers can inflate it.



(a) $\alpha = 0.5$ (solid), $\alpha = 1$ (dotted) (b) $\alpha = 5$ (solid), $\alpha = 10$ (dotted)

Gamma densities with $\lambda = 1$: mean α, variance α. Larger α shifts **and** spreads the density.

Theorem 98. If $\mathrm{Var}(X)$ exists, and $Y = a + bX$, then

$$\mathrm{Var}(Y) = b^2 \,\mathrm{Var}(X).$$

Proof.

$$\mathrm{Var}(Y) = E(Y - E(Y))^2 = E(a + bX - [a + bE(X)])^2 = E(b(X - E(X)))^2 = b^2 E(X - E(X))^2 = b^2 \,\mathrm{Var}(X),$$

$$\mathrm{sd}(Y) = |b| \,\mathrm{sd}(X),$$

which makes a lot of sense (why?). □

HW problem: if $Z = (X - \mu_X)/\sigma_X$, then $E(Z) = 0$ and $\mathrm{Var}(Z) = 1$.

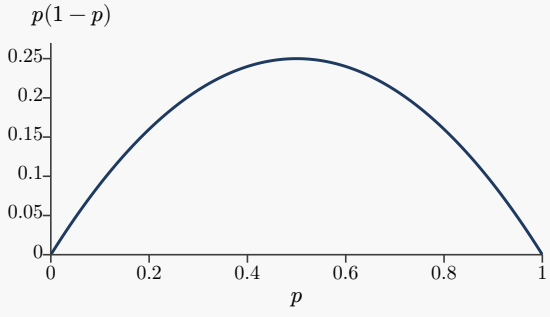
Solution. **Why it makes sense:** shifting by a moves every value by the same amount — the spread does not change. Scaling by b stretches every distance from the mean by $|b|$.

Example 99 (Bernoulli r.v.).

$$\mathrm{Var}(X) = E(X - E(X))^2 = \sum_x (x - E(X))^2 p(x) = \sum_x (x - p)^2 p(x)$$

$$= (1-p)^2 p + (0-p)^2 (1-p) = ((1-p) + p) p(1-p) = p(1-p).$$

Does it make sense?



▸ **Lecture note.** It does: at $p = 0$ or $p = 1$ there is no randomness at all (variance 0); the outcome is most uncertain at $p = 1/2$, where the variance peaks at $1/4$. This is why opinion polls plan their sample size for the worst case $p = 1/2$.

Example 100 (Normal Distribution).

$$\mathrm{Var}(X) = E(X-\mu)^2 = \int_{-\infty}^{\infty} (x-\mu)^2 \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right] \mathrm{d}x.$$

Substitute (to transform to the standard normal) $z = (x-\mu)/\sigma$, $\mathrm{d}x = \sigma\,\mathrm{d}z$:

$$\mathrm{Var}(X) = \sigma^2 \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} z^2 \exp[-z^2/2]\,\mathrm{d}z.$$

Substitute $u = z^2/2$, $z = \sqrt{2}u^{1/2}$, $\mathrm{d}z = \frac{\sqrt{2}}{2}u^{-1/2}\,\mathrm{d}u$, $\int_{-\infty}^{\infty}\mathrm{d}z \Rightarrow 2\int_0^{\infty}\mathrm{d}u$:

$$\mathrm{Var}(X) = \sigma^2 2 \int_0^{\infty} \frac{1}{\sqrt{2\pi}} 2u \exp[-u] \frac{\sqrt{2}}{2} u^{-1/2}\,\mathrm{d}u = \sigma^2 \frac{2}{\sqrt{\pi}} \int_0^{\infty} u^{1/2} e^{-u}\,\mathrm{d}u = \sigma^2 \frac{2}{\sqrt{\pi}}\Gamma(1.5) = \sigma^2 \frac{2}{\sqrt{\pi}}(.5)\Gamma(.5) = \sigma^2.$$

Then, wonderful: the normal pdf is expressed in terms of its population parameters.

Corollary 101. The variance, if it exists, can be given by

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2.$$

Proof.

$$\mathrm{Var}(X) = E(X-\mu)^2 = E(X^2 - 2X\mu + \mu^2) = E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2.$$

□

Example 102 (Uniform(a,b)).

$$E(X) = \int_a^b x f(x)\,\mathrm{d}x = \int_a^b x \frac{1}{b-a}\,\mathrm{d}x = \frac{1}{b-a}\frac{1}{2}x^2\Big|_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{1}{2}(b+a),$$

$$E(X^2) = \frac{1}{b-a}\int_a^b x^2\,\mathrm{d}x = \frac{b^3-a^3}{3(b-a)} = \frac{1}{3}(b^2+ab+a^2),$$

$$\mathrm{Var}(X) = E(X^2) - (E(X))^2 = \frac{1}{3}(b^2+ab+a^2) - \frac{1}{4}(b+a)^2 = \frac{(b-a)^2}{12}.$$

Correction to the original. The original has $E(X^2) = \frac{1}{3}(b^2+ab+b^2)$; the factorisation $b^3 - a^3 = (b-a)(b^2+ab+a^2)$ gives a^2 in the last term. (In Corollary 101, the original's $E(X^2 - 2X\mu - \mu^2)$ should read $+\mu^2$; both are fixed above.)

Theorem 103 (Chebyshev's Inequality). Let X be a r.v. with mean μ and variance σ^2; then

$$P(|X-\mu| \geq t) \leq \frac{\sigma^2}{t^2}, \quad \forall t > 0.$$

Proof. Set $Y = (X-\mu)^2$; then from Markov's inequality, $P(Y \geq t^2) \leq E(Y)/t^2$, $\forall t > 0$, $Y \geq 0$. Then

$$P((X-\mu)^2 \geq t^2) \leq \frac{E(X-\mu)^2}{t^2}, \qquad P(|X-\mu| \geq t) \leq \frac{\sigma^2}{t^2}.$$

Intuition: the probability of falling t-far from the mean increases with σ^2 and decreases with t. □

Corollary 104. If $\mathrm{Var}(X) = 0$, then $P(X = \mu) = 1$.

Proof. Of course, it makes a lot of sense. But rigorously, from Chebyshev, $P(|X-\mu| \geq t) \leq \sigma^2/t^2 = 0$ for all $t > 0$. If $P(X = \mu) \neq 1$, then $P(X \neq \mu) \neq 0$, and $P(|X-\mu| \geq t) \neq 0$ for some $t > 0$ (since $\{X \neq \mu\} = \bigcup_n \{|X-\mu| \geq 1/n\}$), which contradicts the above. □

Example 105 (Normal and Chebyshev). For $X \sim \mathrm{Normal}(0,1)$, we learned before (Example 51) that $P(|X-\mu| > 2\sigma) = 0.0455$. However, from Chebyshev,

$$P(|X-\mu| \geq t) \leq \frac{\sigma^2}{t^2}, \qquad P(|X-\mu| \geq 2\sigma) \leq .25,$$

which is not as tight as the value obtained from the normal density directly, because this is a **distribution-free** result.

k	Chebyshev bound on $P(\lvert X-\mu\rvert \geq k\sigma)$	Exact for the normal
1	≤ 1 (useless)	0.317
2	≤ 0.25	0.0455
3	≤ 0.111	0.0027
4	≤ 0.0625	0.00006

4.2.1 A Model for Measurement Error

- Suppose that we measure a constant x_0.
- Measurements are a r.v. X, with μ and σ.
- The error $X - x_0$ is a r.v.; analyze it!
- Mean squared error (MSE): $\mathrm{MSE} = E(X-x_0)^2$.

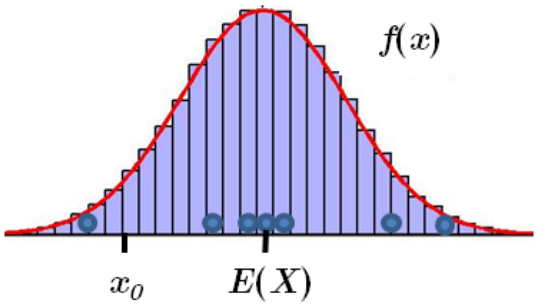


Figure 17: Measurements scatter around $E(X)$, which need not equal the true value x_0.

Theorem 106 (Mean Squared Error (MSE)).

$$\mathrm{MSE} = \mathrm{Variance} + \mathrm{Bias}^2 = \sigma^2 + (\mu - x_0)^2.$$

Proof.

$$\mathrm{MSE} = E(X-x_0)^2 = \mathrm{Var}(X-x_0) + [E(X-x_0)]^2 = \mathrm{Var}(X) + (\mu - x_0)^2.$$

Another proof (common trick): add and subtract μ,

$$\text{MSE} = E((X-\mu)+(\mu-x_0))^2 = E(X-\mu)^2 + 2(\mu-x_0)\underbrace{E(X-\mu)}_{=0} + (\mu-x_0)^2$$
$$= \sigma^2 + (\mu-x_0)^2.$$

□

⋆ **Intuition.** **Variance** is imprecision — shots scattered around their own centre. **Bias** is inaccuracy — the centre itself is off target. A rifle can be precise but biased (tight group, wrong place) or unbiased but imprecise (centred, but spread). MSE charges you for both. This decomposition is the "bias–variance trade-off" of machine learning.

Example 107 (Measurement of the Gravity Constant). Two series of absolute measurements of g by dropping a body 32 times.

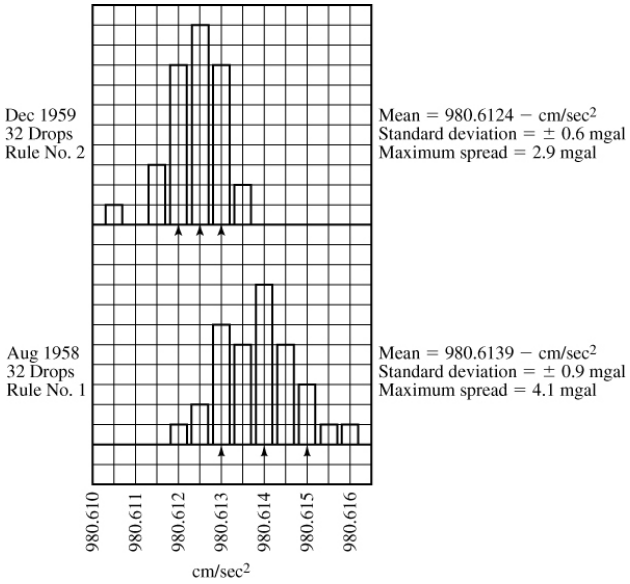


Figure 18: Two series of 32 drops each (1958, 1959): the means agree to 0.0015, but the spreads differ (Rice, 3rd ed.).

Example 108 (Measurement of the Speed of Light). Published measurements of the speed of light, each with its reported uncertainty.

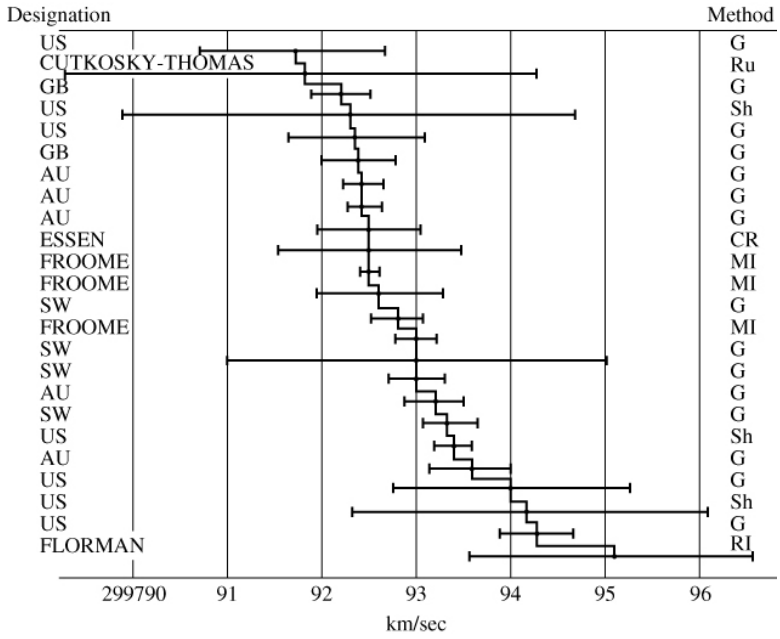


Figure 19: Measurements of the speed of light with their reported error bars, in chronological order (Rice, 3rd ed.). Many error bars do not cover later, more accurate values — the reported "uncertainty" captured the variance but not the bias.

4.3 Covariance & Correlation

This section should have an impact on your way of thinking and reading different situations in life.

Definition 109. If X and Y are two r.v.s with means μ_X and μ_Y,

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)],$$

which can be rewritten as

$$\mathrm{Cov}(X,Y) = E[XY - \mu_X Y - \mu_Y X + \mu_X\mu_Y] = E[XY] - \mu_X\mu_Y - \mu_X\mu_Y + \mu_X\mu_Y = E[XY] - \mu_X\mu_Y.$$

Intuition:

- We need to measure "association".
- $\mathrm{Cov}(X,Y)$ has the units of XY.
- If X and Y are independent, $\mathrm{Cov}(X,Y) = 0$.

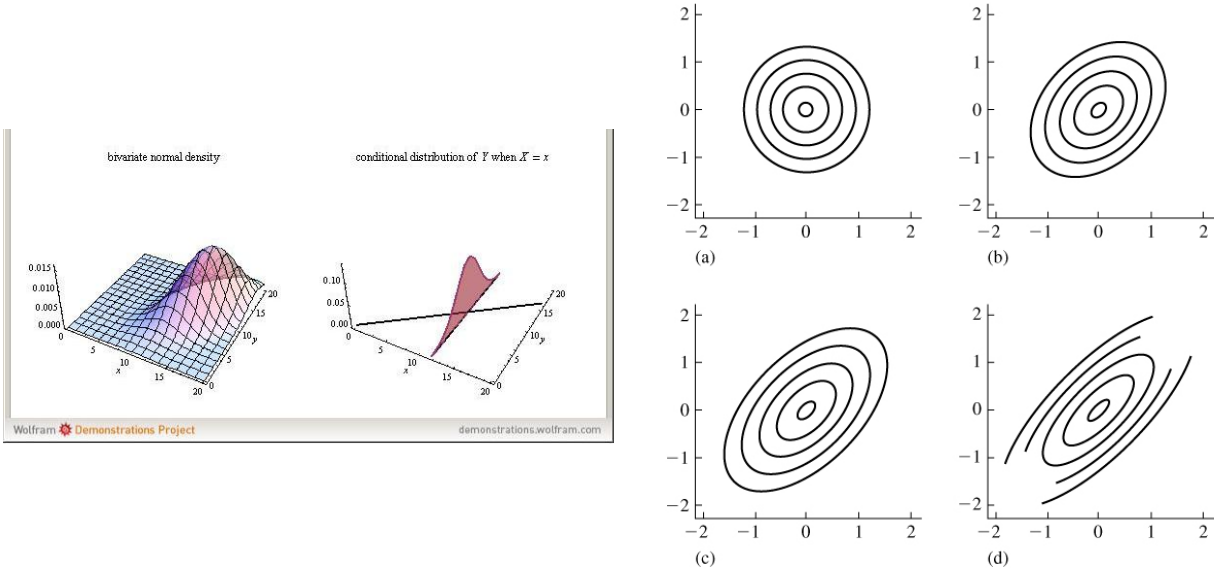


Figure 20: Bivariate normal densities and their contours: (a) $\rho = 0$, (b) $\rho = 0.3$, (c) $\rho = 0.6$, (d) $\rho = 0.9$.

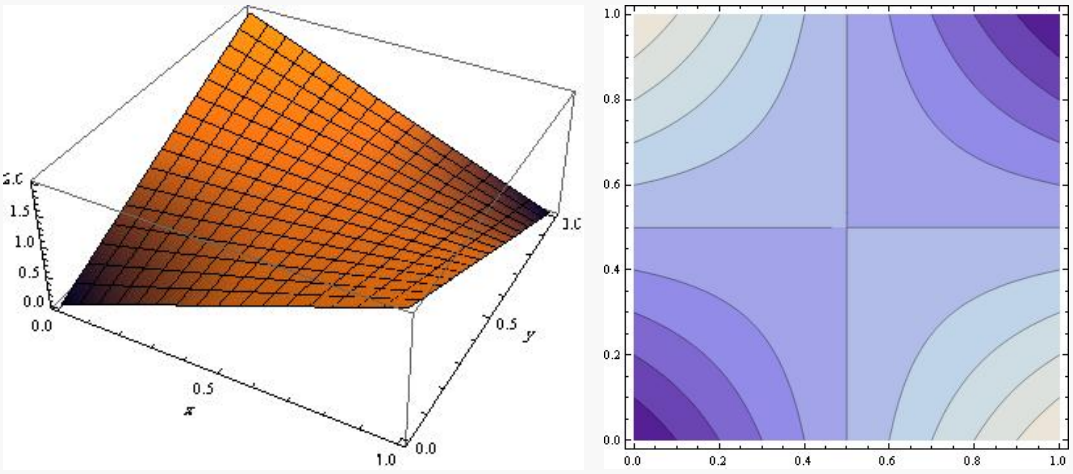
⋆ **Intuition.** Look at the sign of $(X - \mu_X)(Y - \mu_Y)$. Points in the upper-right or lower-left quadrant (both above, or both below, their means) contribute $+$; the other two quadrants contribute $-$. If the cloud tilts up, the $+$ quadrants dominate: $\mathrm{Cov} > 0$. Tilts down: $\mathrm{Cov} < 0$.

Example 110.

$$f(x,y) = 2x + 2y - 4xy, \quad 0 \le x, y \le 1,$$

$$f_X(x) = \int_0^1 (2x + 2y - 4xy)\,\mathrm{d}y = 1,\ 0 \le x \le 1, \quad f_Y(y) = 1,\ 0 \le y \le 1, \quad \mu_X = \mu_Y = \frac{1}{2},$$

$$\mathrm{Cov}(X,Y) = E(XY) - \mu_X\mu_Y = \int_0^1 \int_0^1 xy(2x + 2y - 4xy)\,\mathrm{d}x\,\mathrm{d}y - \frac{1}{4} = \frac{2}{9} - \frac{1}{4} = -\frac{1}{36}.$$



▸ **Lecture note.** Both marginals are Uniform$(0,1)$, yet the variables are dependent — another instance of "marginals do not determine the joint". The density is high near $(1,0)$ and $(0,1)$ and low near $(0,0)$ and $(1,1)$, so large x tends to go with small y: negative covariance.

Lemma 111. Suppose that $U = a_0 + a_1 X$ and $V = b_0 + b_1 Y$; then

$$\operatorname{Cov}(U, V) = a_1 b_1 \operatorname{Cov}(X, Y).$$

Proof. $\mu_U = a_0 + a_1 \mu_X$, $\mu_V = b_0 + b_1 \mu_Y$, so

$$\operatorname{Cov}(U, V) = E[(U - \mu_U)(V - \mu_V)] = E[a_1 (X - \mu_X) b_1 (Y - \mu_Y)] = a_1 b_1 \operatorname{Cov}(X, Y).$$

Intuition: scaling is reflected, since covariance is unit-dependent. □

Theorem 112. Suppose that $U = a_0 + \sum_{i=1}^{n} a_i X_i$ and $V = b_0 + \sum_{j=1}^{m} b_j Y_j$; then

$$\operatorname{Cov}(U, V) = \sum_{i=1}^{n} \sum_{j=1}^{m} a_i b_j \operatorname{Cov}\big(X_i, Y_j\big).$$

Proof. $\mu_U = a_0 + \sum_i a_i \mu_{X_i}$, $\mu_V = b_0 + \sum_j b_j \mu_{Y_j}$, so

$$\operatorname{Cov}(U, V) = E[(U - \mu_U)(V - \mu_V)] = E\left[\left(\sum_i a_i \big(X_i - \mu_{X_i}\big)\right)\left(\sum_j b_j \big(Y_j - \mu_{Y_j}\big)\right)\right]$$

$$= E\left[\sum_i \sum_j a_i b_j \big(X_i - \mu_{X_i}\big)\big(Y_j - \mu_{Y_j}\big)\right] = \sum_i \sum_j a_i b_j E\big[\big(X_i - \mu_{X_i}\big)\big(Y_j - \mu_{Y_j}\big)\big] = \sum_i \sum_j a_i b_j \operatorname{Cov}\big(X_i, Y_j\big).$$

□

Corollary 113. Consider $U = a_0 + \sum_{i=1}^{n} a_i X_i$.

1. In general:
$$\operatorname{Var}(U) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j \operatorname{Cov}\big(X_i, X_j\big) = \sum_{i=1}^{n} a_i^2 \operatorname{Var}(X_i) + 2 \sum_{i>j} a_i a_j \operatorname{Cov}\big(X_i, X_j\big).$$
2. If the X_i are uncorrelated (or independent): $\operatorname{Var}(U) = \sum_{i=1}^{n} a_i^2 \operatorname{Var}(X_i)$.
3. If the X_i are i.i.d. and $a_i = 1$: $\operatorname{Var}(U) = n\sigma^2$.

Proof. Immediate by noticing that $\operatorname{Cov}(U, U) = \operatorname{Var}(U)$. □

! Watch out. $\operatorname{Var}(X - Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) - 2\operatorname{Cov}(X, Y)$ — for independent X, Y the variances **add** even when you subtract. Writing $\operatorname{Var}(X) - \operatorname{Var}(Y)$ is a classic exam error (and can even be negative, which no variance can be).

Example 114 (Variance of Binomial).

$$\operatorname{Var}(X) = E[X^2] - (E[X])^2 = \sum_{k=0}^{n} k^2 \binom{n}{k} p^k (1-p)^{n-k} - (np)^2.$$

However, it is much easier to notice that $X = \sum_{i=1}^{n} I_i$, $I_i \sim$ i.i.d. Bernoulli(p), so

$$\operatorname{Var}(X) = n \operatorname{Var}(I_i) = np(1-p).$$

Definition 115 (Correlation Coefficient). If X and Y are jointly distributed r.v.s with existing means and variances,

$$\rho_{XY} = \frac{\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Intuition:

1. This is dimensionless, hoping that it has a more meaningful figure than the covariance.
2. It is invariant under linear transformation $U = a_0 + a_1 X$, $V = b_0 + b_1 Y$, $a_1, b_1 > 0$:

$$\rho_{UV} = \frac{\operatorname{Cov}(a_0 + a_1 X, b_0 + b_1 Y)}{\sqrt{\operatorname{Var}(a_0 + a_1 X)\operatorname{Var}(b_0 + b_1 Y)}} = \frac{a_1 b_1 \operatorname{Cov}(X,Y)}{\sqrt{a_1^2 \operatorname{Var}(X) b_1^2 \operatorname{Var}(Y)}} = \frac{\operatorname{Cov}(X,Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}} = \rho_{XY}.$$

Theorem 116. $-1 \le \rho \le 1$. Furthermore, $\rho = \pm 1$ **iff** $P(Y = a + bX) = 1$ for some a, b.

Proof.

$$0 \le \operatorname{Var}\left(\frac{X}{\sigma_X} \pm \frac{Y}{\sigma_Y}\right) = \frac{\operatorname{Var}(X)}{\sigma_X^2} + \frac{\operatorname{Var}(Y)}{\sigma_Y^2} \pm 2\frac{\operatorname{Cov}(X,Y)}{\sigma_X \sigma_Y} = 2(1 \pm \rho),$$

so $-1 \le \rho \le 1$. Next, $\operatorname{Var}(X/\sigma_X + Y/\sigma_Y) = 2(1+\rho)$, and

$$\rho = -1 \iff \operatorname{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right) = 0 \iff P\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y} = c\right) = 1,$$

using Corollary 104 for "$\Longrightarrow$" and the variance of a constant for "$\Longleftarrow$". Similarly for $\rho = 1$ with the minus sign. □

Example 117 (Revisit Ex. 110).

$$\rho = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} = \frac{-1/36}{\left(1/\sqrt{12}\right)\left(1/\sqrt{12}\right)} = -\frac{1}{3}.$$

X, Y	$\rho = 0$	$\rho \neq 0$
Dependent	possible	possible
Independent	possible	**impossible**

$$\text{Independence} \Longrightarrow \text{Cov} = 0, \qquad \text{Cov} \neq 0 \Longrightarrow \text{Dependence}.$$

Correlation is a measure of a linear relationship.

Example 118 (Counter Example). X, Y independent:

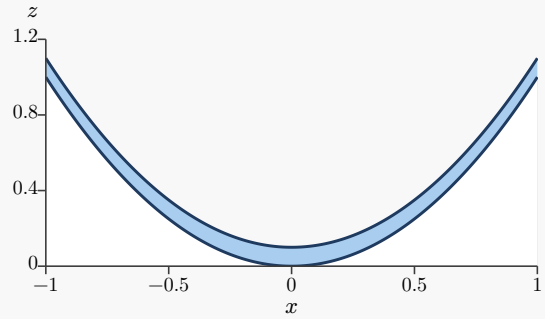
- $X \sim \text{Uniform}(-1, 1)$; $f_X(x) = 1/2$.
- $Y \sim \text{Uniform}(0, 1/10)$; $f_Y(y) = 10$.
- $Z = X^2 + Y$: what are f_{XZ} and $\operatorname{Cov}(X, Z)$?

With $Z = X^2 + Y$, $U = X$:

$$J = \begin{vmatrix} \frac{\partial Z}{\partial X} & \frac{\partial Z}{\partial Y} \\ \frac{\partial U}{\partial X} & \frac{\partial U}{\partial Y} \end{vmatrix} = \begin{vmatrix} 2X & 1 \\ 1 & 0 \end{vmatrix} = -1, \quad |J| = 1,$$

$$f_{UZ}(u,z) = f_X(x) f_Y(y)\, |J|^{-1} = \left(\frac{1}{2}\right)(10)(1) = 5, \quad -1 \le x \le 1, \quad x^2 \le z \le x^2 + .1.$$

It is much easier to say that $f_{Z \mid X}(z \mid x) = 10$ on $x^2 \le z \le x^2 + .1$, and $f_{XZ}(x,z) = f_{Z \mid X}(z \mid x) f_X(x) = 10 \times \frac{1}{2} = 5$.



$$\mathrm{Cov}(X,Z) = E[XZ] - E[X]E[Z] = E[X(X^2+Y)] - 0 \cdot E[Z] = E[X^3] + E[XY] = 0 + E[X]E[Y] = 0.$$

Intuition: there is dependency, yet it is **not linear**.

4.3.1 Observed Correlation Does Not Necessarily Imply Causation

It may be one **or a combination** of the following:

- **Example of A causes B:** many.
- **Example of B causes A.** Observation: (A) the more firemen fighting a fire, (B) the bigger the fire is observed to be. (The big fire causes the many firemen, not the reverse.)
- **Example of (C) causes both.** Observation (Quinn et al., 1999, *Nature*): (A) young children sleeping with the light on (B) are more likely to develop myopia. The study suggested that sleeping with the light on **caused** myopia. However, later studies found that parental myopia (C) is correlated with child myopia (B), and myopic parents (C) were more likely to leave a light on (A) in their children's bedroom.

Correction to the original. The original reads "Later study found that: infants sleeping with the light on caused the development of myopia!! However, they found that …". The causal claim was the 1999 study's; the **later** studies (Zadnik et al. and Gwiazda et al., *Nature* 2000) found no causal effect and identified parental myopia as the common cause. Rewritten above to say so.

▸ **Lecture note.** Before believing "A causes B" from data, run the checklist: (1) could B cause A? (2) is there a C that drives both? (3) is the correlation just chance in a small sample? (4) was the sample selected in a way that creates it? Only a **randomised experiment** — where you assign A by coin toss, so no C can influence it — rules all four out.

4.4 Conditional Expectation and Prediction

4.4.1 Definitions and Examples

Definition 119. The **conditional expectation** is defined as

$$E(Y \mid X = x) = \sum_y y p_Y(y \mid x) \qquad \text{(Disc.)},$$

$$E(Y \mid X = x) = \int y f_{Y \mid X}(y \mid x)\, dy \quad \text{(Cont.)}$$

In general,

$$E(h(Y) \mid X = x) = \sum_y h(y) p_Y(y \mid x) \qquad \text{(Disc.)},$$

$$E(h(Y) \mid X = x) = \int h(y) f_{Y \mid X}(y \mid x)\, dy \quad \text{(Cont.)}$$

Intuition: for a joint distribution $f_{XY}(x, y)$, at each x there is a conditional distribution $f_{Y \mid X}(y \mid x)$, e.g. the **age–salary trend**.

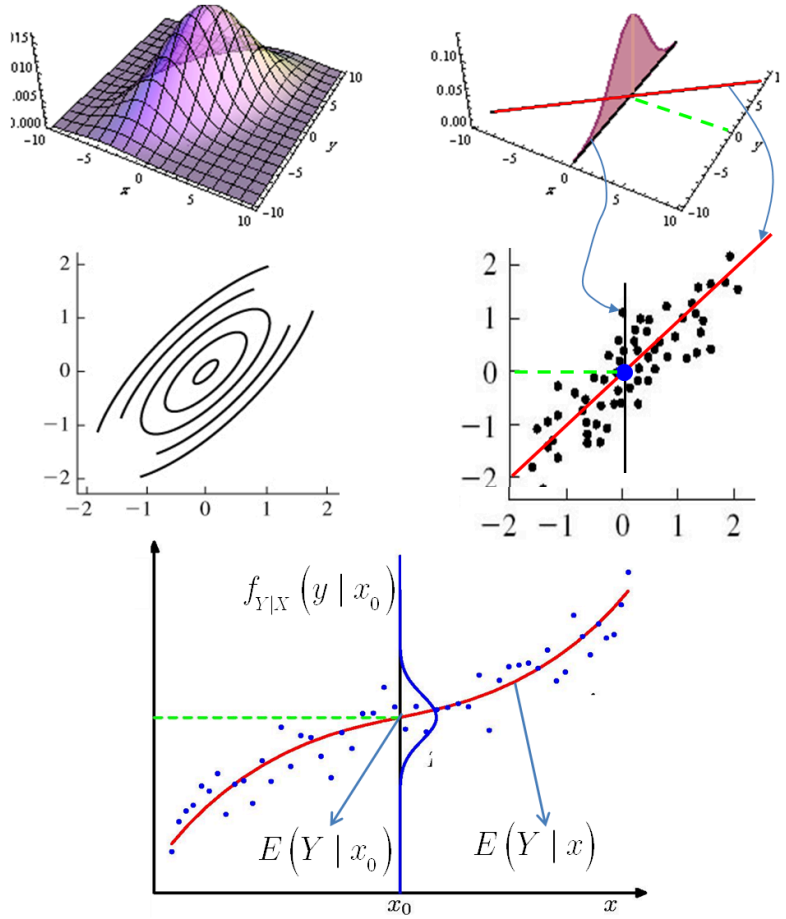


Figure 21: At each x_0 the conditional density $f_{Y \mid X}(y \mid x_0)$ has a mean $E(Y \mid x_0)$; as x_0 moves, those means trace a curve.

This is called the "regression function" $x \mapsto E(Y \mid X = x)$.

Example 120. $N \sim \text{Poisson}(\lambda)$.

- N is the number of observations in $[0, 1]$.

- X is the number of observations in $[0,p]$.
- What is $E[X \mid N=n]$?

$$X = \#\text{ obs. in } [0,p] \sim \text{Poisson}(p\lambda), \quad Y = \#\text{ obs. in } [p,1] \sim \text{Poisson}((1-p)\lambda),$$

$$N = X+Y, \quad X' = X, \quad X = X', \quad Y = N - X.$$

$$\begin{aligned} P_{X'N}(x,n) &= P_{XY}(x,n-x) = P_X(x)P_Y(n-x) \quad \text{(no Jacobian)} \\ &= \frac{(p\lambda)^x e^{-p\lambda}}{x!}\frac{((1-p)\lambda)^{n-x}e^{-(1-p)\lambda}}{(n-x)!} = \lambda^n e^{-\lambda}\frac{p^x(1-p)^{n-x}}{x!(n-x)!}, \end{aligned}$$

and $P_{X' \mid N}(x \mid n)$ is nothing but $P_{X'N}(x,n)$ scaled:

$$P_{X' \mid N}(x \mid n) = \frac{P_{X'N}(x,n)}{P_N(n)} = \frac{\lambda^n e^{-\lambda}p^x(1-p)^{n-x}/(x!(n-x)!)}{\lambda^n e^{-\lambda}/n!} = \binom{n}{x}p^x(1-p)^{n-x} \sim \text{Binomial}(n,p),$$

$$E[X \mid N=n] = np,$$

which is a discrete line in $n = 1, \ldots$. This makes sense because n is fixed. Let's open a Mathematica notebook and understand it visually.

▸ **Lecture note.** Example 120 is Example 70 run backwards: there, $X \mid N \sim \text{Bin}(N,p)$ gave $X \sim \text{Poisson}(\lambda p)$; here, independent Poisson counts in disjoint intervals, conditioned on their total, are binomial. Given n events in $[0,1]$, each lands in $[0,p]$ independently with probability p.

Theorem 121 (Law of Total Expectation). $E(Y) = E[E(Y \mid X)]$. We can also write it as

$$E(Y) = E_X\left[E_{Y \mid X}(Y \mid X)\right].$$

Proof. Discrete case:

$$E_{Y \mid X}(Y \mid X) = \sum_y y p_{Y \mid X}(y \mid x),$$

$$\begin{aligned} E_X\left[E_{Y \mid X}(Y \mid X)\right] &= \sum_x E_{Y \mid X}(Y \mid x)p_X(x) = \sum_x \left(\sum_y y p_{Y \mid X}(y \mid x)\right)p_X(x) \\ &= \sum_y y \sum_x p_{Y \mid X}(y \mid x)p_X(x) = \sum_y y p_Y(y) = E(Y). \end{aligned}$$

The proof for the continuous case is very similar. **Intuition:** see the conditional expectation figure — average the curve $E(Y \mid x)$ against the distribution of X. □

⋆ **Intuition.** To find the average salary in a company, average within each department, then average those department means **weighted by department size**. That is $E(Y) = E[E(Y \mid X)]$ with $X =$ department. The law of total probability, but for means.

Example 122. Consider a unit and its backup:

- each has mean lifetime μ;
- the backup unit may not launch, with probability p;
- what is the mean lifetime of the system?

Consider the launching Bernoulli variable $I \sim \text{Ber}(p)$, where $I = 1$ means the backup fails to launch:

$$T = \begin{cases} T_1, & I = 1 \\ T_1 + T_2, & I = 0 \end{cases} \quad E(T \mid I = 1) = \mu, \quad E(T \mid I = 0) = 2\mu,$$

$$E(T) = \mu p + 2\mu(1-p) = \mu(2-p).$$

More generally,

$$f_T(t) = f_{T \mid I}(t \mid 1)p + f_{T \mid I}(t \mid 0)(1-p) = f_{T_1}(t)p + f_{T_1+T_2}(t)(1-p).$$

This is called a **mixture** of r.v.s (compare Example 47).

Theorem 123 (Variance Decomposition).

$$\mathrm{Var}[Y] = \mathrm{Var}_X\left[E_{Y \mid X}[Y \mid X]\right] + E_X\left[\mathrm{Var}_{Y \mid X}[Y \mid X]\right].$$

Proof.

$$\begin{aligned} \mathrm{Var}_X\left[E_{Y \mid X}[Y \mid X]\right] &= E_X\left[\left(E_{Y \mid X}[Y \mid X]\right)^2\right] - \left(E_X E_{Y \mid X}[Y \mid X]\right)^2 \\ &= E_X\left[\left(E_{Y \mid X}[Y \mid X]\right)^2\right] - (E[Y])^2, \\ E_X\left[\mathrm{Var}_{Y \mid X}[Y \mid X]\right] &= E_X\left[E_{Y \mid X}[Y^2 \mid X] - \left(E_{Y \mid X}[Y \mid X]\right)^2\right] \\ &= E[Y^2] - E_X\left[\left(E_{Y \mid X}[Y \mid X]\right)^2\right]. \end{aligned}$$

The sum is $E[Y^2] - (E[Y])^2 = \mathrm{Var}[Y]$. **Intuition:** see the figure. □

⋆ **Intuition.** Total spread = spread **between** groups (how much the group means differ) + average spread **within** groups. In the figure, the first term is the vertical wiggle of the red curve $E(Y \mid x)$; the second is the average width of the blue conditional densities. This is exactly the decomposition behind ANOVA and R^2 in regression.

Example 124 (Random Sums). The X_i are i.i.d., and independent of N:

$$T = \sum_{i=1}^{N} X_i, \quad T \mid (N = n) = \sum_{i=1}^{n} X_i, \quad E[T \mid N = n] = nE[X],$$

$$E[T] = E_N[NE[X]] = \mu_N \mu_X,$$

which makes sense.

$$\mathrm{Var}[T] = E_N\left[\mathrm{Var}_{T \mid N}[T \mid N]\right] + \mathrm{Var}_N\left[E_{T \mid N}[T \mid N]\right] = E_N[N\sigma_X^2] + \mathrm{Var}_N[N\mu_X] = \mu_N\sigma_X^2 + \mu_X^2\sigma_N^2.$$

A special case would be if N is constant n: $\mathrm{Var}[T] = n\sigma_X^2$, as usual.

Typical values: an insurance company with number of claims $N \sim \mathrm{Poisson}(\lambda = 900)$ and claim value X_i:

$$\mu_N = 900, \quad \sigma_N = 30, \quad \mu_X = \$1000, \quad \sigma_X = \$500,$$

$$\begin{aligned} E[T] &= \mu_N \mu_X = \$900,000, \\ \mathrm{Var}[T] &= \mu_N\sigma_X^2 + \mu_X^2\sigma_N^2 = 900 \times 500^2 + 1000^2 \times 30^2 = 225M + 900M = 1125M, \\ \sigma_T &= \$33,541. \end{aligned}$$

Therefore they should plan on $\mu_T = \$900,000 \pm \$33,541$. If N were fixed, then $\sigma_T^2 = 900 \times 500^2$ and $\sigma_T = \$15,000 \ll \$33,541$.

▸ **Lecture note.** The striking part: 80% of the risk ($900M$ of $1125M$) comes from not knowing **how many** claims, not from not knowing **how big**. An insurer that only models claim sizes underestimates its risk by more than half.

4.4.2 Prediction

Let's predict a r.v. by some constant c:

$$\begin{aligned}\text{MSE} &= E(Y-c)^2 = E((Y-E(Y)) + (E(Y)-c))^2 \\ &= E(Y-E(Y))^2 + (E(Y)-c)^2 + 2\underbrace{E[(Y-E(Y))(E(Y)-c)]}_{=0} = \underbrace{\text{Var}[Y]}_{\text{irreducible error}} + (E(Y)-c)^2,\end{aligned}$$

$$c_{\min} = \arg\min_c[\text{MSE}] = E(Y).$$

If we replace E by $E_{Y\mid X}$, then $c = h(X) = E_{Y\mid X}(Y \mid X)$ minimizes $E_{Y\mid X}[(Y-h(X))^2 \mid X]$, and therefore minimizes

$$\text{MSE} = E(Y-h(X))^2 = E_X E_{Y\mid X}[(Y-h(X))^2 \mid X],$$

which is the regression function. **This is what Machine Learning is about!**

▸ **Lecture note.** Read it as a two-line recipe. With no information, the best guess (in squared error) is the mean, and you cannot beat $\text{Var}(Y)$. With information X, the best guess is $E(Y \mid X)$, and the remaining error is $E[\text{Var}(Y \mid X)]$ — which by Theorem 123 is smaller than $\text{Var}(Y)$ by exactly $\text{Var}(E(Y \mid X))$, the part of Y that X explains. A machine-learning model is an estimate of $E(Y \mid X)$ from data.

4.5 The Moment-Generating Function

Definition 125 (Moments).

r^{th} moment $= E[X^r]$, 1st moment = mean (location),

r^{th} central moment $= E[(X-E[X])^r]$, 1st central moment $= 0$,

2nd central moment = variance (dispersion), 3rd central moment $\to$ skewness (asymmetry),

4th central moment $\to$ kurtosis (flatness / tails).

It is clear that if f_X is symmetric,

- its point of symmetry is $E[X]$;
- $X - E[X]$ will be symmetric around 0;
- $E[(X-E[X])^r]$, for odd r, will be 0;
- the rth normalized moment $= E[(X-E[X])^r]/\sigma^r$.

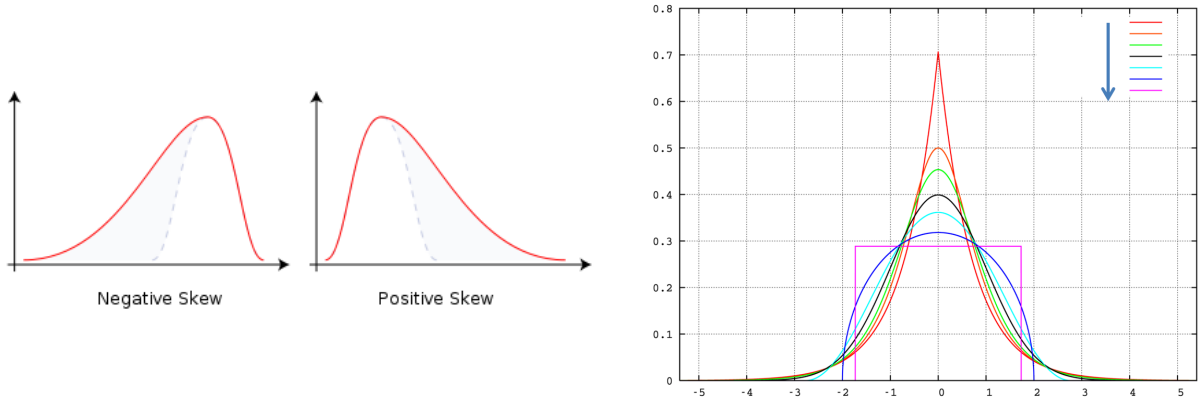


Figure 22: Left: negative and positive skew. Right: symmetric densities with unit variance and different kurtosis — from peaked and heavy-tailed (top) to flat (the uniform box).

▸ **Lecture note.** Skewness is the 3rd **normalized** moment $E[(X-\mu)^3]/\sigma^3$; kurtosis is the 4th, $E[(X-\mu)^4]/\sigma^4$, which equals 3 for every normal. "Excess kurtosis" subtracts 3. High kurtosis means more mass in the **tails** (and the peak) than a normal of the same variance — rare extreme events are more common, which is what matters in finance and risk.

Definition 126. The **moment-generating function** (mgf) $M_X(t)$ of a r.v. X is given by

$$M_X(t) = E[e^{tX}] = \sum_x e^{tx} p_X(x) \quad \text{(Disc.)}, \qquad M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x)\,\mathrm{d}x \quad \text{(Cont.)}$$

- It may not exist, e.g. if $X \sim$ Cauchy, since f_X fades out slowly.
- If it exists (on an open interval around 0) it uniquely defines f_X.
- The **characteristic function** always exists,

$$\varphi_X(t) = \int_{-\infty}^{\infty} e^{itx} f_X(x)\,\mathrm{d}x = \mathcal{F}\{f_X\},$$

  which is the Fourier transform of f_X. This is because $|e^{itx}| \le 1$.
- Many nice properties of M and φ, and connections to the $\mathcal{L}$ and $\mathcal{F}$ transforms, in the "Signals and Systems" course.

Proposition 127. If the mgf exists in an open interval containing zero, then

$$M^{(r)}(0) = E[X^r].$$

Proof.

$$M(t) = \int_{-\infty}^{\infty} e^{tx} f(x)\,\mathrm{d}x, \quad M'(t) = \frac{\mathrm{d}}{\mathrm{d}t}\int_{-\infty}^{\infty} e^{tx} f(x)\,\mathrm{d}x = \int_{-\infty}^{\infty} x e^{tx} f(x)\,\mathrm{d}x,$$

$$M''(t) = \int_{-\infty}^{\infty} x^2 e^{tx} f(x)\,\mathrm{d}x, \quad \ldots, \quad M^{(r)}(t) = \int_{-\infty}^{\infty} x^r e^{tx} f(x)\,\mathrm{d}x, \quad M^{(r)}(0) = \int_{-\infty}^{\infty} x^r f(x)\,\mathrm{d}x = E[X^r].$$

□

Therefore:

$$\text{pdf} \iff \text{mgf} \overset{\text{Taylor}}{\iff} \text{mgf derivatives} = \text{pdf moments}, \qquad M(t) = M(0) + \sum_r \frac{1}{r!} t^r M^{(r)}(0) \quad \text{(Taylor series)}.$$

Example 128 (Poisson).

$$M(t) = \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k}{k!} e^{-\lambda} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t-1)},$$

$$M'(t) = \lambda e^t e^{\lambda(e^t-1)}, \quad E[X] = M'(0) = \lambda,$$

$$M''(t) = \lambda e^t e^{\lambda(e^t-1)} + \lambda^2 e^{2t} e^{\lambda(e^t-1)}, \quad E[X^2] = M''(0) = \lambda + \lambda^2,$$

$$\mathrm{Var}[X] = E[X^2] - (E[X])^2 = \lambda.$$

Example 129 (Standard Normal).

$$M(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-x^2/2} \, \mathrm{d}x = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t^2/2 - (x-t)^2/2} \, \mathrm{d}x \quad \text{(completing the square)}$$

$$= e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2/2} \, \mathrm{d}x = e^{t^2/2},$$

$$M'(t) = te^{t^2/2}, \quad E[X] = M'(0) = 0, \quad M''(t) = e^{t^2/2} + t^2 e^{t^2/2}, \quad E[X^2] = M''(0) = 1, \quad \mathrm{Var}[X] = 1.$$

Lemma 130. If $Y = a + bX$, then $M_Y(t) = e^{at} M_X(bt)$.

Proof. $M_Y(t) = E[e^{tY}] = E[e^{at+btX}] = e^{at} E[e^{btX}] = e^{at} M_X(bt)$. □

Example 131. If $Y \sim N(\mu, \sigma^2)$, then $Y = \mu + \sigma Z$, and

$$M_Y(t) = e^{\mu t} M_Z(\sigma t) = e^{\mu t} e^{\sigma^2 t^2/2}.$$

Lemma 132. If X and Y are independent r.v.s and $Z = X + Y$, then $M_Z(t) = M_X(t) M_Y(t)$.

Proof. $M_Z(t) = E[e^{tZ}] = E[e^{tX+tY}] = E[e^{tX} e^{tY}] = E[e^{tX}] E[e^{tY}] = M_X(t) M_Y(t)$, using Corollary 93. □

Example 133 (Sum of Poissons λ_1, λ_2).

$$M_Z(t) = M_X(t) M_Y(t) = e^{\lambda_1(e^t-1)} e^{\lambda_2(e^t-1)} = e^{(\lambda_1+\lambda_2)(e^t-1)},$$

so $Z \sim \text{Poisson}(\lambda_1 + \lambda_2)$ — the promise made in Example 41.

▸ **Lecture note.** The mgf turns the hard operation (convolution of densities, §3.6.1) into an easy one (multiplication). The same two lines show: a sum of independent normals is normal ($e^{\mu_1 t + \sigma_1^2 t^2/2} e^{\mu_2 t + \sigma_2^2 t^2/2}$ is again a normal mgf), and a sum of n independent Exp(λ) is Gamma(n, λ). And it is the engine of the CLT proof in Chapter 5.

4.6 Approximate Methods

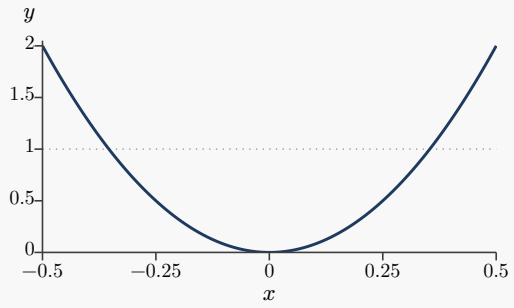
- X is a r.v., and $Y = g(X)$.
- We know μ_X and σ_X (or $\hat{\mu}_X$ and $\hat{\sigma}_X$).
- What are μ_Y and σ_Y, even approximately?

Delta methods (propagation of error):

$$Y = g(X) = g(\mu_X) + (X-\mu_X)g'(\mu_X) + \frac{1}{2!}(X-\mu_X)^2 g''(\mu_X) + \cdots \quad \text{(Taylor series)}$$
$$\approx g(\mu_X) + (X-\mu_X)g'(\mu_X) \quad \text{(1st-order approx.)},$$
$$\mu_Y \approx g(\mu_X), \qquad \sigma_Y^2 \approx \sigma_X^2 (g'(\mu_X))^2;$$
$$Y \approx g(\mu_X) + (X-\mu_X)g'(\mu_X) + \frac{1}{2!}(X-\mu_X)^2 g''(\mu_X) \quad \text{(2nd-order approx.)}, \qquad \mu_Y \approx g(\mu_X) + \frac{1}{2}\sigma_X^2 g''(\mu_X).$$

Example 134 (Very Nonlinear).

- $X \sim \text{Uniform}(-\frac{1}{2}, \frac{1}{2})$.
- $Y = g(X) = 8X^2$.



$$\mu_X = 0, \quad \sigma_X^2 = \frac{1}{12}, \quad \mu_Y = \int_{-1/2}^{1/2} 8x^2\,\mathrm{d}x = \frac{2}{3}, \quad \sigma_Y^2 = E[Y^2] - \mu_Y^2 = \int_{-1/2}^{1/2} (8x^2)^2\,\mathrm{d}x - \left(\frac{2}{3}\right)^2 = \frac{16}{45}.$$

However, by using the 1st-order approximation,

$$\mu_Y \approx g(\mu_X) = 8\mu_X^2 = 0, \qquad \sigma_Y^2 \approx \sigma_X^2 (g'(\mu_X))^2 = \frac{1}{12}(0) = 0.$$

By using the 2nd-order approximation,

$$\mu_Y \approx g(\mu_X) + \frac{1}{2}\sigma_X^2 g''(\mu_X) = 0 + \frac{1}{2} \times \frac{1}{12} \times 16 = \frac{2}{3}. \quad (4.1)$$

This is exact because (4.1) is exact! Indeed,

$$g(x) = g(x_0) + (x-x_0)g'(x_0) + \frac{1}{2!}(x-x_0)^2 g''(x_0) + \cdots = 8x_0^2 + (x-x_0)16x_0 + \frac{1}{2}(x-x_0)^2 16 = 8x^2.$$

If $X \sim \text{Uniform}(0.3, 0.5)$, then

$$\mu_X = .4, \quad \sigma_X^2 = \int_{.3}^{.5} x^2(5)\,\mathrm{d}x - (.4)^2 = 3.3333 \times 10^{-3},$$

$$\mu_Y = \int_{.3}^{.5} (8x^2)(5)\,\mathrm{d}x = 1.3067, \quad \sigma_Y^2 = \int_{.3}^{.5} (8x^2)^2(5)\,\mathrm{d}x - (1.3067)^2 = 0.13702.$$

By using the 1st-order approximation,

$$\mu_Y \approx g(\mu_X) = 8(.4)^2 = 1.28, \quad \sigma_Y^2 \approx \sigma_X^2 (g'(\mu_X))^2 = \frac{1}{300}(16 \times .4)^2 = 0.13653.$$

▸ **Lecture note.** The lesson: the delta method is excellent when g is **nearly linear over the range where X lives** (the second case: X in $[0.3, 0.5]$, where the parabola is nearly straight —

errors under 2%), and useless when X sits on a turning point of g (the first case: $g'(0) = 0$, so the first-order variance is 0 while the truth is 0.36). Always check $g'(\mu_X) \neq 0$ and that σ_X is small compared with the curvature scale.

Chapter summary

- **Mean:** $E(X) = \sum xp(x)$ or $\int xf(x)\,dx$, defined only if $E|X| < \infty$ (Cauchy has none). $E[g(X)] = \int g(x)f(x)\,dx$ — no need for $f_{g(X)}$. Linearity $E(a + \sum b_i X_i) = a + \sum b_i E(X_i)$ needs **no** independence.
- **Variance:** $\mathrm{Var}(X) = E(X-\mu)^2 = E(X^2) - \mu^2$; $\mathrm{Var}(a+bX) = b^2\,\mathrm{Var}(X)$.
- **Inequalities:** Markov $P(X \geq t) \leq EX/t$; Chebyshev $P(|X-\mu| \geq t) \leq \sigma^2/t^2$.
- **MSE = variance + bias².**
- **Covariance** $E(XY) - \mu_X\mu_Y$; $\mathrm{Var}(\sum a_i X_i) = \sum\sum a_i a_j \mathrm{Cov}(X_i, X_j)$. **Correlation** $\rho \in [-1,1]$, $|\rho| = 1$ iff linear; $\rho = 0$ does **not** imply independence.
- **Conditional:** $E(Y) = E[E(Y \mid X)]$; $\mathrm{Var}(Y) = \mathrm{Var}[E(Y \mid X)] + E[\mathrm{Var}(Y \mid X)]$; the best predictor of Y from X is $E(Y \mid X)$.
- **mgf:** $M(t) = Ee^{tX}$, $M^{(r)}(0) = EX^r$; $M_{a+bX}(t) = e^{at}M(bt)$; independent sums multiply mgfs.
- **Delta method:** $\mu_Y \approx g(\mu_X)$, $\sigma_Y \approx |g'(\mu_X)|\,\sigma_X$.

Distribution	Mean	Variance	mgf
$\mathrm{Ber}(p)$	p	$p(1-p)$	$1-p+pe^t$
$\mathrm{Bin}(n,p)$	np	$np(1-p)$	$(1-p+pe^t)^n$
$\mathrm{Geom}(p)$	$1/p$	$(1-p)/p^2$	$pe^t/(1-(1-p)e^t)$
$\mathrm{Poisson}(\lambda)$	λ	λ	$e^{\lambda(e^t-1)}$
$\mathrm{U}(a,b)$	$(a+b)/2$	$(b-a)^2/12$	$(e^{tb}-e^{ta})/t(b-a)$
$\mathrm{Exp}(\lambda)$	$1/\lambda$	$1/\lambda^2$	$\lambda/(\lambda-t)$, $t<\lambda$
$\mathrm{Gamma}(\alpha,\lambda)$	α/λ	α/λ^2	$(\lambda/(\lambda-t))^\alpha$
$N(\mu,\sigma^2)$	μ	σ^2	$e^{\mu t+\sigma^2t^2/2}$
$\mathrm{Beta}(a,b)$	$a/(a+b)$	$ab/((a+b)^2(a+b+1))$	—

Practice problems — Chapter 4

4.1 Find $E(X)$ and $\mathrm{Var}(X)$ for the number of heads in 3 fair tosses, directly from the pmf, and check against np and $np(1-p)$.

4.2 $X \sim \mathrm{Exp}(\lambda)$. Find $E(X^2)$ using the gamma trick of Example 86, then $\mathrm{Var}(X)$.

4.3 A die is rolled; you win X^2 dollars. Find the expected win.

4.4 X, Y independent with $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$. Find $\mathrm{Var}(2X - 3Y + 7)$.

4.5 In Example 110, find $\mathrm{Var}(X+Y)$.

4.6 Let $X \sim \mathrm{U}(0,1)$ and $Y = X^2$. Find $\mathrm{Cov}(X,Y)$ and ρ_{XY}. Is the relationship linear?

4.7 Use the mgf to find the mean and variance of $\mathrm{Exp}(\lambda)$.

4.8 A store gets $N \sim \mathrm{Poisson}(50)$ customers a day; each spends an amount with mean 20 and standard deviation 10, independently. Find the mean and standard deviation of daily revenue.

4.9 Use the delta method to approximate the mean and SD of $Y = \sqrt{X}$ when $\mu_X = 100$, $\sigma_X = 10$.

CHAPTER 5

Limit Theorems

We are getting into Statistics.
Great concepts and intuition are here.

▸ **Lecture note.** Two theorems carry the chapter. The **Law of Large Numbers** says the sample mean $\overline{X}_n$ gets close to μ. The **Central Limit Theorem** says **how** close, and with what shape: $\overline{X}_n$ is approximately $N(\mu, \sigma^2/n)$, whatever the distribution of a single X. Together they justify almost every statistical procedure you will learn next.

5.1 The Law of Large Numbers

- It is always believed, **subjectively**, that tossing a fair coin will produce **ultimately** a 0.5 heads proportion.
- The mathematician John Kerrich tried it in prison (in a WWII internment camp); he got 5067 heads out of 10,000 tosses.
- The LLN is a mathematical formulation of large sums.
- In particular, for coin tossing:

$$X_i \sim \text{Bernoulli}(0.5) \quad \text{(i.i.d.)}, \qquad \frac{1}{n}\sum_{i=1}^{n} X_i \quad \text{“approaches”} \quad 0.5.$$

- What is “approaches”, formally?

Definition 135 (Convergence in Probability). Let $Z_1, ..., Z_n$ be a sequence of r.v.s (s.r.v.). We say that Z_n **converges in probability** to α if

$$\lim_{n\to\infty} P(|Z_n - \alpha| > \varepsilon) = 0 \quad \forall \varepsilon > 0.$$

This is written in different ways:

$$Z_n \xrightarrow{p} \alpha, \qquad P(|Z_n - \alpha| > \varepsilon) \to 0 \quad \forall \varepsilon > 0, \text{as } n \to \infty.$$

▸ **Lecture note.** Read it as a game. You pick a tolerance ε (say 0.01). Then the chance that Z_n misses α by more than your tolerance can be made as small as you like by taking n large enough. It does **not** say Z_n eventually equals α, nor that every single sequence settles — only that big misses become improbable.

Theorem 136 (Weak Law of Large Numbers). If $X_1, ..., X_n$ is a s.r.v., independent with **existing** and common μ and σ^2 (but not necessarily identical), then $\overline{X}_n (= S_n/n) \xrightarrow{p} \mu$.

Proof.

$$\overline{X}_n = \frac{S_n}{n} = \frac{1}{n}\sum_{i=1}^{n} X_i, \quad E\left[\overline{X}_n\right] = \mu, \quad \operatorname{Var}\left[\overline{X}_n\right] = \frac{\sigma^2}{n},$$

$$P\Big(|\overline{X}_n - \mu| > \varepsilon\Big) \le \frac{\operatorname{Var}\left[\overline{X}_n\right]}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \quad \forall \varepsilon > 0 \quad \text{(Chebyshev's ineq.)},$$

$$\lim_{n\to\infty} P\Big(|\overline{X}_n - \mu| > \varepsilon\Big) = 0 \quad \forall \varepsilon > 0.$$

□

Note. The proof only uses $\operatorname{Var}\left(\overline{X}_n\right) = \sigma^2/n$, which needs the X_i to be **uncorrelated**, not fully independent. And the word "existing" (red in the original) matters: for the Cauchy there is no μ, and the law fails — see Example 138.

Example 137 (Repeated Measurements). A special case of the WLLN is when the X_i are i.i.d.:

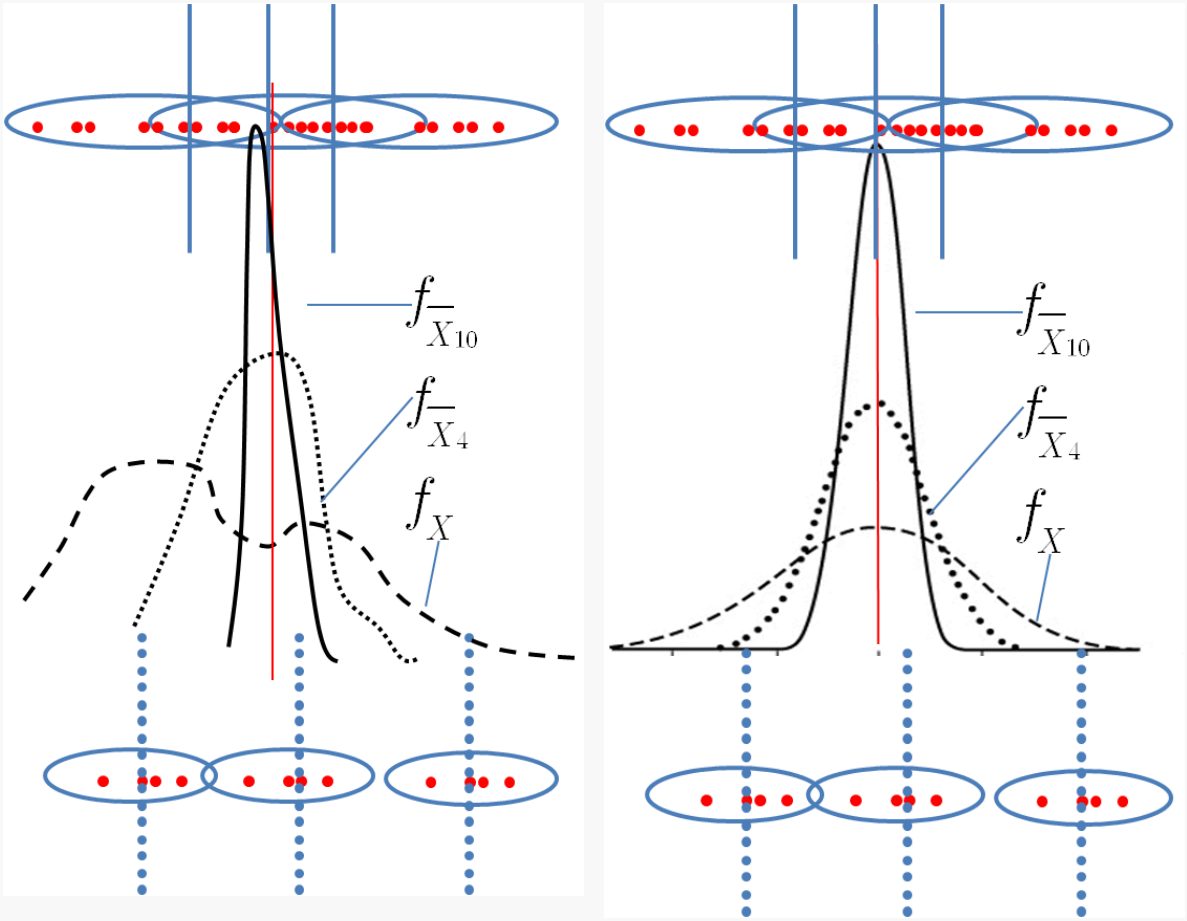
$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

For a particular sample $x_1, ..., x_n$, $\overline{X}_n$ becomes a number, not a r.v.:

$$\overline{X}_n(x_1, ..., x_n) = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

$$X \overset{\text{Sample}_1}{\to} x_1, x_2, ..., x_n, \qquad X \overset{\text{Sample}_2}{\to} x_1, x_2, ..., x_n, \qquad \vdots$$

Let's see the meaning of the WLLN for $\overline{X}_n$:



Each ellipse is one sample; its mean is one draw of $\overline{X}_n$. Whatever the shape of f_X (left: skewed and bimodal; right: symmetric), the densities $f_{\overline{X}_4}$ and $f_{\overline{X}_{10}}$ concentrate around μ (red line).

Example 138 (Normal vs. Cauchy).

- $X \sim N(0,1)$, $Y \sim$ Cauchy.
- One sample (500 obs.) from each.
- $G_n = \frac{1}{n}\sum_{i=1}^{n} x_i$, $C_n = \frac{1}{n}\sum_{i=1}^{n} y_i$, $n = 1, ..., 500$.

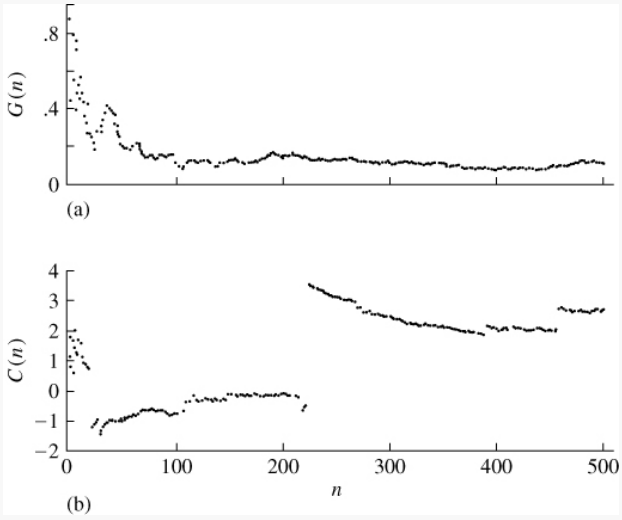


Figure 23: (a) Running mean of normal data settles at 0. (b) Running mean of Cauchy data keeps jumping — a single huge observation (near $n = 220$) moves the whole average (Rice, 3rd ed.).

Example 139 (Estimation of Moments). $m_r = E[X^r]$. Define

$$\hat{m}_r = \frac{1}{n}\sum_{i=1}^{n} X_i^r.$$

Then

$$E[\hat{m}_r] = \frac{1}{n}\sum_{i=1}^{n} E[X^r] = m_r, \qquad \hat{m}_r \xrightarrow{p} m_r.$$

Example 140 (Monte Carlo Integration). How do we calculate the integral $I = \int_a^b g(x)\,\mathrm{d}x$? Let $X \sim \text{Uniform}(a, b)$; generate a sample x_i, $i = 1, ..., n$. Define

$$\hat{I} = \frac{1}{n}\sum_{i=1}^{n} g(X_i),$$

$$E\left[\hat{I}\right] = E[g(X)] = \int_a^b g(x) f_X(x)\,\mathrm{d}x = \int_a^b g(x)\frac{1}{b-a}\,\mathrm{d}x = \frac{1}{b-a}I,$$

$$\hat{I} \xrightarrow{p} \frac{1}{b-a}I, \qquad I \approx (b-a)\hat{I}.$$

▸ **Lecture note.** Try it with $g(x) = 4\sqrt{1-x^2}$ on $[0, 1]$, whose integral is π. The error of $\hat{I}$ shrinks like $\sigma/\sqrt{n}$ (Chapter 4) **regardless of the dimension** of the integral — which is why Monte Carlo beats grid methods for integrals over many variables, and why it powers physics simulation, finance and Bayesian statistics. Appendix A builds the tools.

5.2 Convergence in Distribution and the Central Limit Theorem (CLT)

Definition 141. Let $X_1, ..., X_n$ be a s.r.v. with cdfs $F_1, ..., F_n$, and let X be another r.v. with cdf F. We say that X_n **converges in distribution** to X if

$$\lim_{n\to\infty} F_n(x) = F(x),$$

except at discontinuities of F. This can be written as

$$X_n \xrightarrow{d} X, \quad F_n(x) \to F(x) \text{ as } n \to \infty, \quad \lim_{n\to\infty} P(X_n \le x) = P(X \le x).$$

Theorem 142. For the setting above, if $M_n(t) \to M(t)$, then $F_n(x) \to F(x)$ at all points of continuity.

Proof. Omitted. □

Theorem 143 (CLT). Let $X_1, ..., X_n$ be i.i.d., with common μ, σ^2, F, M. Then the standardized version

$$Z_n = \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\sum_i (X_i - \mu)/\sigma}{\sqrt{n}}$$

converges in distribution to a standard normal $N(0,1)$, i.e. $Z_n \xrightarrow{d} Z$.

Before the rigorous proof, notice:

- This is regardless of F!!!
- If the WLLN shows that $\overline{X}_n$ **goes** to μ (in probability), the CLT shows how it **fluctuates** around μ (i.e. its distribution and rate).
- It is more precise than Chebyshev (Ex. 105).
- There are several other versions of the CLT.

⋆ **Intuition.** Each X_i is a small random push. Adding many independent pushes washes out every detail of their individual shape — skew, bumps, flat tops — except two numbers: the average push μ and its spread σ. What survives is a bell. This is why measurement errors (sums of many tiny disturbances), heights (many genes), and sample means all look normal.

Example 144. $X \sim \text{Uniform}\left(-\sqrt{3}, \sqrt{3}\right)$; then $\mu = 0$, $\sigma = 1$, and

$$Z_n = \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{S_n}{\sqrt{n}}.$$

Obtain 1000 samples, each with 12 observations ($n = 12$). We have 1000 values of Z_{12}. Notice that

$$Z_{n_{\max}} = \frac{nX_{\max}}{\sqrt{n}} = \sqrt{n}X_{\max} = \sqrt{12}\sqrt{3} = 6.$$

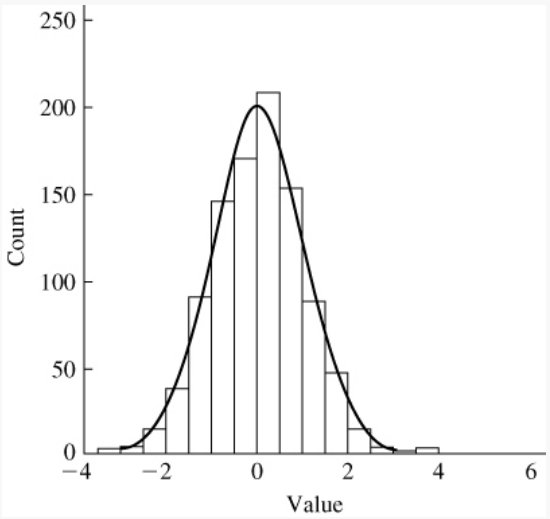


Figure 24: Histogram of 1000 values of Z_{12} with the $N(0,1)$ density: already indistinguishable from normal at $n = 12$ (Rice, 3rd ed.).

▸ **Lecture note.** So Z_{12} can never exceed 6 in absolute value — a normal can — but values beyond 4 have normal probability 6×10^{-5} anyway. The old computer trick "add 12 uniforms on $(0,1)$ and subtract 6" to make an approximately $N(0,1)$ number is exactly this example.

Example 145 (Measurement Error).

$$\begin{aligned} P\big(|\overline{X}_n - \mu| < c\big) &= P\big(-c < \overline{X}_n - \mu < c\big) \\ &= P\left(\frac{-c}{\sigma/\sqrt{n}} < \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} < \frac{c}{\sigma/\sqrt{n}}\right) \\ &\approx \Phi\left(\frac{c}{\sigma/\sqrt{n}}\right) - \Phi\left(\frac{-c}{\sigma/\sqrt{n}}\right). \end{aligned}$$

▸ **Lecture note.** Turn it around to **plan** an experiment: to be 95% sure the average is within c of the truth, set $c/(\sigma/\sqrt{n}) = 1.96$, i.e. $n = (1.96\sigma/c)^2$. Halving the error needs **four times** the data — the $\sqrt{n}$ law.

Example 146. $X \sim \text{Binomial}(n, p)$, $n = 100$, $p = 0.5$. E.g.,

$$P(X \geq 60) = \sum_{k=60}^{100} \binom{n}{k} p^k (1-p)^{100-k}$$

is computationally expensive. However,

$$X = \sum_{i=1}^{n} I_i \quad (I_i \sim \text{Bernoulli}(p)), \qquad \frac{X - np}{\sqrt{p(1-p)}\sqrt{n}} \xrightarrow{d} N(0,1).$$

Then

$$P(X \geq 60) = P\left(\frac{X - np}{\sqrt{p(1-p)}\sqrt{n}} \geq \frac{60 - 100/2}{\sqrt{\frac{1}{2}\cdot\frac{1}{2}}\sqrt{100}}\right) \approx P(Z \geq 2) = 1 - \Phi(2) = 0.0228.$$

▸ **Lecture note.** **Continuity correction.** The exact answer is 0.0284. The approximation is off by 20% because X is discrete: $\{X \geq 60\}$ is the same event as $\{X \geq 59.5\}$, and a continuous curve should start at the half-integer. Using 59.5 gives $P(Z \geq 1.9) = 0.0287$ — almost exact. Use the correction whenever you approximate a discrete r.v. by the normal.

Proof. (**Proof of the CLT.**) Let $Y_i = (X_i - \mu)/\sigma$, so $E[Y] = 0$, $E[Y^2] = 1$, and

$$Z_n = \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\sum_i (X_i - \mu)/\sigma}{\sqrt{n}} = \frac{\sum_i Y_i}{\sqrt{n}}.$$

By Lemmas 132 and 130,

$$M_{\sum_i Y_i}(t) = (M_Y(t))^n, \qquad M_{Z_n}(t) = M_{\sum_i Y_i}\big(t/\sqrt{n}\big) = \big(M_Y\big(t/\sqrt{n}\big)\big)^n.$$

Expand M_Y in a Taylor series:

$$\begin{aligned} M_Y\left(\frac{t}{\sqrt{n}}\right) &= \underbrace{M_Y(0)}_{E[Y^0]=1} + \left(\frac{t}{\sqrt{n}}\right)\underbrace{M_Y'(0)}_{E[Y]=0} + \frac{1}{2!}\left(\frac{t}{\sqrt{n}}\right)^2 \underbrace{M_Y''(0)}_{E[Y^2]=1} + \sum_{k=3}^{\infty} \frac{M_Y^{(k)}(0)}{k!}\left(\frac{t}{\sqrt{n}}\right)^k \\ &= 1 + \left(\frac{t}{\sqrt{n}}\right)^2 \left[\frac{1}{2} + \sum_{k=3}^{\infty} \frac{M_Y^{(k)}(0)}{k!}\left(\frac{t}{\sqrt{n}}\right)^{k-2}\right] = 1 + \frac{1}{n}t^2\left(\frac{1}{2} + \underbrace{r_n}_{\to 0 \text{ as } n\to\infty}\right). \end{aligned}$$

Therefore,

$$M_{Z_n}(t) = \left(1 + \frac{1}{n}t^2\left(\frac{1}{2} + r_n\right)\right)^n, \qquad \lim_{n\to\infty} M_{Z_n}(t) = e^{t^2/2},$$

using $(1 + a_n/n)^n \to e^a$ when $a_n \to a$. This is the mgf of $N(0,1)$ (Example 129), and Theorem 142 completes the proof. □

Chapter summary

- **Convergence in probability** $Z_n \xrightarrow{p} \alpha$: $P(|Z_n - \alpha| > \varepsilon) \to 0$ for every $\varepsilon > 0$.
- **WLLN:** $\overline{X}_n \xrightarrow{p} \mu$ when μ and σ^2 exist; proof: Chebyshev with $\operatorname{Var}\left(\overline{X}_n\right) = \sigma^2/n$. Fails for the Cauchy.
- Consequences: sample moments estimate population moments; Monte Carlo integration $\int_a^b g \approx (b-a) \cdot \text{mean of } g(X_i)$.
- **Convergence in distribution** $X_n \xrightarrow{d} X$: $F_n(x) \to F(x)$ at continuity points; implied by convergence of mgfs.
- **CLT:** $\left(\overline{X}_n - \mu\right)/(\sigma/\sqrt{n}) \xrightarrow{d} N(0,1)$ for any i.i.d. with finite variance. In practice: $\overline{X}_n \approx N(\mu, \sigma^2/n)$ and $S_n \approx N(n\mu, n\sigma^2)$.
- Normal approximation to the binomial: $X \approx N(np, np(1-p))$, with continuity correction.

Practice problems — Chapter 5

5.1 A fair coin is tossed n times. Use Chebyshev to find n that guarantees $P\left(|\overline{X}_n - 0.5| > 0.05\right) \le 0.05$. Then use the CLT. Compare.

5.2 The lifetime of a bulb is Exp with mean 1000 h. Using the CLT, approximate the probability that the average lifetime of 100 bulbs exceeds 1100 h.

5.3 A die is rolled 600 times. Approximate the probability of between 90 and 110 sixes (inclusive), with continuity correction.

5.4 Write the Monte Carlo estimator of $\int_0^1 e^{-x^2}\,\mathrm{d}x$ and give its standard deviation for n samples in terms of $\operatorname{Var}\left(e^{-U^2}\right)$.

5.5 Explain in two sentences why the average of Cauchy observations does not converge.

APPENDIX A

Simulation

A very nice practical chapter.

This chapter follows Knuth (1997, Vol. 2, Ch. 3) and DeGroot and Schervish (2002, Ch. 11).

▸ **Lecture note.** The original's code is MATLAB. Each listing is kept as is, and followed by an equivalent in Python (NumPy) — added, so the examples can be run without a MATLAB licence.

A.1 Generating r.v. by Simulation

Starting from the 1920s, people thought of generating random numbers from ready-made tables. John von Neumann in 1946 (although it is very deterministic, it looks like good scrambling that carries no physical significance) — the **middle-square** method:

1. start with a number, e.g. 5772156649;
2. square it: 33317792380594909201;
3. take the middle: 7923805949.

Such methods are called **pseudorandom** or **quasirandom**.

Current methods are of two steps.

First: generate a Uniform$(0,1)$ random number using the **linear congruential** method:

$$\begin{aligned} m,&\text{ the modulus;} && 0 < m,\\ a,&\text{ the multiplier;} && 0 \le a < m,\\ c,&\text{ the increment;} && 0 \le c < m,\\ X_0,&\text{ the seed;} && 0 \le X_0 < m, \end{aligned}$$

$$X_{n+1} = (aX_n + c) \bmod m, \quad n \ge 0.$$

E.g., $m = 10$, $X_0 = a = c = 7$: $X_n = 7, 6, 9, 0, 7, 6, 9, 0, \ldots$ The choice of the parameters is a matter of research. Typical values are $a = 7^5$, $c = 0$, $m = 2^{31} - 1$. Then $U_n = X_n/m$ is (approximately) Uniform$(0,1)$.

▸ **Lecture note.** The toy example repeats with period 4 — the sequence can never be longer than m, and a bad choice of a, c makes it much shorter. With $a = 7^5 = 16807$, $c = 0$, $m = 2^{31} - 1$ (the Park–Miller "minimal standard") the period is $m - 1 \approx 2.1 \times 10^9$. Modern libraries use better generators (e.g. the Mersenne Twister or PCG), but the principle — a deterministic recursion that **looks** random — is the same. Same seed, same sequence: that is a feature, because it makes simulations reproducible.

Second: use the generated uniform r.v. to convert it to any other r.v. by one of the following two methods:

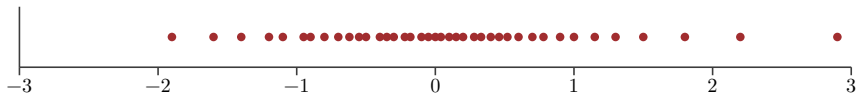
- **Transformation methods** (end of Ch. 2), if the cdf is known: $U \sim \text{Uniform}(0,1)$, $X = F^{-1}(U) \Longrightarrow F_X = F$ (Proposition 53).
- **Rejection method**, if the cdf cannot be found in closed form (only the pdf is known).

▸ **Lecture note. Rejection method in one paragraph** (the original refers to it without details). Suppose $f(x) \le c\,g(x)$ for a density g you can already sample. Draw $Y \sim g$ and $U \sim \text{U}(0,1)$; **accept** $X = Y$ if $U \le f(Y)/(c\,g(Y))$, otherwise try again. Geometrically: throw points uniformly under the curve $c\,g$ and keep those that fall under f — their x-coordinates have density f. On average c tries are needed per accepted value, so choose g close to f.

MATLAB Code A.1

```
n=100; mu=0; sigma=1;
x = random('normal', mu, sigma, [n,1]);
plot(x, zeros([length(x),1]), '.r');
```

```
import numpy as np, matplotlib.pyplot as plt
rng = np.random.default_rng(0)
x = rng.normal(0, 1, 100)
plt.plot(x, np.zeros_like(x), '.r'); plt.show()
```



Normal data on a line: dense in the middle, sparse in the tails. The density is visible only as the **crowding** of points — which motivates the histogram.

A.2 Histograms

Let's define first the indicator function

$$I_{(c)} = \begin{cases} 1 \text{ if } c \text{ is } T \\ 0 \text{ if } c \text{ is } F, \end{cases} \qquad I_{(c)} \sim \text{Bernoulli}(\Pr(c)).$$

For data $x_1, ..., x_n$, divide the data range T into K equal regions of equal width Δ (so that $K = T/\Delta$):

$$T_k = [t_0 + \Delta k, t_0 + \Delta(k+1)[= [t_k, t_{k+1}[, \quad k = 0, ..., K-1.$$

Notice: decreasing Δ increases K.

We have three versions of the histogram:

$$N_k = \sum_{i=1}^{n} I_{(X_i \in T_k)} \qquad \text{(counts)},$$

$$R_k = \frac{N_k}{n} \qquad \text{(relative counts)},$$

$$f_k = \frac{N_k}{\Delta n} \qquad \text{(normalized)}.$$

$$\text{Area(under } N_k) = \sum_k \Delta N_k = \Delta \sum_k N_k = \Delta n.$$

$$R_k \xrightarrow{p} \Pr(X \in T_k), \qquad f_k = \frac{\Pr(X \in T_k)}{\Delta} \quad (\text{for large } n) \approx \frac{f_X(t_k)\Delta}{\Delta} = f_X(t_k) \quad (\text{for small } \Delta).$$

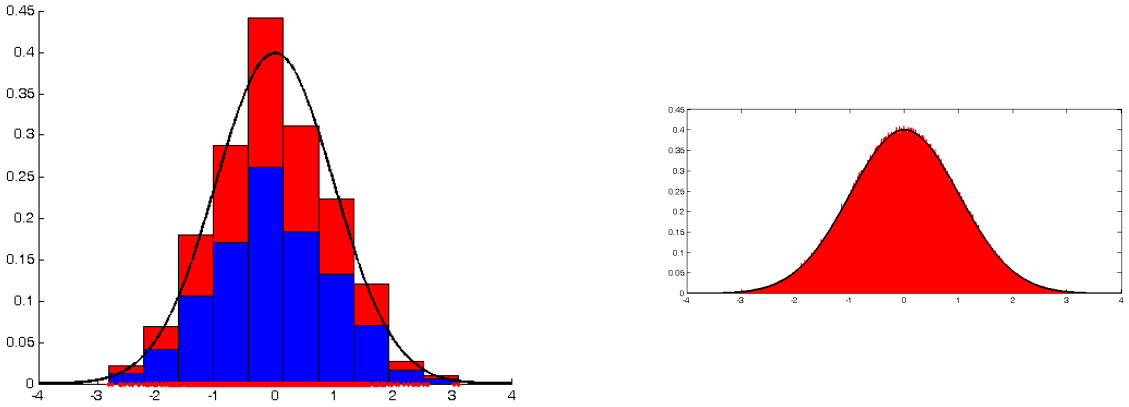


Figure 25: Left: $n = 100$ normal observations; relative counts R_k (blue) and normalized f_k (red), with the $N(0,1)$ density. Only f_k is on the same scale as the density. Right: with a very large n and a small Δ, f_k is the density.

▸ **Lecture note.** Why f_k approximates the density: N_k is $\text{Bin}(n, p_k)$ with $p_k = \Pr(X \in T_k)$ (Example 59), so $R_k = N_k/n \to p_k$ by the WLLN; and $p_k \approx f_X(t_k)\Delta$ for small bins. Two limits, two requirements: n large (so R_k is close to p_k) **and** Δ small (so p_k/Δ is close to f). Shrinking Δ with n fixed makes each bin hold few points and the histogram gets noisy — the bias–variance trade-off of Theorem 106 again.

MATLAB Code A.2

```
n=1000; mu=0; sigma=1;
x = random('normal', mu, sigma, [n,1]);
figure; hold on;
[N, xout] = hist(x);
bar(xout, N/(n*(xout(2)-xout(1))), 'barwidth', 1, 'facecolor', 'r');
bar(xout, N/n, 'barwidth', 1, 'facecolor', 'b');
z = -4:.01:4;
y = 1/(sqrt(2*pi)*sigma) * exp(-(z-mu).^2 / (2*(sigma^2)));
plot(z, y, 'k', 'LineWidth', 2);
plot(x, zeros([length(x),1]), '.r');
```

```
x = rng.normal(0, 1, 1000)
N, edges = np.histogram(x, bins=10); d = edges[1] - edges[0]; mid = edges[:-1] + d/2
plt.bar(mid, N/(len(x)*d), width=d, color='r'); plt.bar(mid, N/len(x), width=d, color='b')
z = np.linspace(-4, 4, 801); plt.plot(z, np.exp(-z**2/2)/np.sqrt(2*np.pi), 'k', lw=2); plt.show()
```

Correction to the original. The original density line is `1/(sqrt(2*pi*sigma))`; the normal constant is `1/(sqrt(2*pi)*sigma)` (the two agree only for $\sigma = 1$). Fixed above.

A.3 Population Parameters

$$\alpha = \int_R g(x) f_X(x)\,\mathrm{d}x, \qquad \hat{\alpha} = \frac{1}{M}\sum_{m=1}^{M} I_{(X_m \in R)} g(X_m),$$

$$I_{(X_m \in R)} g(X_m) = \begin{cases} g(X_m) & X_m \in R \\ 0 & X_m \notin R \end{cases}, \qquad E\left[I_{(X_m \in R)} g(X_m)\right] = \int_R g(x) f_X(x)\,\mathrm{d}x, \qquad \hat{\alpha} \xrightarrow{p} \alpha.$$

MATLAB Code A.3: Monte Carlo (MC) Simulation

```
P = {x1, ..., xM};   % Data Generated
S = 0;
for m = 1:M
    if (R(xm))
        S = S + g(xm);
    end
end
ret = S/M;
```

```
P = rng.normal(0, 1, M)              # data generated
ret = np.mean(np.where(R(P), g(P), 0))
```

General case:

$$\int_R g(x) f_X(x)\,\mathrm{d}x \approx \frac{1}{M}\sum_{m=1}^{M} I_{(X_m\in R)} g(X_m).$$

Special cases:

$$\int g(x) f_X(x)\,\mathrm{d}x \approx \frac{1}{M}\sum_{m=1}^{M} g(X_m) \qquad (\mu_g),$$

$$\int x f_X(x)\,\mathrm{d}x \approx \frac{1}{M}\sum_{m=1}^{M} x_m \qquad (\mu_X),$$

$$\int_R f_X(x)\,\mathrm{d}x \approx \frac{1}{M}\sum_{m=1}^{M} I_{(X_m\in R)} \quad (\Pr(X\in R)).$$

Other cases: $F^{-1}(0.5) \approx y^{(M/2)}$ (median), the middle of the sorted values.

> **General rule:** generate a pseudo-population $P = \{x_1, ..., x_M\}$ for very large M, then treat the data as if it is the population to calculate your function.

Different variances and MC size M.

$$X, \mu_X, \sigma_X, ..., \alpha_X, \qquad Y, \mu_Y, \sigma_Y, ..., \alpha_Y,$$
$$P_k, \hat{\mu}_k, \hat{\sigma}_k, ..., \hat{\alpha}_k, \qquad \mathrm{sd}(\hat{\alpha}), \hat{\mathrm{sd}}(\hat{\alpha}).$$

In each MC repetition k, we get a pseudo-population P_k, and hence a different histogram and $\hat{\alpha}_k$ (because of limited M):

$$\hat{\mathrm{sd}}(\hat{\alpha}) = \sqrt{\frac{1}{K-1}\sum_k \left(\hat{\alpha}_k - \overline{\hat{\alpha}}\right)^2}.$$

In the case of $\hat{\alpha} = \frac{1}{M}\sum_{m=1}^{M} y_m$, we can get $\mathrm{sd}(\hat{\alpha})$ from a single MC repetition by

$$\mathrm{sd}(\hat{\alpha}) = \frac{1}{\sqrt{M}}\,\mathrm{sd}(y), \qquad \hat{\mathrm{sd}}(\hat{\alpha}) = \frac{1}{\sqrt{M}}\sqrt{\frac{1}{M-1}\sum_m (y_m - \overline{y})^2}.$$

MATLAB Code A.4: MC variation

```
M=1000; mu=0; sigma=1; K=100;
muhat = zeros([1,K]);
for k = 1:K               % repeating MC
    P = random('normal', mu, sigma, [M,1]);
    S = 0;
    for m = 1:M
        S = S + P(m);
    end
    muhat(k) = S/M;
end
% Shorter:
P = random('normal', mu, sigma, [M,K]);
muhat = mean(P, 1);
```

```
muhat = rng.normal(0, 1, (1000, 100)).mean(axis=0)   # K = 100 MC repetitions
print(muhat.std(ddof=1), 1/np.sqrt(1000))            # both about 0.0316
```

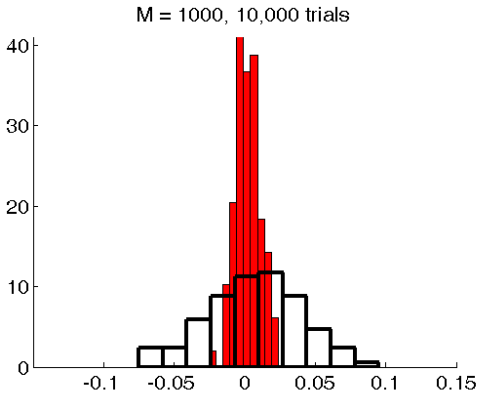


Figure 26: 10 000 repetitions of $\hat{\mu}$: with $M = 1000$ (red) the estimates cluster within about ± 0.03 of 0; with a smaller M (outlined) they spread more — $\mathrm{sd}(\hat{\mu}) = \sigma/\sqrt{M}$.

A.4 Statistics

Example, an order statistic: $Y = X^{(n)}$ $(Y = Y(n))$. In MC simulation, for each trial m we generate a dataset

$$D_m = \{x_1, ..., x_n\}, \quad m = 1, ..., M,$$

from which we calculate a single value of our statistic, $y_m = x^{(n)}$. Then $X, \mu_X, \sigma_X, ..., \alpha_X$; $Y, \mu_Y, \sigma_Y, ..., \alpha_Y$; $P_k, \hat{\mu}_k, \hat{\sigma}_k, ..., \hat{\alpha}_k$; $\mathrm{sd}(\hat{\alpha})$, $\hat{\mathrm{sd}}(\hat{\alpha})$ are for a particular n.

MATLAB Code A.5: Order Statistics MC

```
M=1000; n=100; mu=0; sigma=1;
y = zeros([M,1]);
for m = 1:M
    D = random('normal', mu, sigma, [n,1]);
    y(m) = max(D);
end
```

```
y = rng.normal(0, 1, (1000, 100)).max(axis=1)        # M = 1000 datasets of n = 100
```

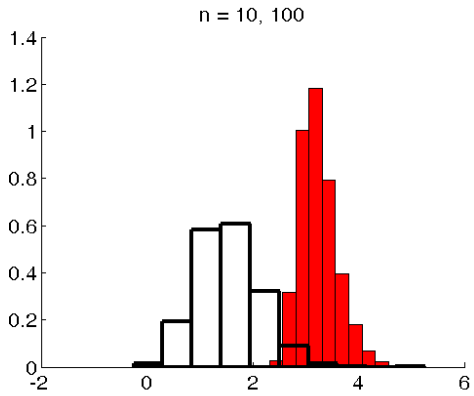


Figure 27: The distribution of the sample maximum for $n = 10$ (outlined) and $n = 100$ (red): it moves right and narrows as n grows. Compare with the exact density $nf(x)F(x)^{n-1}$ of §3.7.

A.5 Conditional Probability

We will condition on a non-zero-probability event e:

$$\alpha = \int_R g(x) f_X(x \mid e) = \frac{\int_R g(x) f_{XE}(x, e)}{\Pr(e)}, \quad \hat{\alpha} = \frac{\frac{1}{M}\sum_{m=1}^{M} I_{(e_m \ \& \ X_m \in R)} g(X_m)}{\frac{1}{M}\sum_{m=1}^{M} I_{(e_m)}} = \frac{\sum_{m=1}^{M} I_{(e_m \ \& \ X_m \in R)} g(X_m)}{\sum_{m=1}^{M} I_{(e_m)}}.$$

It is as if we generate a pseudo-population $P = \{x_1, ..., x_M\}$ ($|P| = M$), then make up the new dataset of size $\sum_{m=1}^{M} I_{(e_m)}$:

$$P' = \{x \mid x \in P \ \ \& \ \ e(x) = T\}.$$

MATLAB Code A.6: Algorithm: Cond. Prob. MC

```
P = {x1, ..., xM};   % Data Generated
S = 0; Mprim = 0;
for m = 1:M
    if (e(xm))
        Mprim = Mprim + 1;
        if (R(xm))
            S = S + g(xm);
        end
    end
end
ret = S/Mprim;
```

```
keep = e(P)                                          # condition on the event
ret = np.sum(np.where(R(P[keep]), g(P[keep]), 0)) / keep.sum()
```

Example (conditional MC — added). Check Example 20 by simulation: draw two balls without replacement from $\{R, R, R, B\}$, 10^5 times; among draws whose first ball is red, the fraction with a red second ball should be $P(R_2 \mid R_1) = 2/3$.

```
urn = np.array([1, 1, 1, 0])
draws = np.array([rng.permutation(urn)[:2] for _ in range(100_000)])
first_red = draws[:, 0] == 1
print(draws[first_red, 1].mean())                    # about 0.667
```

▸ **Lecture note.** Conditional MC is "throw away the rows where e did not happen". Its cost: only a fraction $\Pr(e)$ of the samples survive, so the effective sample size is $M \Pr(e)$. For rare

events ($\Pr(e) = 10^{-4}$) you need enormous M — which is why specialised methods (importance sampling, MCMC) exist.

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